They Say
They Wore Them

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# They Say
# They Wore Them

*The Hidden Psychology of Clear Aligner Compliance and the Clinical Protocols That Make Treatment Work*

ENZO PASCIUTI ACADEMY

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INTRODUCTION

# Why I Wrote This Book

There's a moment during aligner treatment when we realize the problem isn't the planned tooth movement. The patient isn't wearing their aligners as agreed, skips their orthodontic elastics, or stops wearing their retainer. We can explain once again what they need to do. But if that explanation hasn't worked so far, why would it work today?

I wrote this book to answer a question that has stayed with me for years: **What can I actually do when a patient doesn't follow instructions?** I wanted to move beyond simply recognizing the problem and find a way to address it during the appointment, when the patient is in front of me and I have to decide what to say and what next step to suggest.

For a while, I thought the difficulties I was having were mostly my fault. Maybe I didn't know how to communicate effectively. In Italy, at least in the conversations I had with other orthodontists, we talked a lot about tooth movements, protocols, and clinical outcomes, but little about what to say when a patient didn't follow instructions.

Reading discussions in Facebook groups for orthodontists working with clear aligners made me realize I wasn't alone. Even clinicians with far more experience than I had were facing the same challenges and looking for better ways to communicate with patients and get them actively involved in treatment. Many practiced in the United States, where considerable emphasis is placed on patient communication. Yet even they couldn't take patient cooperation for granted. Some things, it seems, are the same everywhere. Recognizing my own struggles in other people's

experiences helped me feel less alone and gave me the push to share what has worked in my practice over the years.

The scripts you'll read grew out of eighteen years of experience with aligners. They aren't phrases I dreamed up at a desk. They're the result of real conversations, successful attempts, and failed attempts. Over time, I changed the questions I asked, the explanations I gave, and the ways I presented the consequences of a choice, trying to understand what actually helped a patient get back on track with treatment.

I also noticed that many of those conversations kept recurring. The patients and their explanations changed, but I often found myself facing the same problems and having to decide all over again how to address them. So I began organizing the responses that had worked, trying to understand not just which words to use, but when to say them, what to check, and what to do at the next appointment. That's how the scripts became part of protocols: pathways that help my team and me stay on the same course. They don't eliminate the need to assess each case individually, but they give us a clearer starting point and leave fewer decisions to the fatigue or rush of the day.

In these pages, you'll also find a few communication tactics. Sometimes I use what I would honestly call a *white lie*: a simple reason that appeals to what matters to that patient, even though it isn't a scientific explanation of the problem. I've chosen to say so openly because I don't want those phrases mistaken for clinical evidence. They must never be used to conceal a risk, distort a diagnosis, or prevent the patient from making an informed choice. Their purpose is to help the patient do something useful that they already know they should do but struggle to keep doing over time.

**No script works for everyone every time, and none guarantees 100% adherence.** The strategies I suggest need to be chosen and adapted to the social and cultural context in which we work. You don't need to use them all or repeat them word for word: they can give you an idea for how to handle a situation you haven't been able to resolve until now.

I'd like this book to be a tool you keep close at hand in your daily practice: a collection of ideas to choose from, adapt, and put to the test. **The goal is to leave the appointment with one more way to help the patient stay on track and bring treatment to a conclusion you both agree on.**

CHAPTER

# 1

# The Day I Realized Moving Teeth Wasn't Enough

*"The only true voyage of discovery would be not to visit strange lands but to possess other eyes."*

— MARCEL PROUST

If you've decided to read this book, you probably already use clear aligners or are seriously considering adding them to your clinical practice. Either way, before I tell you my story, I'd like to ask you a question.

Have you ever gone home after a day at work with a case you couldn't stop thinking about?

Maybe it was a treatment you had planned with extreme care. You'd spent hours on the plan, checked every detail, revised the digital treatment setup several times, and when you finally clicked the approval button, you were convinced you'd done an

excellent job. Then the patient comes back a few months later. You ask how things are going, and they give you the answer every orthodontist hears at least a dozen times a day:

*"Great, Doctor."*

You get them settled in the chair, begin the exam, and within seconds you realize something is off. I've had that feeling many times myself. For years, though, whenever I faced a situation like that, my reaction was always the same: look for a technical explanation. I thought I'd made a mistake in the treatment plan, underestimated a tooth movement, chosen an ineffective attachment, or designed biomechanics that weren't sophisticated enough. I was convinced that if I studied more, I'd find the answer.

I didn't know I was asking the wrong question.

## Let's Start at the Beginning

When I think back to the young man I was more than twenty years ago, I realize I held a belief that was as simple as it was reassuring: to become a good orthodontist, I just had to keep learning. It wasn't a naive or superficial thought. On the contrary, it drove everything I did. It gave me an extraordinary sense of control, because it meant my future depended almost entirely on how much effort I was willing to put into my education.

If something didn't work as I'd expected, the explanation always seemed the same to me: I hadn't learned everything I needed to know yet. If I ran into a difficulty, it meant there was a book I hadn't read, a conference I hadn't attended, or a colleague I could still learn something from. Even when I saw others getting better results than I was, I told myself they had simply discovered something I didn't know yet and that, sooner or later, I'd come to understand it too.

Looking back now, I realize that belief stayed with me through much of my career and, deep down, is still with me. I believe that devoting time to study is one of the greatest ways an orthodontist can show respect for their patients. What has changed over time isn't how much I value education, but what the word learning means to me. At first, I was convinced that learning meant accumulating technical knowledge. It would take me many years to understand that there was another kind of learning, perhaps even more important, that no one had ever told me about.

When I was taking my first steps in the profession, orthodontics was undergoing a profound transformation. Of all the innovations of those years, one captured my attention completely: clear aligners. Some people regarded them with curiosity,

others with deep skepticism, and still others considered them little more than an esthetic alternative to brackets on the social six. I, on the other hand, immediately had the sense that something much more important was happening. For the first time, it wasn't just a treatment tool that was changing: the very way orthodontists thought was changing.

With fixed braces, you could continually adjust course during treatment. With aligners, you had to envision the outcome before you even began. It was a radical shift in perspective that forced you to plan for the future rather than simply manage the present, and it was probably this intellectual challenge that made me fall in love with them right away.

Deep down, I've always been drawn to difficult things. Not because I liked making life harder for myself, but because whenever I encountered something that seemed impossible, the same question would arise in me: *"How can I make it work?"*

## Searching for the Answer

It was that curiosity that drove me to immerse myself in the world of aligners. I underlined every scientific article, read every book with a pencil in hand, and treated every conference as an opportunity I couldn't miss. If I heard about a colleague who had developed an interesting approach to biomechanics or a speaker who could tackle a problem from a different perspective, I immediately started looking into it, wherever they were. Over the years, that spirit took me across much of the world. I wasn't interested in collecting certificates to hang on the wall. I wanted to come home with at least one insight that, on Monday morning, could make even one treatment better than I would have delivered the previous Friday.

Every trip was an investment of time, energy, and money. It meant leaving the practice, rearranging my work, giving up time with my family, spending hours in airports, enduring endless flights, and often getting home Sunday evening only to be back at the practice Monday morning. Yet I never saw any of it as a sacrifice. It was simply the price I was happy to pay to keep growing.

Deep down, I was convinced that if I kept gathering all those fragments of experience, sooner or later I'd complete the puzzle. I truly believed that somewhere there was a formula that could make every treatment perfectly predictable, and that my job was simply to find it.

Then I began to experience something I think everyone who genuinely loves their work has in common: the more I learned, the more questions I had. Every answer inevitably raised another question, every certainty was challenged by the

next patient, and whenever I thought I had finally understood a subject in depth, a case came along that forced me to rethink everything.

Of course, back then I didn't take all this in stride the way I do today. Whenever a treatment took a different course from the one I had envisioned, my reaction was almost automatic: I was convinced the mistake was mine. If the outcome didn't match what I had planned, it meant I still lacked some skill. In my mind, the solution was always the same: study more, take another course, read another book, board another plane to listen to someone who might have already found the answer I was still looking for.

That belief had helped me grow, but it also concealed a small trap: I thought every problem must have a technical solution. Without realizing it, I was looking for biomechanical answers to questions that belonged to an entirely different realm.

## I Must Be the Problem

Meanwhile, my practice was growing, my cases were becoming more numerous and complex, and the first invitations to speak at conferences began to arrive. I loved teaching because preparing a lecture forced me to organize my thoughts and understand what I was doing even better. Yet just as my career seemed to offer more and more validation from the outside, I still had a feeling inside that was hard to explain.

The first time someone gave it a name, I wasn't in a medical office, at a conference, or in a session with a psychologist. I was at home, having dinner with two friends.

Alice and Federico are the founders of Geldis, a company born from an idea closely connected to my work: creating a gel specifically for the daily cleaning of aligners. Over time, though, our relationship had grown well beyond work, and we had become friends.

I don't remember exactly how the conversation started. We were probably talking about work, courses, or one of the many conferences where I spoke. At some point, I told them something I had always considered a quirk of my personality.

I explained that, despite my years of experience, I still didn't feel completely comfortable stepping onto a stage. It wasn't fear of public speaking, and it wasn't uncertainty about the material, either: I prepared every presentation with almost obsessive care and knew the subjects I covered inside and out. It was something else.

I could be standing before a thousand or two thousand people who had come to hear me—colleagues who had paid for a course, traveled, and devoted a day of their lives to learning something from me. During breaks, they would stop me, ask ques-

tions, and tell me how something I'd suggested had changed the way they planned treatments. Yet part of me kept wondering whether I really deserved to be there. When I finished telling them, Alice looked at me as if the explanation were obvious.

*"Enzo, you have impostor syndrome."*

Those words struck me because I had always thought of an impostor as someone who deliberately deceives others, claims skills they don't have, or occupies a position they don't deserve. I wasn't trying to deceive anyone. I studied, worked, and prepared every talk with almost obsessive care. But Alice wasn't saying I was an impostor. She was saying that, despite all the evidence to the contrary, part of me still feared I was one.

Over the following days, I began reading about it. I learned that the phenomenon had been described in the late 1970s by psychologists Pauline Clance and Suzanne Imes in accomplished, highly capable women who, despite what they had achieved, struggled to recognize their success as a result of their own abilities. They might attribute their successes to luck, good timing, help from others, or the extraordinary effort they had made to compensate for perceived shortcomings, while difficulties took on disproportionate weight and seemed to confirm their feared inadequacy at once.

I recognized that pattern immediately. An invitation to a conference could be due to my relationship with a company; a successful lecture, to the almost obsessive preparation I had put into it; compliments from colleagues, to their generosity. I could always find an explanation that diminished my accomplishments. A clinical difficulty, on the other hand, needed no interpretation: in my mind, it was my fault.

Today, the literature tends to use the term impostor phenomenon because it isn't an illness or a psychiatric diagnosis, but a recurring way of interpreting oneself. It has also been observed among physicians, dentists, and other health care professionals—fields where high standards and real responsibilities make it essential to keep questioning yourself.

That mindset had a positive side, too. It had pushed me to study, travel, learn from more experienced professionals, and not settle for the first answers I found. The problem arose when self-criticism stopped helping me understand a difficulty and became a judgment on my professional identity.

There is an enormous difference between asking why a tooth movement didn't occur and asking whether we're really capable of using aligners. The first is a clinical question: it allows us to gather information, examine the causes, and adjust course. The second turns a single episode into a verdict on our entire worth as professionals.

That was when I understood how much that feeling had shaped my work in the practice, too. When a treatment didn't go as planned, I focused almost all my attention on what I could control directly: my own work. What the patient told me remained in the background as an accepted fact, and if they said they had followed all my instructions, the conclusion seemed inevitable: I had made a mistake.

It was like assessing whether a drug worked without knowing whether the patient had taken it. In my daily practice, I risked changing the treatment before checking that basic premise.

## The Answer Was Right in Front of Me

That realization led me to look more closely at certain cases that still didn't add up. The more experience I gained, the more I realized that the cases I couldn't explain were almost never the most complex ones. I would treat extremely difficult malocclusions that progressed exactly as I had planned, while seemingly simple treatments appeared to stall for no logical reason.

At first, I dismissed it all as a string of coincidences. Then I began paying closer attention and, almost without realizing it, started looking elsewhere. For years, I had watched almost exclusively the teeth, the scans, the attachments, the biomechanics, and the planned movements. All my attention was focused there because I was convinced that was where all the answers lay. Instead, I realized that the variable I kept looking for wasn't on the computer screen; it was sitting in the chair, right in front of me.

It wasn't the teeth. It was the people.

I remember one morning perfectly. I had two follow-up appointments scheduled back-to-back. The patients were two young men about the same age, with very similar malocclusions, nearly identical treatment plans, and the same amount of time since their last visits. Clinically, the two cases looked like carbon copies.

The first came in. He greeted me enthusiastically and, before I could even ask how things were going, pulled a bag of orthodontic elastics out of his pocket.

> *"Doc, I think I've gone through more than expected. Could you give me a few extra packs?"*

I examined him and found exactly what I expected to see. Treatment was going according to plan. A few minutes later, the second young man came in. He was polite, too, and smiling. When I asked him the same question, he answered with the same confidence:

> *"Everything's great. I wear my elastics all the time."*

I examined him, and it took me only a few seconds to see that treatment wasn't progressing as I'd expected. Of course, that didn't prove he wasn't wearing his orthodontic elastics; there could be several explanations. Until then, though, my mind would have automatically started looking for a biomechanical cause. This time, I caught myself asking a different question: what if the problem wasn't the treatment plan?

I didn't immediately look for an answer. I simply kept observing, and over the following months I began to recognize similar situations with surprising frequency. The faces, ages, and malocclusions changed, but certain behaviors seemed to repeat. Some patients were extremely meticulous, while others couldn't remember an instruction they had received only a short time earlier; some asked questions until they understood exactly what to do, while others nodded during the visit, only to return for their next appointment without having followed the instructions; some parents made their child's treatment a shared project, while others seemed to lose all interest in how it was going after the first appointment.

That was when I began to understand that trust and verification weren't incompatible. The relationship between clinician and patient necessarily rests on trust, and we can't turn every appointment into an interrogation, but checking doesn't mean accusing. A sound clinical decision can't depend solely on information we haven't been able to verify.

Small incidents began to take on a different meaning: an aligner I expected to have been worn for weeks that looked new, a pack of orthodontic elastics that was still unopened, a patient who couldn't remember which aligner they were wearing or who came to an appointment without bringing it. None of those details, on its own, was enough to establish how well the patient had followed the instructions, but taken together, they forced me to ask how many times I had looked for a biomechanical solution to a behavioral problem.

## Beginning to See the People

From that point on, I began to look at patients differently. Of course, I still analyzed photographs, scans, occlusal relationships, and planned movements, but more and more of my attention was shifting elsewhere.

I noticed that people to whom I gave exactly the same instructions reacted in completely different ways. Some wanted to know every detail, while others seemed annoyed by explanations that went on too long; some needed reassurance, while others mainly wanted to know how long it would take and what would happen if

something didn't work. Some followed instructions with almost obsessive precision, while others, despite understanding them perfectly, adapted them to fit their lives until they became something quite different from what I had prescribed.

During my years at university and in specialty training, I had studied anatomy, biomechanics, craniofacial growth, diagnosis, cephalometrics, materials, and treatment planning. I had learned to measure angles with almost obsessive precision and to spend hours discussing the inclination of a root or planning a movement of a few tenths of a millimeter. But no one had ever taught me how to recognize the patient who would nod throughout the appointment and then follow only half the instructions they had been given, or how to understand why two people to whom I had explained exactly the same thing could behave so differently.

It was as if all our training stopped right at the door to the practice. Beyond that point lay territory left almost entirely to personal experience, made up of trial, error, and intuition.

That was when I began to suspect something that, at the time, felt almost heretical: perhaps the predictability of a treatment depends on more than the quality of its plan. Perhaps there is a variable no software can calculate, one that doesn't appear in a cephalometric analysis, can't be measured with calipers, and can't be planned in front of a screen, yet is present in every single treatment we undertake.

That variable sat in front of me every day. It had a face, a history, a personality, beliefs, fears, and its own way of interpreting what happened to it. In other words, it had a mind.

I had spent years trying to make the biomechanics of my treatments increasingly predictable. Now I was beginning to understand that I had looked at only part of the problem. I could plan with great precision what I wanted the teeth to do, but between that plan and the final result was still a person who, every day, would have to make decisions that could help achieve it, slow it down, or change it.

This realization didn't immediately give me answers. It gave me something more important: a new area to explore. For years, when a treatment wasn't progressing, my first question had been *"What did I do wrong?"* I still ask myself that question today, because it would be dangerous to stop, but it can't be the only one, and above all, it can't come before an even more important question:

What really happened?

Only after piecing together what happened can we determine how much responsibility lies with the treatment plan, how much with biology, and how much, if any, with the patient's behavior.

I had spent years studying how teeth move. Now I finally understood that, to truly understand what happens during treatment, I would have to start studying the people who live with those treatments every day, too.

And that was a far more complex world than I had imagined.

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CHAPTER

# 2

# Understanding the Patient

*"When dealing with people, remember you are not dealing with creatures of logic, but creatures of emotion."*

— DALE CARNEGIE

I was used to speaking to very different patients using almost the same words every time. I explained the treatment, outlined the goals, and gave precise instructions. If they understood, I considered my job done.

Over time, I have come to understand that understanding a piece of information does not mean giving it the same importance. If different people listen, decide, and react in different ways, why should they all be expected to cooperate in the same way?

Some patients want to know every detail, while others feel overwhelmed when they receive too much information. Some make decisions quickly, while others need time, reassurance, and confirmation before making a choice. Some people mainly

want to know how long treatment will last; others ask to see photographs, simulations, and X-rays; still others seem most interested in how the treatment will fit into their daily lives.

Effective communication, then, is not just about knowing what to say. It is also about understanding how to say it to the person in front of us.

Over time, psychologists, behavioral researchers, management experts, and trainers have developed numerous models to describe these differences. Some grew out of scientific research on personality; others were created as practical tools to improve leadership, teamwork, sales, or relationship management.

None of these models can fully capture the complexity of a human being. We are shaped by our histories, our upbringing, our experiences, the circumstances we find ourselves in, our emotions at the moment, and the people we are interacting with. The same person may behave very differently at work, at home, during a financial negotiation, or while sitting in a dental chair.

It would therefore be simplistic to divide everyone into rigid categories. Yet the ability to simplify is also the strength of some behavioral models. They are not meant to tell us who a person really is or to provide a psychological diagnosis, but they can help us recognize certain prevailing tendencies and more quickly choose the language, pace, and amount of information best suited to the person we are speaking with.

One of the best-known and most widely used of these models is DISC.

Its origins are generally traced to the American psychologist William Moulton Marston, who published *Emotions of Normal People* in 1928, a book in which he analyzed how people respond to the environment they perceive around them.

People may tend to respond more actively or more reflectively, focus primarily on results or on relationships, and seek control, approval, stability, or precision to varying degrees. These tendencies give rise to four styles now commonly identified by the letters D, I, S, and C: Dominance, Influence, Steadiness, and Conscientiousness.

DISC has gradually come to be used in hiring, leadership, team building, sales, and conflict management, but it should not be treated as an absolute truth about human personality: it is, rather, a practical map of behavioral and communication styles.

Modern psychology, in fact, does not recognize a universal classification of four personality types. The model with the strongest scientific support is the **Big Five**, which describes people along five broad dimensions: extraversion, agreeableness, conscientiousness, neuroticism or emotional instability, and openness to experi-

ence. The fundamental difference is that the Big Five does not place each person in a rigid category: we all have these traits to varying degrees, and their combination helps shape our individual profile.

The Big Five therefore has a stronger scientific basis for describing personality. DISC, by contrast, offers a more accessible language for observing how people interact, and that is precisely what interests us. Our goal is not to conduct a complete psychological assessment of a patient or give them a questionnaire before every orthodontic appointment. We need a simple lens that can help us recognize certain communication styles and adapt how we talk with them.

In fact, a person rarely fits just one of the four profiles perfectly. More often, they show a combination of traits, with one or two predominant styles, and their behavior may change depending on the situation. A normally sociable patient may become extremely analytical when facing surgery; a usually cautious person may become dominant when they think their child's treatment is progressing too slowly; a very detail-oriented patient may suddenly become emotional when told an extraction is being considered.

Our task, then, is not to stick a label on anyone, but to observe. What questions do they ask? How quickly do they want to get to the point? Do they mainly need data, reassurance, control, or a personal connection? How do they express doubt? How do they raise an objection? The answers to these questions can help us choose the most effective way to talk with them.

## The Dominance-Style Patient

The Dominance-style patient is focused on results. They like to be in control, make decisions quickly, and tend to assess situations in terms of goals, timelines, costs, and consequences.

They often come into the office with a fairly clear idea of what they want. They may have seen other professionals, read up on the subject, or decided in advance what result they would consider acceptable. They are not necessarily aggressive, but they communicate directly and may have little patience for explanations they see as unnecessarily long.

During the appointment, they tend to get straight to the point and may ask questions such as "*How long will it take?*" "*What's the fastest approach?*" "*Can we avoid surgery?*" "*What's the best solution?*" "*How much will it cost me in total?*" or "*What happens if we don't get the result?*"

These questions may sound blunt, but we should not immediately interpret them as an attack. Very often, they are the patient's way of checking whether the clinician is knowledgeable, confident, and clear. A Dominance-style patient does not just want an answer: they want to know whether the person in front of them can guide the treatment without hesitation.

With this patient, it helps to start with the conclusion and then explain the reasoning. Instead of opening with a lengthy description of the malocclusion, we can say:

> *"In your case, we have two realistic options. The first is simpler, but it involves a compromise in the result. The second requires a more involved treatment, but it would also allow us to correct the skeletal problem. Let me quickly explain the advantages and limitations of each."*

A Dominance-style patient appreciates clear alternatives. That does not mean letting them choose a clinically inappropriate treatment, but involving them in the decision by making a clear distinction between what is possible and what is not. Whenever we can, it helps to connect our instructions to concrete consequences:

> *"Treatment will take approximately eighteen months, provided you wear the aligners as instructed. If you wear them less than prescribed, treatment will take longer, and we may need to replan certain stages."*

This kind of wording is generally more effective than a vague instruction such as *"try to wear them as much as possible"* because the first links a behavior to a consequence, while the second leaves much more room for interpretation.

When a Dominance-style patient raises an objection, it is important not to turn it into a contest. If they ask, *"Isn't there a way to do it faster?"* a defensive answer such as "*No, because biology takes time, and you need to trust me"* risks turning the conversation into a confrontation. Instead, we can respond:

> *"We can cut down on some scheduling delays, but we can't speed up the biological process beyond a certain point without increasing the risk. In your case, this is the fastest safe approach. The factor we can do the most to control is how consistently you wear your aligners."*

That way, we maintain our authority without turning it into a power struggle. With this type of patient, then, it is best to avoid lengthy introductions, vague answers, absolute promises, and a paternalistic tone. Being direct does not mean being cold; it means respecting their need for clarity.

## The Influence-Style Patient

The Influence-style patient is generally focused on relationships, emotions, and the overall experience. They communicate easily, tell stories about their life, share how they feel, and tend to form a personal connection with the clinician and the team quickly.

They are often the patient who comes in smiling, chats with the front desk staff, comments on the office, and tells us how they came to find the practice. During the appointment, they may jump quickly from one subject to another and get caught up in their enthusiasm.

Their questions often concern not just the treatment, but how it will fit into their life: "Will they be very noticeable?" "What do I do when I go out to dinner?" "Can I take pictures without them showing?" "Will I be able to speak normally?" "Do you know other people who have worn them without any problems?" or "What will my smile look like at the end?"

The Influence-style patient may accept the plan quickly because they feel involved and trust the clinician, but that initial ease should not be mistaken for certainty that they have taken in all the instructions. A pleasant conversation can leave both people feeling that everything is perfectly clear, while some important information may have faded into the background.

With this patient, it helps to connect our instructions to their actual experience and the result they want, using simple language, examples they can easily picture, and clear goals. We can involve them, show them photographs, let them see their progress, and use the personal connection as a resource, without letting that connection take the place of clarity.

If we need to explain why wearing aligners consistently is important, for example, we might say:

> *"The most important thing you can do to achieve the smile we've planned is to wear your aligners consistently. If you wear them less than prescribed, your teeth may start to fall behind the planned sequence, and we may have to stop and replan part of your treatment."*

With an Influence-style patient, it is especially useful to make sure their enthusiasm has turned into a concrete understanding. At the end of the explanation, we can ask them to summarize the key points or give them just a few very clear instructions, rather than overwhelming them with details they probably will not remember. Otherwise, we risk mistaking a good relationship for good communication. They are two different things.

## The Steadiness-Style Patient

The Steadiness-style patient tends to seek security, continuity, and harmony. They generally do not like pressure, sudden changes, or situations in which they feel they have to make a decision too quickly.

They may be the patient who listens attentively, rarely interrupts, and seems extremely cooperative during the appointment. They are often kind, patient, and respectful of the clinician's instructions, but that very willingness can lead us to overestimate how convinced or comfortable they really are with what we are proposing.

When they have a concern, they may not voice an objection right away. They may nod, thank us, and leave the office without expressing a worry that surfaces only later, perhaps in a call to the front desk or a conversation with a family member.

With this type of patient, it helps to slow the pace of the conversation, explain what will happen next, and reduce uncertainty. A statement such as:

> *"You don't have to decide anything today. I'll explain the options and the steps involved in treatment. Then you can take the time you need to think it over, and if any questions come up, we can talk through them together."*

can be much more effective than asking for an immediate decision.

During treatment, it is important to create space for patients to tell us about things that are not going well, too. Questions like "Everything going okay?" make it easy to give a positive answer. It may be more useful to ask, "What was the hardest thing to manage this week?" or "Was any part of the instructions I gave you especially difficult to follow?"

That way, we are not suggesting that the patient has done something wrong; we are making it normal to have encountered a difficulty.

With a Steadiness-style patient, reassurance must not turn into minimizing concerns. If a particular condition is essential to treatment, we need to communicate it clearly, but without creating unnecessary pressure. Steadiness does not mean having no rules; it means making those rules understandable and predictable.

## The Conscientiousness-Style Patient

The Conscientiousness-style patient tends to seek precision, logic, and consistency. They want to understand the reasons behind decisions and may ask very detailed questions before they feel sufficiently confident.

They are often the patient who arrives for the first appointment having read extensively, brings previous X-rays or records, compares the information they receive, and wants to know not only what we will do, but why we chose that particular approach.

Their questions may be extremely specific: "How predictable is this movement?" "Why are you using this attachment?" "How likely is it that I'll need a refinement?" "How can we know the tooth will move exactly as the simulation shows?"

It is natural to feel tested by these questions, but often that is not the patient's intent. A Conscientiousness-style patient uses information to reduce uncertainty. The vaguer our answers seem, the more questions they will ask.

With this patient, then, it helps to make our clinical reasoning clear and distinguish what we know from what we cannot predict with certainty:

> *"These movements are the most predictable, while these teeth have a greater chance of needing a refinement. That's why we'll monitor certain specific measurements during treatment."*

An answer like this may be more reassuring than an absolute promise, because it does not try to erase uncertainty artificially. Instead, it shows that we know where uncertainty may arise and how we intend to manage it. If the patient asks:

> *"How can you guarantee that the result will look exactly like the simulation?"*

a sales-oriented answer would be: *"Don't worry, that's exactly how it will turn out."* A more accurate and probably more effective answer might be:

> *"The simulation shows the planned result, not a mathematical guarantee. Biology and how consistently you follow our instructions can affect how your teeth respond. Our job is to monitor progress and intervene when the actual tooth movement differs from what we planned."*

With a Conscientiousness-style patient, being honest about limitations increases our credibility. If we do not know the answer to a very specific question, it is better to say so and check than to improvise. We should therefore avoid approximate answers, absolute promises, unverifiable data, and irritation at their questions. Precision in communication is especially important because this patient tends to remember what was said and compare it with what happens.

## A Map, Not a Label

At this point, it might seem that all we need to do is observe a patient for a few minutes, assign them a letter, and automatically apply a communication style. But that is not how it works.

DISC should not become a new form of stereotyping. A patient who asks a lot of questions is not necessarily a Conscientiousness-style patient: they may simply be frightened. Someone who wants to decide quickly is not always a Dominance-style patient: they may be short on time or may already have had other consultations. A person who talks a lot is not necessarily an Influence-style patient, and someone who stays quiet does not automatically fit the Steadiness profile.

The four styles should be used as working hypotheses, not diagnoses. We can begin a conversation with one approach and adjust it as we observe the patient's response. If we give them a summary and they keep asking for details, they probably want a more thorough explanation; if we begin with a long explanation and notice impatience, we may need to get to the conclusions more quickly.

Effective communication, then, is not a fixed script but a continuous process of adaptation: DISC had taught me to explain the same concept in different ways, but it had also shown me its limitations. A person's communication style does not let us predict how consistently they will follow instructions. A Dominance-style patient may follow instructions rigorously or resist the rules; a Conscientiousness-style patient may be impeccable or use every detail to shift responsibility away from themselves.

For this reason, DISC will remain with us in the pages that follow, but it will remain what it should be: a map for communication, not a way to classify patients. It will help us choose our words, the level of detail, and the pace of the conversation; our starting point will always be the behavior we observe.

We can learn to communicate better, but no personality model can tell us for certain what a patient will do after they leave the office with a box of aligners in hand and we are not there to watch them.

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CHAPTER

# 3

# Is the Patient Lying?

*"The confidence that people have in their beliefs is not a measure of the quality of the evidence but of the coherence of the story the mind has managed to construct."*

— DANIEL KAHNEMAN

*"If my patient tells me they always wear their aligners, why isn't treatment progressing as planned?"*

I asked myself that question many times during follow-up visits. The patient assured me they had followed the instructions, but what I observed clinically told a different story. For a long time, my first reaction in these situations was to look for a problem with the treatment. As I have already described, only over the years did I begin to consider another possibility: that part of the explanation might lie in the patient's behavior as well as in the biomechanics.

I still remember very clearly one of the cases that helped change my perspective.

Laura was about thirty. She was a lawyer, a mother, and, to all appearances, a very conscientious patient. We had begun treatment together with the goal of correcting a Class II malocclusion through maxillary distalization and the use of orthodontic elastics.

During treatment, the aligners were working well: the anterior crowding was resolving, and the fit was excellent. The Class II malocclusion, however, was not improving. At every follow-up visit, I asked Laura if she was wearing her orthodontic elastics, and she answered: *"Yes, absolutely. All the time."*

For almost two years, we saw no significant correction of the Class II malocclusion. I kept looking for the explanation in the treatment: I reviewed the biomechanics, modified the treatment plan, and ended up doing four refinements. Meanwhile, the upper incisors kept moving forward. Then, at a follow-up visit, I found that Laura had finally reached Class I. I asked her what had changed. She told me she was getting married in nine months and wanted to finish treatment before the wedding. That was why she had decided to wear her orthodontic elastics *for real*. Only then did she explain that until that point she had worn them mostly at night and very little during the day.

For me, that admission was more important than the correction of the Class II malocclusion. I had spent a long time looking for a biomechanical explanation for a problem that depended, in large part, on Laura's behavior. She knew my instructions and had always assured me she was following them. That was when I began to ask myself a simple but uncomfortable question:

**Had Laura lied to me?**

At first glance, it seemed so. Yet calling it a lie did not help me understand what had happened. Why had a patient who knew the instructions given me, visit after visit, an answer so different from her actual habits?

She might have chosen not to tell me the truth. But there might also have been something I had not considered about the way she remembered and described her habits. From then on, I began to ask myself: **why might what a patient reports differ from what they do?**

If we immediately assume the patient is lying, we will look for a way to prove them wrong. If instead we try to understand how they arrived at that answer, we have a better chance of finding out what is really happening and helping them change. In the pages that follow, we will explore some of the mechanisms that can create this gap between behavior and the account of it. They often operate at the same time: each may help explain part of what we observe.

## Reconstructive Memory

We tend to imagine memory as a vast archive where events are stored in an orderly way, ready to be retrieved when needed. But research tells us it does not work that way. Remembering does not mean opening a drawer and pulling out a perfect copy of the past. It means reconstructing the past from fragments of experience, prior knowledge, expectations, and personal meanings.

If someone asked us how many hours we slept each night over the past month, we would be unlikely to answer precisely. We would probably remember a few particular nights and use those to estimate an average.

The same thing can happen when we ask a patient how many hours they wore their aligners in the preceding weeks. Putting them in and taking them out are actions repeated every day: it is hard to remember exactly how long each interval lasted. The patient may therefore give us an inaccurate estimate without any intention of deceiving us.

Frederic Bartlett described remembering as a reconstructive process. Later, Elizabeth Loftus's studies showed that information received after an event can influence how it is remembered. Daniel Schacter described several memory distortions as aspects of how memory normally works.

When we ask a patient to tell us how they behaved in the preceding weeks, then, we are gathering a reconstruction, not an exact account of every day. That does not automatically make their answer false. It makes it human.

## Cognitive Dissonance

Reconstructive memory helps us understand why a patient might remember their behavior inaccurately. But there are situations in which that explanation alone seems insufficient. The patient is sitting across from us. Some tooth movements have not occurred as planned. We ask how things are going with the aligners, and the answer comes without hesitation:

*"I wore them all the time."*

At moments like these, it is natural to wonder whether the patient is trying to convince us or, at least in part, themselves. That question brings us to **cognitive dissonance**, described by Leon Festinger in 1957. We tend to seek consistency between how we see ourselves and what we do. When the two conflict, we may feel tension and look for a way to reduce it.

Imagine a patient who considers themselves conscientious, reliable, and cooperative. During treatment, however, they forget to wear their aligners several times, spend too long eating, leave them out during dinner with friends, or decide not to wear them for an entire evening. They then have to reconcile two thoughts: *I am a responsible person* and *on some occasions I did not follow the instructions*. They may acknowledge those occasions and decide to change their behavior. Or they may downplay their importance in their own mind:

*"After all, I wore them almost all the time." Or "It was just one evening."*

The exceptions thus become part of an account that allows the patient to continue seeing themselves as cooperative. They may not do this consciously: when they speak to us, they might sincerely believe they followed the instructions much more closely than they actually did.

## Social Desirability Bias

There is another factor to consider. The patient is not merely trying to remember what they did: they are talking to us, and they know what answer we expect. **Social desirability bias** is the tendency to present ourselves favorably to others, especially when a question concerns behavior for which we might feel judged.

The patient knows the instructions they were given. They know how many hours they should wear their aligners and that orthodontic elastics are part of the treatment. Admitting they have not followed those instructions can be uncomfortable. So when we ask how things are going, the desire to seem like a good patient can influence their answer. They do not need to make up a story. A small adjustment may be enough: *"pretty often"* becomes *"almost all the time,"* and *"almost all the time"* becomes *"all the time."*

This is a well-known phenomenon in studies that rely on people's reports of their own behavior. For us, it has a practical consequence: a question like *"You're wearing your aligners, right?"* already suggests the expected answer and can make it harder for the patient to tell us what is really happening.

## Self-Serving Bias

Another mechanism concerns how we explain outcomes to ourselves. **Self-serving bias** is the tendency to attribute successes to our own actions and failures to external causes.

When treatment is going well, the patient may think: *"I've done a good job following the instructions."* When a planned tooth movement does not occur, they may instead think: *"These aligners aren't working."*

In this case, what changes is not necessarily our memory of what we did, but how we interpret what happened. Attributing a difficulty to an external cause can protect us from the idea that we contributed to the problem. Of course, sometimes the cause really does lie in the treatment: a movement may have been planned poorly, an attachment may be inadequate, or the biological response may differ from what was expected. This mechanism does not, therefore, give us license to automatically blame the patient.

It does, however, help me look again at what happened in cases like Laura's. I tended to look for an error in my treatment plan; she may have looked for one in the aligners. So I modified the treatment, while she may have remained convinced that she was following the instructions as asked.

## Confirmation Bias

In 1960, Peter Wason described another phenomenon that would become central to the study of human reasoning: **confirmation bias**, the tendency to seek out and interpret information that fits what we already believe.

Patients, too, may view their treatment through this filter. Someone convinced they have followed the treatment plan extremely closely will tend to give more weight to the days they wore their aligners exactly as instructed, the sacrifice they made during a dinner, or the times they put them back in immediately after lunch. The weekend they left them on the nightstand, the long Sunday lunch, or the vacation when their schedule became less strict will figure much less prominently.

The opposite can also happen. A patient convinced that treatment is not working will tend to focus on the tooth that is still rotated or the small imperfection that remains, while all the tooth movements already completed gradually fade into the background.

The initial belief thus becomes the filter through which we interpret what happens. And of course, this applies to us as clinicians, too. If we are convinced the problem is biomechanical, we will look mainly for evidence that supports that explanation. If we are convinced the patient is not following the treatment plan, we run the opposite risk.

Understanding confirmation bias, then, is not about proving the patient wrong. It is also about reminding ourselves how easily we could be wrong.

## Unrealistic Optimism and Temporal Discounting

Then there is a question every orthodontist has asked at least once: why does a patient keep acting against their own interests despite knowing the rules of treatment perfectly well? **Unrealistic optimism** can help us understand part of the answer.

In the 1980s, Neil Weinstein observed that we tend to consider some negative events less likely to happen to us than to others and, conversely, to see positive events as relatively more likely. In other words, we know the risk, but it tends to feel more remote when it concerns us.

The patient knows that wearing their aligners less than required can slow treatment, but thinks a few hours less will not really make a difference in their case. They know that not wearing their orthodontic elastics can prevent certain tooth movements, but imagine they will still be able to catch up. They know retention is important, but consider it unlikely that their own teeth will shift significantly. So it is not necessarily that the patient has failed to understand the information: they may have understood it perfectly. What changes is the weight the patient gives to the possible consequence.

Another mechanism can come into play here, one well described in studies of **temporal discounting**, or the devaluation of future outcomes: we tend to give greater weight to immediate costs and benefits than to consequences farther in the future. For the patient, this means continually weighing a small discomfort now against a future benefit. Taking out their aligners for an evening, not putting them back in immediately after dinner, or skipping their orthodontic elastics brings a small but immediate benefit; a possible delay in treatment, by contrast, seems distant, uncertain, and hard to perceive at the moment the decision is made.

It is the interaction between these two mechanisms that gives a future consequence so little weight: not only is it far away, but the patient may also be convinced that it probably will not happen to them.

This point will be especially important when we begin developing the protocols, because it will force us to distinguish between informing the patient of a consequence and making that consequence truly matter when they have to make a decision.

## So, Is the Patient Lying?

At this point, we can return to the question we started with: **is the patient lying?** Sometimes, yes. It would be naive to build an entire chapter around cognitive mechanisms to avoid that possibility. Some people know perfectly well that they have not

followed the instructions and deliberately choose to tell us otherwise, perhaps to avoid a reprimand, protect their image, or avoid taking responsibility. But it would be just as naive to stop there.

Between objective reality and the account we give of it lies a much more complex space, where memory reconstructs what happened; dissonance helps us preserve a consistent image of ourselves; social desirability bias changes what we are willing to admit to others; self-serving bias shapes how we assign responsibility; confirmation bias favors information that best fits what we already believe; and unrealistic optimism and temporal discounting can make the future consequences of our decisions seem less relevant. These mechanisms do not automatically turn a false answer into a true one, but they help us understand why the gap between actual and reported behavior can be much more complex than a simple choice between honesty and lying.

And it is precisely this distinction that changes how we approach the patient. If our goal is to establish who is right, every inconsistency becomes the start of a confrontation: we show the patient what we observe, the patient defends what they remember, we press the point, and they dig in. The harder we try to prove that they are not telling the truth, the more we risk pushing them to defend the account they have just given. Our goal, though, is not to get a confession, but to get them to behave differently from that point forward.

That is probably the most important implication of everything we have seen in this chapter. When a patient says they have worn their aligners as instructed but the clinical picture suggests otherwise, we do not necessarily need to decide whether they are lying, remembering inaccurately, or unconsciously protecting their self-image. Instead, we need to figure out how to bring the conversation back to what will happen **from today on**.

Had Laura told me the truth in the end? Had she deliberately lied to me for months, or was she genuinely convinced that the way she wore her orthodontic elastics was close enough to what I had asked of her? I do not know, and years later I realized that perhaps it is not even the most important question. What I do know is that when the wedding turned a distant, abstract consequence into a concrete, near-term goal that mattered to her personally, her behavior changed. The orthodontic elastics were the same, the biomechanics were the same, and I was the same orthodontist: **what had changed was what following the treatment plan meant to her.**

And that is where understanding the patient's mind stops being merely interesting and starts becoming clinically useful. When adherence breaks down, the consequences extend beyond the clinical result: they mean more time, more

appointments, more refinements, and more resources spent to achieve the same result. In the next chapter, then, we will look at poor adherence from another, much more concrete perspective: **what is its real cost?**

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CHAPTER

# 4

# The Hidden Cost of Poor Patient Adherence

*"Beware of little expenses; a small leak will sink a great ship."*

— BENJAMIN FRANKLIN

Some patients I remember because of the complexity of their malocclusion, others because the result was particularly difficult to achieve, and still others because, when I think back on their treatment, the first thing that comes to mind is not their teeth but the time it took. Not necessarily because the treatment was long in absolute terms: some complex cases take time, and that is part of orthodontics. I mean the very different feeling you get when you reach the end, look back, and realize that a considerable amount of that time would not have been necessary.

A new scan because the aligners had stopped fitting properly, a new treatment plan, another series of aligners, an unplanned appointment to deliver them, and a few extra checkups. Then perhaps another scan, because the new sequence had not produced what we expected, either. Meanwhile, more months had passed, and the treatment, which on paper should have been nearing completion, continued to take up space on the practice schedule.

On their own, these steps seem almost harmless. A scan takes only a few minutes, an extra checkup can be squeezed in between two appointments, preparing a new treatment plan is part of our job, and when we use packages that allow us to request additional series of aligners for a set period, even producing new trays can seem to have no real cost. Only when we start adding it all up does the picture change: how many hours of chair time did we spend on that patient beyond what we had planned? How many times did we reopen the case, revise the prescription, review a new digital treatment setup, or add an appointment that should not have been needed? And above all, how much of that extra work was truly unavoidable, and how much had become necessary because the patient had not followed the treatment plan?

Until then, I had viewed the problem mainly through individual clinical episodes. I was seeing patients whose aligners lost tracking, treatments that dragged on, and refinements that piled up, but I had not yet tried to look at the phenomenon as a whole. In the previous chapter, we saw why what a person remembers and reports about their behavior may not match exactly what actually happened; at this point, though, the question changes. If we stop relying solely on what patients tell us and start measuring what they actually do, **how large is that gap?**

## How Closely Do Our Patients Really Follow the Treatment Plan?

In orthodontics, we are used to the word *compliance*, although in recent years the literature has tended to favor *adherence*, a less paternalistic term describing the extent to which a patient's behavior matches the instructions they have received.

The problem is that there is still no universally accepted definition of orthodontic adherence. One of the most comprehensive reviews on the subject, published in 2024 by Van der Bie and colleagues, identified 3,284 papers and included 60 in the final analysis. The authors showed how fragmented the literature is and how difficult it is to compare studies that measure different behaviors using different tools. In practice, though, three broad areas emerge through which we can assess adherence: wearing appliances as instructed, keeping appointments, and maintaining adequate oral hygiene during treatment.

This means it probably does not make much sense to divide patients simply into those who follow instructions and those who do not. A patient may wear their aligners very consistently but forget their orthodontic elastics, may be meticulous about oral hygiene but miss appointments, or may start treatment enthusiastically and gradually become less consistent after many months. Adherence, then, is not an unchanging personal trait but a set of behaviors that can change over the course of treatment.

For a very long time, the simplest way to find out how much a patient wore a removable appliance was to ask them: *"How many hours do you wear it?"*, *"Do you wear it every night?"*, *"Are you wearing your elastics?"* Today, we can compare those answers with data from systems that objectively record wear time, and when we do, a gap emerges that deserves our attention.

In 2017, Al-Moghrabi and colleagues published a systematic review and meta-analysis of compliance with removable orthodontic appliances, comparing the wear time prescribed by the orthodontist, the time reported by the patient, and the time recorded objectively. The results showed a fairly clear pattern: actual adherence fell short of what was prescribed, and patients tended to overestimate their wear time. In the six studies that compared prescribed wear time with objectively recorded wear time, patients wore the orthodontic appliance an average of **5.81 fewer hours per day** than prescribed. Even more relevant to our discussion was the gap between what patients reported and what the devices recorded: reported wear time was **5.02 hours per day higher**.

Five hours.

A difference that, repeated every day for weeks or months, completely changes the conditions under which orthodontic tooth movement is supposed to take place. Van der Bie's 2024 review confirms how difficult it is to assess adherence based solely on what patients report and emphasizes that objective measurements are more reliable when available.

The situation becomes even more complicated during adolescence. A review published in 2025 on adolescents' adherence to removable orthodontic appliances found considerable variation in adherence and identified numerous factors that can affect it, including motivation, forgetfulness, discomfort, family support, and social pressures. In one of the included studies, about 40% of adolescents wore their vacuum-formed retainer for fewer than 18 hours a day during a phase when they were supposed to wear it for 22 hours. The point is not to regard adolescents as less reliable patients, but to recognize how difficult it is to ask someone to maintain the same behavior every day, for months or even years. On paper, *"wear it every night"*

seems like a very simple instruction; in real life, it is a much more demanding request than those four words suggest.

With aligners, too, the ability to measure wear time objectively lets us compare what we prescribe with what patients actually do. In 2026, Sabbagh and colleagues published a randomized controlled study in which wear time was recorded by microsensors built into the aligners. Forty-three adult patients, who had been instructed to wear their aligners for twenty hours a day, were divided into two groups: patients in the first group did not know the data collected would be used to assess their compliance, while those in the second group were explicitly informed of this.

Of the participants, **67.5% wore their aligners for between ten and twenty hours a day**, below the prescribed twenty hours. But perhaps even more interesting was the effect of monitoring: knowing that their wear time was being tracked increased patients' average aligner wear time by **4.41 hours per day**. The appliance, the instructions, and the treatment were the same in both groups; the only difference was whether patients knew their behavior would be measured.

This finding introduces a principle we will return to in the protocols later: fully understanding an instruction does not necessarily mean following it. The context in which we ask patients to do something, how we check whether they do it, and how we make the consequences visible can all affect what they actually do.

### When Adherence Changes the Treatment

At this point, it would be easy to conclude that if a patient follows the treatment plan less closely, treatment inevitably takes longer. The relationship, though, is not so simple, because the length of orthodontic treatment depends on the complexity of the malocclusion, the individual's biological response, the planned tooth movements, treatment choices, treatment planning, and many other variables unrelated to patient behavior. Van der Bie's review also emphasizes that the relationship between the degree of adherence and treatment outcomes is not yet fully understood.

There is, however, evidence that certain patient behaviors are associated with treatment duration. Beckwith and colleagues analyzed 140 treatments completed at five orthodontic practices, examining thirty-one variables. Six factors together explained more than half of the observed variation in treatment duration, and three of them were related to patient behavior: missed appointments, brackets or bands that had to be replaced, and oral hygiene problems. A few years later, Skidmore and colleagues analyzed 366 patients treated with fixed appliances and again found an association between longer treatment and behaviors such as not wearing orthodontic elastics enough, inadequate oral hygiene, and broken brackets.

These studies were conducted predominantly on fixed appliances, and it would be wrong to apply their findings automatically to aligner treatment. The principle that emerges remains important, though: treatment duration depends not only on what we plan and do, but also on what patients do during the weeks between appointments. **In aligner treatment, this matters particularly because daily wear time is one of the conditions on which we build the entire treatment plan.** And this is where we encounter something extremely familiar to anyone who works with aligners every day: **refinements.**

## Not All Refinements Are the Same

Before linking refinements to adherence, we need to make a distinction: not every refinement results from poor adherence, and needing a new series of aligners does not necessarily mean treatment has failed. Some tooth movements are less predictable, patients' biological responses differ, and biomechanical limitations, treatment-planning problems, and many other clinical factors can make a refinement entirely necessary even for a patient who follows the plan impeccably. We are interested in a much more specific group: **avoidable refinements**, particularly those made necessary because the patient's actual behavior differed from what we had based the treatment plan on.

This distinction is essential, especially given how common refinements are regardless of their cause. Kravitz and colleagues analyzed 500 patients treated with Invisalign at two orthodontic practices and found an average of **2.5 refinement scans per patient**. Only 6% completed treatment without an additional scan, while the actual average treatment duration was 22.8 months, 5.1 months longer than initially projected. In addition, 17.2% of patients had to switch to fixed appliances to complete treatment.

These numbers **do not tell us how many of those refinements resulted from poor adherence**, because the study was not designed to determine that. They do, however, show how wide the gap can be between the treatment we envision at the outset and what is actually needed to complete it. In 2025, an international consensus on the indications and limitations of aligner treatment called attention to this very problem: more than 10% of patients in Kravitz's study required five or more refinements, increasing the time and resources needed to complete treatment.

Another study helps us understand why the number of refinements deserves attention. Abu Arqub and colleagues analyzed 355 patients treated with aligners, examining the relationship between initial complexity, number of refinements,

treatment duration, and outcome. The greatest improvement occurred with the **first refinement**. Among patients who had not yet reached the improvement threshold set by the authors, the likelihood of reaching it decreased progressively as the number of refinements increased; in the sample analyzed, after more than three refinements, no additional patients were observed reaching that improvement threshold for the first time.

It is important not to turn this finding into a claim the study does not make. **It does not mean that treatment becomes pointless after three refinements, and it does not mean a fourth refinement should be avoided.** A complex case may require further stages that are entirely justified. The finding does, however, suggest something extremely interesting: more refinements do not guarantee a proportional clinical benefit, and this is precisely where the clinical issue begins to intersect with the economic one.

### The Hidden Cost of Refinements

For me, this changes how I think about a phrase we can say very easily in day-to-day practice: *"Let's do another scan."* Clinically, it may seem like a relatively small decision; from an operational and financial standpoint, though, it is anything but.

Many aligner systems offer packages that, after the initial purchase of treatment, let us request additional series for a set period without paying a new production fee each time. This model is extremely convenient, especially for complex treatments, but that very feature can unconsciously change how we perceive the cost of a refinement. If no new invoice arrives from the company after another scan, it is easy to feel that the new series is essentially free. Patients may see it the same way: if they need more aligners, more aligners will simply arrive.

That is where the misconception begins.

The price of the new series is only part of what is involved when we decide to do a refinement. We need to understand why the aligners lost tracking, perform a new scan, update the records when necessary, reconsider the prescription, review and approve the new digital treatment setup, arrange delivery of the aligners, and schedule additional checkups. Much of this cost does not appear on any invoice, because it takes the form of the most valuable and limited resource we have in our practices: **time**.

When we talk about financial cost, though, we need to return to the earlier distinction. A clinically necessary refinement is part of the work required to complete treatment; an avoidable refinement, by contrast, requires resources we could have saved. The time spent may not appear as a separate line item in the practice's

accounts, and we will not receive an invoice at the end of the month to remind us of it, but that does not mean it has no cost.

An hour spent getting a treatment back on track could be an hour in which we might have conducted an initial consultation, started a new case, or performed a scheduled procedure. When the same thing happens with dozens of patients, those hours begin to add up. If we assumed, for example, that an avoidable refinement required even just one additional hour of clinical and administrative resources in total, **fifty refinements would mean fifty hours of work**; if each required two hours, that would become one hundred.

It would be wrong, though, to assign those hours a universal monetary value. A practice that generates 300 euros per hour will not have the same opportunity cost as one that generates 1,000 euros per hour, just as a scan handled entirely by staff will have a different impact from a procedure that directly involves the orthodontist. The principle, though, does not change: **every avoidable refinement consumes some of the practice's capacity that can no longer be used for something else.**

When I searched the literature for a figure that would establish the average cost of an avoidable refinement, I could not find one, and a universal figure probably could not exist, because it depends on how the practice is organized, staffing costs, the aligner system used, the time spent on treatment planning, and the value of clinical time. The literature does, however, allow us to trace the chain that leads to the problem: actual adherence may be lower than both prescribed and reported adherence; some patient behaviors are associated with longer treatments; and refinements are extremely common in aligner treatment, while their benefit does not necessarily grow in proportion to their number. What no article can tell us is **how much all this costs in our own practices: we have to calculate that ourselves.**

The point, then, is not to determine which pricing model is best, but to understand a much more important distinction: **the price of a new series of aligners and the cost of managing that new phase of treatment are not the same thing.** The first may already be included in the package purchased; the second remains every time the practice has to step in again.

## The Real Price of Poor Adherence

At this point, poor adherence takes on a different meaning. It is no longer simply a matter of a patient not following our instructions, but a variable capable of changing the conditions on which we built the treatment plan. If patients wear their aligners less than necessary, do not wear their orthodontic elastics as instructed, or repeat-

edly miss appointments, the plan we designed remains the same, but the conditions under which it is supposed to work do not.

When this leads to a loss of tracking, incomplete tooth movements, or the need to redesign the treatment, part of the treatment has to be made up. The number of appointments, the time spent with the patient, the overall treatment duration, and the resources needed to achieve a result we had planned to achieve sooner and under different conditions all increase. But perhaps the most important point to emerge from this chapter is not about cost: it is about **predictability**.

If we know that actual adherence may fall short of what we prescribe, that what patients report may overstate what they actually do, and that even knowing they are being monitored can change aligner wear time by several hours a day, it becomes difficult to keep treating adherence as something we can simply prescribe and then take for granted. This does not mean relieving patients of responsibility, but recognizing that adherence is a **clinical variable** and, like every variable that can affect the outcome, it must be considered, observed, and managed.

That shift in perspective changed the question I ask myself when a patient is not following the treatment plan. No longer just *"Why aren't they doing what I told them?"* but *"If I know this behavior can occur, what can I do to spot it sooner, make it less likely, and intervene when it does?"*

I cannot know for certain how many hours a patient will wear their aligners once they leave my office, nor can I be there when they have to decide whether to put on their orthodontic elastics, switch to the next aligner, or follow a particular instruction. But I can decide how to prepare them for those choices, how to check what is happening during treatment, and, above all, how to intervene when I recognize one of the behaviors I have seen recur again and again over the years.

And this is where the problem of adherence finally meets the central theme of this book: recognizing recurring behaviors allows us to prepare and replace improvisation with a method. **That idea gave rise to the protocols we will look at in the coming chapters.**

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CHAPTER

# 5

# Introducing the Protocols

*"There is always a well-known solution to every human problem—neat, plausible, and wrong."*

— H. L. MENCKEN

Sooner or later, we will see another patient who forgets their orthodontic elastics, wears their aligners less than they need to, or insists they followed every instruction while the course of treatment tells a different story.

The question is no longer whether it will happen, but whether we will improvise again when it does. Every recurring behavior gives us a chance to prepare a response that the whole practice can recognize, assess, and follow: a protocol.

For many years, though, I kept improvising. When something was not working, I searched for the right words, tried to explain it better, added information, showed photographs, stressed the consequences, or changed my approach completely depending on the person in front of me. Sometimes it worked very well, sometimes

decidedly less so; with some patients I could bring about an immediate change, with others I ended up in endless conversations, and with still others I left the appointment feeling that I had explained the problem perfectly without managing to change a thing. At the next appointment, if the behavior happened again, I essentially started from scratch.

Over time, I realized there was something paradoxical about working this way. In orthodontics, we use protocols for practically everything: we standardize record collection, diagnosis, planning, clinical procedures, follow-up visits, and many of our biomechanical decisions because we know that when a situation tends to recur, deciding in advance how to handle it reduces variability and increases predictability. Yet when the variable that deviates from the plan is the patient's behavior, we still very often react case by case, even though these situations, too, recur with surprising regularity.

The names, ages, malocclusions, and excuses change, but the patterns underlying many adherence problems are far fewer than we might think. That was probably the most important discovery: **problematic behaviors are numerous, but they are not endless and, above all, they are not entirely unpredictable**. If we learn to recognize the patterns that tend to recur, our response, too, can stop being improvised and start being planned.

That need gave rise to the protocols.

## Before the Protocols: Planning for Adherence

In Chapter 2, we saw that communicating effectively means more than knowing what to say. It also means adapting our language, pace, and amount of information to the person in front of us. This can help us make an explanation clearer, more credible, and more meaningful. Even perfectly tailored communication has a limit, though: it can help a patient understand, but it cannot guarantee what they will do.

A patient may understand perfectly well how important it is to wear their aligners, wear their orthodontic elastics, or follow the retention plan and still do something different a few hours later. They may even leave the office sincerely convinced that, from then on, they will do exactly what we asked, only to slip gradually back into their old habits when everyday life takes over. **Informing someone and changing their behavior are not the same thing.**

Psychologists have long studied this gap between intention and behavior. Among the models most relevant to our discussion are the *implementation intentions* studied by Peter Gollwitzer, in which a general intention becomes a much more concrete

plan linking a particular situation to a specific action: not simply *"I need to remember to do this,"* but *"When this situation comes up, I'll take this action."* It is a particularly useful distinction in orthodontics, because many adherence problems arise not from a lack of intention to follow instructions, but from the difficulty of turning that intention into a sufficiently consistent daily routine.

That, though, is only part of the problem. Making a behavior easier to carry out can help when patients want to follow the plan but struggle to turn that intention into a habit; in other situations, we must first make the requested behavior matter enough to patients that they genuinely decide to change it. This brings us to another important element in how I developed my protocols: **persuasion**.

Oddly enough, this was not a new subject for me at all. Long before I became an orthodontist, I was fascinated by magic, sleight of hand, mentalism, and hypnosis. I wanted to understand why two people could watch the same scene and perceive different things, how attention could be directed toward certain details, and how much the context in which information is presented could change its interpretation. Over the years, that curiosity also led me to study communication and persuasion more deeply, at first out of personal interest and, only much later, to wonder whether some of those principles might have practical applications in my relationships with patients, too.

The word *persuasion* can make us uneasy when used in connection with medicine, as though it necessarily implied a desire to convince someone to do something against their own interests. But I believe the problem is not persuasion itself; it is the purpose for which it is used. Every communication changes the listener's perception to some degree: choosing one word instead of another, presenting a consequence in terms of what can be gained or lost, showing a photograph, using a number, or asking a question already creates a frame through which the other person will interpret what we are saying.

Persuasion becomes ethically problematic when it is used to conceal information, distort facts, exploit a vulnerability, or steer someone toward a choice that benefits us but not them. It can mean something entirely different when it is used to help a patient maintain a behavior they themselves have agreed is necessary to achieve the treatment goal they chose. If someone freely decides to begin aligner treatment and knows that the outcome will also depend on their own behavior, helping them wear their aligners for the required amount of time, wear their orthodontic elastics, or follow their retention plan correctly does not take away their freedom. It means recognizing the enormous gap between knowing what we

should do and actually managing to do it every day—a gap filled with habits, lapses of memory, immediate rewards, resistance, and rationalizations.

That is precisely the gap the protocols must address.

## The Three Motivators

When I began thinking this way, I realized I did not need to come up with a different way to convince each patient to follow instructions every time. I first had to understand **what really mattered to the person in front of me**, because not everyone wants the same things with the same intensity and, above all, not everyone gives the consequences of a behavior the same weight.

Over time, I began to recognize three major motivators that now recur throughout my protocols: **time, the outcome, and the investment**. These are not three persuasive techniques, but three different ways a consequence can become genuinely meaningful to a patient.

The first motivator is **time**, a limited resource that is easy to understand and easy to connect directly to the consequences of our actions. In orthodontics, it has another distinctive feature: patients often perceive the length of treatment as something that depends mainly on the clinician, when many treatments have a behavioral component that can change it substantially. For some people, finishing six months earlier or six months later makes an enormous difference: there may be a wedding, a trip, a move, college, or simply the desire to be done with aligners at last. In these cases, making the connection between what patients do today and what might happen in the months ahead clear can mean far more than another explanation of the biomechanics.

The second motivator is **the outcome**. Some patients place such a high value on the final result that they will accept a longer course of treatment without much difficulty, as long as it allows them to achieve exactly the goal they want. In these cases, repeatedly talking about additional months may have very little effect. It becomes much more important to connect the requested behavior to the quality of the result they hope to achieve, helping them understand that a particular tooth movement might not be fully achieved or that insufficient adherence might force us to accept a compromise we had not originally planned.

The third motivator is the **investment**, by which I do not mean only money. Orthodontic treatment certainly involves a financial investment, but it also involves months or years of time, appointments, family scheduling, daily attention, sacrifices, and trust. Everything the patient has already devoted to treatment has value,

and that investment can become particularly meaningful when a behavior threatens to undermine what has already been accomplished or force us to repeat part of the process.

Time, the outcome, and the investment will play a role in practically every protocol we encounter later, but we will not necessarily use them all at once, and no single motivator is universally better than the others. **The clinical context and the person in front of us** will suggest which consequence may matter most at that moment. The general principle, then, is this: **communication can influence behavior only when it connects what we ask patients to do with a consequence they consider genuinely important**.

## The Basic Structure of a Protocol

When I began putting these principles together, I noticed that, beyond any particular technique, the conversations that worked best tended to follow a fairly consistent structure. It was not a matter of finding especially clever words each time, but of moving through a series of steps in an order that made confrontation less likely and concrete change more likely.

The first step is to **reduce defensiveness** by presenting the problem through clinical observations, specific questions, and verifiable evidence rather than judgments about the person. The second is to allow patients, when possible, to **save face without continuing the problematic behavior**: we do not necessarily need them to admit they were wrong or that they lied, but we do need them to recognize that something must change. The third is to make **the consequences concrete**, connecting the behavior to the effects it can actually have and choosing, based on the person, which of the three motivators—time, the outcome, or the investment—matters most. The fourth step is to **reduce ambiguity about future behavior**, turning a vague *"you need to follow the instructions more consistently"* into something specific enough to observe and verify. Finally, we must **make the behavior easier and check whether it works**, reducing the obstacles that can prevent patients from turning intention into action and arranging feedback soon enough to tell whether the change we made is actually working.

This structure should not become a rigid framework. The purpose of a protocol is not to get the same response from every patient, which is impossible, but to make **our response** more predictable when we face problems we have encountered before. In the pages ahead, we will look at a series of psychological and communication principles we can use to build scripts suited to the clinical situation and the

person in front of us. **We will then see them put into practice in the protocols in later chapters.**

## 1. Reduce Defensiveness

In difficult conversations, I used to start with what seemed most logical to me: the problem. If a patient was not wearing their orthodontic elastics, I talked about the elastics; if they were not wearing their aligners enough, I talked about the hours; if they were not wearing their retainer, I explained the risk of relapse. From my point of view, this was a perfectly rational approach, but over time I realized that the first words of a conversation could profoundly affect how the patient interpreted everything I said afterward.

If we open an appointment by saying *"You're not doing what I asked you to do,"* we have already assigned the roles in the conversation without saying so: we are the accuser, and the patient is the one who has to defend themselves. We may be entirely right from a clinical standpoint, but from that point on, some of the conversation's energy will go toward assigning responsibility, justifying what happened, or protecting the patient's self-image.

If instead we begin by saying *"Let's look together at how your teeth are responding compared with what we planned"* or *"One tooth movement isn't progressing as we expected, and I'd like to figure out what we can do to get it back on track,"* we have not changed the clinical problem or concealed what we think, but we have created a different context for addressing it.

The request does not begin when we say it aloud. It begins with how we set up the conversation in which we make it. The principle of **framing**, studied by Tversky and Kahneman, describes how the way information or a problem is presented can influence how it is perceived and the decisions that follow, even when the underlying facts remain the same. That is why, when I need to address insufficient adherence, I now try to start with the treatment rather than with blame. This does not mean avoiding a difficult conversation. It means creating conditions that give the conversation a better chance of producing change rather than a defensive response.

The same principle applies when a patient raises an objection. One of the most natural mistakes in clinical communication is to try immediately to correct what we have been told. If patients say something is uncomfortable, we are tempted to explain that it should not be all that uncomfortable; if they say they cannot do something, we start showing them that they actually could; if they tell us they have always worn their aligners, we try to explain that what we are seeing proves otherwise.

The problem is that the moment we try to prove that another person's perception is wrong, we almost automatically force them to defend it. If a patient says *"I can't wear my elastics because they're too uncomfortable"* and we immediately reply *"That can't be right. All my other patients wear them,"* the conversation quickly stops being about the elastics and starts being about whether their experience is legitimate. The patient has to convince us that the discomfort is real, while we try to convince them that it should be tolerable. In the end, we may have argued for ten minutes without getting any closer to the behavior we want to encourage.

If framing concerns the lens through which we present a problem, **reframing** means trying to change that lens, giving the same fact a different meaning without having to deny it. If part of treatment is uncomfortable, I can acknowledge that it is and then connect that discomfort to the role it plays in reaching the goal. If a behavior takes effort, I can acknowledge that too and help the patient see that effort in a different light.

**An objection that has been acknowledged needs less defending**, and only then can we begin to work on the meaning we give it. This does not mean agreeing with the patient about everything. It means choosing carefully what is worth debating, because our goal is not to win an argument but to create conditions that make the necessary behavior more likely.

## 2. Save Face Without Continuing the Behavior

In the chapter on the patient's mind, we saw how difficult it can be to acknowledge behavior that conflicts with our self-image. This principle becomes particularly important when we need to address adherence.

Telling someone *"you're an uncooperative patient"* turns a behavior into a personal trait. The patient now has to do more than discuss how many hours they wore their aligners: they have to defend their identity as a responsible person. From that point on, anything we say may be perceived as a personal judgment.

That is why I gradually learned to separate the person from the behavior. A patient may be exceptionally punctual for appointments but forget their orthodontic elastics, maintain impeccable oral hygiene but wear their aligners incorrectly, or follow instructions perfectly for a year and then go through a period when keeping up their routine becomes harder. Labeling them broadly as *"nonadherent"* does not help us understand which behavior we need to change and, more importantly, may increase their need to defend themselves.

When possible, I prefer to do the opposite and start with something the patient is already doing well. If they wear their aligners correctly but forget their orthodontic elastics, I can acknowledge the part of the treatment plan they are following and use that ability as a starting point for the other part. This is not flattery, and we should not credit patients with qualities they do not have. It means recognizing a real behavior and building on it to address what is missing. The implicit message is not *"you're incapable of following a treatment plan,"* but *"you've already shown you can manage one demanding part of treatment; we need to bring that same consistency to this part, too."* **Protecting the person does not mean protecting the behavior we need to change.**

The same approach becomes even more important when what a patient tells us does not match what we observe clinically. We ask, *"Have you been wearing them?"* and the patient answers, *"Yes."* We try to be more specific: *"How many hours?" "All the time, practically all day."* At that point, faced with something that does not seem clinically possible, we might be tempted to respond, *"That can't be right, because otherwise this tooth would have moved."* The problem is that, from that moment on, the goal of the conversation changes: instead of looking for a solution to the clinical problem, we try to prove that the patient did not follow instructions, and the patient tries to prove otherwise.

But we do not necessarily need a confession to change the behavior. If what we observe is enough to tell us that something needs to change, we can use that information to decide what to do without forcing the patient to admit they made a mistake or gave us inaccurate information. **The protocol should give us a way out of the choice between proving the patient wrong and accepting their version of events:** we can recognize that the behavior needs to change without first settling who is right about the past.

This does not mean ignoring what happened or accepting any explanation. It simply means distinguishing the information we need to make a clinical decision from the information we seek only to assign responsibility. Once I have gathered the information I need, I prefer to shift the conversation from the past to the future. We do not always need to establish with absolute precision how many hours a patient wore their aligners three weeks ago, but we certainly need to establish how they should wear them starting tomorrow and how we will check whether the change is working.

**The past helps us make a decision, not win an argument.**

## 3. Make the Consequences Concrete

Once we have reduced the patient's need to defend themselves, we need to make it clear why that behavior must change. This is where the three motivators truly come into play: **time, the outcome, and the investment**.

One of the things that changed how I talk with patients most was realizing that a reason a patient gives can mean far more than one I give them. For years, I tried to explain why following the treatment plan mattered, when it often would have been more useful to pause first and understand what really mattered to the person in front of me. The difference may seem small, but it changes the balance of the conversation: in the first case, I tell the patient why they should want something; in the second, I try to understand what they already want and connect the necessary behavior to that goal.

This principle is close to what the literature on social influence describes in terms of **commitment and consistency**: people tend to prefer behaviors consistent with positions, goals, or commitments they see as their own. This does not mean trapping patients with their own words, but using what they themselves consider important as a reference point for what they do next. If a patient tells me their priority is to finish before their wedding, I do not need to convince them that time matters; they have just told me so. If they explain that they do not care how long treatment takes as long as the result is perfect, emphasizing the duration means using the wrong motivator. If, instead, they feel particularly strongly about **the investment they have already made in treatment**, connecting adherence to the possibility of protecting what they have achieved will probably be more meaningful. **Before trying to change a behavior, then, we need to understand which consequence truly matters to the person in front of us.**

At that point, we need to make it concrete. As orthodontists, we inevitably tend to describe problems using the categories we learned during our training: we see tracking, torque, anchorage, relapse, stability, and movements that have not been expressed, because these are the concepts through which we interpret treatment. Patients, though, do not experience treatment through the same categories. Saying *"this canine isn't expressing the planned movement correctly"* may be clinically impeccable, but it does not necessarily create a meaningful picture in the mind of the person listening. Explaining that, if the movement continues to lag behind, we may have to pause the aligner sequence, take a new scan, and add several months to treatment makes the same problem much more concrete. **Time, the outcome, and**

**the investment do not change the clinical reality, but they allow us to translate it into something patients can assess in terms that matter to them.**

This step is particularly important because many orthodontic consequences are far off. A patient who decides not to put their aligners back in after dinner tonight does not immediately see a loss of tracking; one who forgets their orthodontic elastics does not notice that night the movement that will not occur. The benefit of following the treatment plan is often delayed, while the benefit of not doing so is immediate: they can eat more comfortably, they do not have to look for their orthodontic elastics, and they can go to bed without thinking about treatment. Making a consequence concrete therefore means narrowing that gap, mentally connecting today's behavior to something the patient wants to achieve or avoid tomorrow.

How we present that consequence can also change what it means to the patient. One of the central principles of *prospect theory* is that people do not weigh gains and losses in a perfectly symmetrical way and that, in many decision-making contexts, a loss can carry particular psychological weight. We can explain to patients that wearing their orthodontic elastics correctly will allow them to keep moving toward the planned outcome, or help them understand that inconsistent wear risks undermining a correction they have already achieved; we can show them the benefit of finishing on schedule, or make the possibility of losing some of the time they have already invested more visible.

We must use this principle responsibly. Making a consequence more apparent does not necessarily mean giving every possible scenario equal weight when we describe it to a patient: we can choose to emphasize the one we think will best help them understand the importance of the requested behavior and, in some cases, present a possible consequence more starkly precisely to reduce the risk that they will underestimate it. If a loss of tracking could require a new scan, or inconsistent adherence could lengthen treatment, we can use those possibilities to make the message more concrete, even though we cannot know for certain that they will occur. **A consequence does not have to be certain for us to communicate it, but it must remain plausible enough to be clinically defensible.**

Making a consequence concrete also means choosing carefully how to represent it. For a long time, I was convinced that a more detailed explanation was necessarily a better one. Over the years, though, I have learned that patients rarely remember everything we say during an appointment, especially when the information is extensive, technical, or far removed from their everyday experience. They are much more likely to remember an image, a comparison, a phrase, or a number that helped turn an abstract concept into something immediately understandable. That

is why photographs, scans, metaphors, comparisons, and even simple numerical references can be extremely useful tools.

Not every number we use in a conversation has to represent a scientific measurement. Sometimes a number simply serves a communication purpose: it makes a proportion more intuitive, gives a consequence a sense of scale, or helps the patient remember the message. **What matters is distinguishing a simplification used to communicate from clinical data presented as scientifically measured.** If I show a scan or cite an actual figure, that information must, of course, be accurate; if I use a number, a metaphor, or a simplification to make a concept understandable, that is a communication tool.

Our goal is not to convey the full complexity of our clinical reasoning to the patient, but to make sure the important idea stays with them after the appointment and comes back to mind a few hours later, when they have to decide what to do.

## 4. Reduce Ambiguity About Future Behavior

Once we understand why something needs to change, we need to establish **what, specifically, will change**. Expressions such as *"almost always," "as much as I can," "only sometimes," "practically all day,"* or *"I can't"* are perfectly natural in everyday language, but they do a poor job of describing clinically relevant behavior.

To us, *"wearing the aligners"* may mean wearing them twenty-two hours a day; to the patient, it may mean wearing them for most of the day, carefully putting them back in before the appointment, or wearing them whenever they could. Clinician and patient can therefore use the same words while assigning them very different meanings.

For a long time, my response was to challenge those statements. If what I observed clinically suggested something different, I tried to convince the patient that their *"always"* was not really always. Today I prefer to do something much simpler: I try to **get the patient to be specific**. I do not debate the meaning of *"always"*; I walk through a typical day. I do not try to determine whether *"I can't"* is true or false; I try to understand exactly when the difficulty arises.

If a patient tells me they always wear their aligners, I can ask what time they eat breakfast, how long the aligners stay out, what happens at lunch, whether they drink coffee during the afternoon, how they handle dinner, and what time they put the aligners back in before bed. If, instead, they tell me they cannot wear their orthodontic elastics, I can try to find out when the problem occurs: at work, while they are talking, in the evening, or when they go out. I can ask whether they forget

them, do not have them with them, or find the discomfort too much after a few hours. I am not interrogating them to catch them in a contradiction. I am turning a broad, subjective assessment into a sequence of behaviors we can observe and, if necessary, change. **When the problem becomes specific, it often becomes more manageable, too.**

The same principle should apply when we define what needs to happen from that point forward. The more important an instruction is to the outcome, the less it should depend on personal interpretation. *"Try to wear your elastics as much as possible"* leaves enormous room for ambiguity. Specifying when they need to be put on, when they can be taken off, and what to do if the patient forgets them turns a general recommendation into an observable behavior. This is where the *implementation intentions* we discussed at the beginning of the chapter come in handy. Telling a patient *"you need to remember your elastics"* leaves them with the task of remembering every time. Working with them to create a rule such as *"when I finish brushing my teeth at night, I put on my elastics"* instead links the behavior to a recognizable situation. We have not changed the instruction, but we have made it easier to turn into a routine.

Reducing ambiguity also means paying attention to how we phrase our own instructions. In a therapeutic relationship, there is inevitably a difference in expertise: patients come to us precisely because they expect us to know what is necessary and what is optional. Yet, in our understandable effort not to sound authoritarian, we sometimes make something seem negotiable when, clinically, it is not.

Phrases like *"maybe give it a try," "it would be better," "try to do it,"* or *"let's see if you can"* are perfectly appropriate when we are genuinely suggesting something optional, but they become problematic when they concern a behavior we consider necessary for treatment to work. Being authoritative does not mean being harsh or using our role as clinicians to demand blind obedience. It means taking responsibility for stating clearly when a particular condition is necessary to achieve the goal we have agreed on.

This becomes especially important when patients understandably look for an easier solution. If they ask whether they can wear their aligners for fewer hours, stop wearing orthodontic elastics, or cut back on an instruction we consider necessary, the desire to maintain a pleasant relationship can lead us to make something seem negotiable when, clinically, it is not. The approach I have found most useful is to distinguish **what** from **how**. The *what* is a matter of clinical instruction: if a particular behavior is necessary for treatment, there is no point pretending otherwise. The *how*, on the other hand, can often be worked out with the patient, giving them

room to decide how to organize their day, build a routine, and fit that behavior into their daily life.

Patients do not get to decide whether a clinically necessary part of treatment is necessary, but they can help decide how to make it work with their daily schedule. I consider this balance essential: **being authoritative about the clinical need while giving patients autonomy over the behavioral strategy**.

## 5. Make the Behavior Easier and Check Its Effect

Defining exactly what patients need to do does not yet mean they will be able to do it. The problem is not always a lack of motivation: a behavior may be hard to fit into a routine, require too many decisions, depend on remembering at the wrong moment, or need to happen somewhere the patient does not have what they need.

In these cases, increasing the pressure in our conversations does relatively little, while changing the environment in which the behavior needs to happen may be much more useful. This is one of the ideas behind **choice architecture**: every decision takes place in a context, and how that context is organized can make certain actions easier or harder.

The principle of *nudging* fits within this approach: small interventions that make a useful behavior easier without ruling out alternatives. A reminder, a sooner follow-up check, a supply of orthodontic elastics kept in the right place, or a routine tied to an existing habit can narrow the gap between a stated intention and everyday behavior.

A patient who forgets their orthodontic elastics before bed, for example, could link putting them on to something they already do every night, such as brushing their teeth or setting their phone on the nightstand. Someone who tends to leave their aligners at home could organize their routine so they always have what they need with them. In these cases, we have not necessarily increased motivation; we have reduced the friction that kept intention from turning into action. **Our goal, then, is not just to persuade patients to do something, but to make that behavior easy enough to fit into their daily lives.**

At this point, though, one element is still missing: **feedback**. One of the problems with orthodontic treatment is that many consequences emerge slowly. Patients can wear their aligners less than they need to for several days without seeing any obvious difference, and they can forget their orthodontic elastics for weeks without immediately noticing what is happening. By the time the problem becomes clinically

visible, the behavior that caused it may be well in the past, making the connection between cause and effect much less intuitive.

That is why showing an earlier photograph, comparing two scans, pointing out a spot where the aligner no longer fits properly, or showing that a movement has fallen behind can be much more effective than repeating the same recommendation yet again. Feedback makes visible something that would otherwise remain abstract and helps patients connect what they do every day with what is happening during treatment.

Feedback, though, must not become evidence in a trial. We do not show the scan to say *"See? I was right, wasn't I?"*, because at that point we would simply have turned a clinical tool into a weapon in the discussion. We use it to make the connection between what is happening and what needs to change visible.

But feedback is not just for the patient. **It also helps us check whether the assumption behind our intervention was correct.** If we have identified a problematic behavior, tried to change it, and, after an appropriate interval, treatment still is not responding as we expected, we cannot simply conclude that the patient is still not following instructions. We need to be willing to reconsider our assumption and ask whether there is a clinical, biological, or biomechanical problem we failed to recognize.

Feedback therefore closes the loop on the protocol and, at the same time, can start it again. It shifts the conversation away from personal trust and toward observable evidence: we do not have to decide whether to believe the patient. We can look together at what is happening, decide what to change, and check soon enough whether the new strategy is working. **A protocol does not end when we have told the patient what to do, but when we have verified that what we decided together is producing the effect we wanted.**

## Shared Responsibility, Distinct Roles

Orthodontic treatment is inevitably a collaboration in which clinician and patient take part in the same process with different responsibilities. We are responsible for diagnosis, treatment planning, clinical decisions, follow-up visits, and any changes needed during treatment. The patient has different responsibilities, including following instructions, wearing aligners and orthodontic elastics correctly, keeping appointments, and telling us when something is not working.

Shared responsibility does not mean that responsibilities are indistinguishable. Simply saying *"the outcome depends on both of us"* is accurate only if we can then

clarify which part depends on us and which depends on the patient. This helps us avoid two opposite extremes: automatically blaming the patient for every problem or taking responsibility ourselves for behaviors over which we have no direct control.

That is also one reason protocols should begin **before a problem arises**. Defining roles when everything is going well is much easier than trying to establish them once the aligners have already stopped tracking and clinician and patient are trying to figure out who is responsible.

The same goes for consequences. If a particular behavior could require a new scan, change the timeline, or affect how we need to proceed, it is better for the patient to know that before the situation arises. **A consequence explained after a problem occurs can feel like a punishment; the same consequence explained in advance becomes a condition of treatment.**

## A Protocol Must Belong to the Practice

Finally, there is one point I consider essential: a protocol cannot exist only in the orthodontist's head, because over the course of treatment, the patient speaks with the front desk staff, assistants, hygienists, other clinicians, and everyone else involved in their care. If each of them communicates a different rule, it quickly stops being a rule.

Not everyone needs to use exactly the same words. Repeating a script too rigidly could seem artificial and keep each team member from communicating naturally. But it is essential that **the meaning of the answer remain the same**.

The front desk staff needs to know which appointments can be rescheduled, which scheduling consequences need to be communicated, and which decisions are solely the clinician's to make. Assistants need to understand what the instructions mean and not inadvertently turn a necessary condition into a mere suggestion. The clinician, for their part, cannot contradict a position in front of the patient that the team has correctly taken based on the shared protocol.

The patient may speak with different people, but should still sense **one consistent clinical direction**. If the clinician presents an instruction as essential and an assistant presents it as optional, we cannot be surprised when the patient chooses the interpretation that requires less effort. Likewise, if a rule or consequence changes depending on whom the patient speaks with, that rule inevitably becomes negotiable.

Team consistency, then, does not mean everyone repeating the same phrases. It means agreeing on the same conditions, the same limits, and the same direction. **A protocol truly becomes a protocol only when it stops being the clinician's personal response and becomes the consistent way the entire practice handles a familiar situation.**

## From Principles to Strategies

None of the principles described in this chapter is a magic formula, and none is enough on its own to build a protocol. We will not need to use framing, loss aversion, nudging, feedback, and every other tool at the same time in every conversation. A conversation built that way would feel artificial and would probably be ineffective, too.

Each strategy should draw only on the elements that are useful for the problem in front of us. In some cases, our main task will be to reduce defensiveness; in others, it will be to make a consequence more concrete, reduce ambiguity, make a behavior easier, or state a clinical requirement more clearly. Cognitive psychology and behavioral economics do not give us a script to recite. They give us tools that can help us understand why some responses make change easier while others are more likely to lead to conflict.

The way we communicate must also adapt to the person in front of us. A direct patient may need a brief explanation focused on the decision at hand, while a more analytical patient will want more detailed information and a fuller account of each step. Someone who is less confident may need a more gradual approach and more reassurance. What should not change is the logic of the protocol: **reduce defensiveness, protect the person without protecting the behavior, make consequences concrete, define the future behavior precisely, make it easier to carry out, and check its effect**.

The protocols presented in the following pages grew out of situations I encountered repeatedly in clinical practice, responses that did not work at first, and changes made over time. They should not be treated as universal formulas that always produce the same reaction. Human behavior inevitably varies, and no sequence of communication can guarantee that a patient will follow instructions. Their value lies not in eliminating that variability, but in helping us approach similar situations with a recognizable method—one we can observe, discuss, and improve. **An improvised response disappears with the conversation in which it was used; a repeatable method, by contrast, can help the entire practice learn.**

That is why each protocol in the chapters that follow will have a recognizable structure. We will start with the clinical situation and the signs that help us identify it, pinpoint the behavior that needs to change and the common mistakes that can increase patient resistance, then define the steps to take, the questions to ask, the points to check, and the consequences to explain in advance. The clinical dialogue will show how to put that structure into concrete words, while the **"Why This Protocol Works"** box will let us return, when useful, to the psychological principles we have just discussed.

This distinction between procedure and explanation matters. Colleagues need to know what to say and do, but they should also understand why a particular sequence may be more effective than an accusation, a vague reminder, or simply repeating instructions. Only by understanding the mechanism can we change the words without changing the protocol's function.

The goal, then, is not to put together a collection of *"clever lines"* or to turn the orthodontist into an expert in persuasion. I want to arrive at something much simpler and more useful: when faced with a patient who is not following instructions, I want our first thought to gradually stop being *"How do I explain it to them one more time?"* and become *"What is keeping this person from doing what they already know they need to do? What consequence actually matters to them? And what protocol can I use to make that behavior more likely to change?"*

For many years, we have planned tooth movement with extreme precision while leaving the conversations needed to protect that plan to improvisation. We repeated instructions, motivated patients, and tried to find a new way to put it each time. When the problem came up again, we started over.

Protocols change that perspective. They do not try to make patient behavior predictable, and they certainly cannot control it. But they allow us to make **our response** more predictable. If certain behaviors follow recognizable patterns, we can learn to recognize them. If we can recognize them, we can prepare to manage them. And if we can prepare to manage them, our response can finally stop being an improvisation and become a method.

**This is where the most practical part of this book begins.**

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CHAPTER

# 6

# The Patient Who Doesn't Wear Their Aligners

*"Well done is better than well said."*

— BENJAMIN FRANKLIN

Sometimes, looking at the aligners in the patient's mouth is enough to see that things are not going as planned. But not every loss of tracking means the same thing, and, more importantly, not every case calls for the same clinical response.

If a single tooth is not tracking perfectly with the aligner, I do not automatically assume the patient is not following instructions, and I do not necessarily stop the sequence to take a new scan. We know that some movements are less predictable than others: a maxillary lateral incisor that needs a complex rotation, a difficult extrusion, or another particularly demanding movement may lag behind even when a patient wears their aligners impeccably. If that movement is not essential

to continuing treatment and its failure to occur will not interfere with subsequent movements, I often prefer to continue with the planned sequence and address it later during the refinement phase.

The situation changes when the loss of tracking no longer involves just one tooth but starts to become widespread. **That is why I normally schedule the first check during the sequence after about two months.** I consider that long enough to get a sense of how the patient is actually wearing their aligners and to begin seeing any loss of tracking, but short enough that, in most cases, I can intervene before the discrepancy becomes too significant. If I notice at that visit that several teeth are starting to lag behind their planned positions, we can often still restore tracking by changing the patient's behavior or temporarily slowing the sequence, without having to stop everything and take a new scan.

When I see at this first check that the loss of tracking no longer involves a single movement but is progressively affecting more teeth, I inevitably begin to suspect strongly that the patient is not wearing their aligners for enough hours. That is precisely when one of the most common communication problems in aligner treatment begins. To understand what is happening, I start with a very simple question:

*"How are things going with your aligners? What time of day is it hardest for you to put them back in?"*

That question helps me more than a general *"Are you able to wear them all the time?"*, to which the patient will almost inevitably say yes. So I try to get a clearer picture of what happens during a typical day, asking when they take the aligners out, how long meals last, what happens at work, and what happens when they go out in the evening or on weekends. I pay particular attention to the time between the end of a meal and when the aligners actually go back in, because it is often then, rather than during the meal itself, that the hours without aligners add up.

I do not approach this as an interrogation. I try to work with the patient to identify times when, without realizing it, they may be leaving their aligners out longer than they think. In the end, though, the patient may still be absolutely certain of their answer:

*"No, Doctor. I really do wear them all the time."*

At that point, I have two pieces of information that seem hard to reconcile: what I see clinically suggests that the teeth are not tracking with the aligners as they should, while the person in front of me insists with great conviction that they have followed every instruction to the letter.

## The Behavior We Need to Change

The behavior we want to change is not simply *"following instructions more closely."* We need the patient to wear their aligners for the required number of hours, especially by reducing the times during the day when the aligners stay out of their mouth longer than they realize.

This matters because the problem very often does not come from a conscious decision to wear the aligners too little. It comes from the sum of many seemingly insignificant stretches of time: breakfast that takes longer than expected, coffee after lunch, a phone call before putting them back in, drinks before dinner, a dinner that runs long, or simply leaving them on the table and remembering them an hour later. Taken individually, these episodes seem negligible; added up over the course of a day, they can substantially reduce actual wear time.

That is why our first goal is not to determine whether the patient is telling the truth, but to turn a general *"I wear them all the time"* into a concrete enough account of their day to help us understand where the problem might lie.

## The Mistake I Made: Trying to Convince the Patient

When faced with these situations, I chose the approach that seemed simplest and most logical to me. If I saw a widespread loss of tracking, I explained to the patient that they probably were not wearing their aligners enough and stressed the need to increase their wear time. I showed them what I was seeing clinically and said something like:

> *"Look, you can clearly see here that your teeth aren't tracking with the aligners. You need to try to wear them more."*

The problem was that I almost always got the same answer:

> *"But I do wear them."*

I could see the loss of tracking and kept pressing the point, while the patient insisted they had followed the instructions. The more I tried to convince them that the problem was their adherence, the more they felt the need to defend their behavior. After a few minutes, we were no longer discussing how to get treatment back on track, but which of us was right.

This dynamic can become even more pronounced when the patient in front of us is a very self-assured adult, perhaps someone with a prominent professional role who is used to making decisions, managing other people, and standing firmly by

their views. Telling that patient outright, *"That's not true. You aren't wearing your aligners,"* calls into question not only their adherence but also their credibility.

At that point, the clinical problem risks turning into a personal confrontation in which we both try, more or less consciously, to prove we are right. The paradox is that even if I won that argument, I would not necessarily have achieved what really matters to me. My goal should not be to convince the patient that they made a mistake in the preceding weeks, but to help them wear their aligners correctly from that point on.

It took me a while to understand this, and I probably needed one particular patient to change my perspective completely.

## The Patient I Didn't Believe

I had treated Chantal's husband with aligners first, followed by their two children. All three had followed instructions closely, and their treatments had ended without any particular difficulties. So when she decided to straighten her teeth, too, I had no reason to expect anything different. Hers was, at least on the surface, a very simple case, with some crowding mainly around the lower incisors and nothing that, from a biomechanical standpoint, suggested any particular difficulty.

We started treatment, and at the first few checkups I realized that something was not going as planned: her incisors were moving much less than I expected, and the aligners were progressively losing tracking. She worked in fashion, traveled extensively, and had an especially busy professional life full of events, drinks before dinner, and other social occasions when it is easy to forget about aligners. For that reason, the explanation seemed almost obvious to me: she simply was not wearing them enough.

So I began stressing the importance of following the instructions, reminding her that she needed to wear the aligners for the prescribed number of hours and explaining that, if things continued this way, her teeth would not track as planned. But she kept telling me the same thing:

*"Enzo, I do wear them."*

I kept looking at those incisors, and I kept not believing her. This went on for more than six months, until Chantal told me she would be moving to London for work in a few months and asked me to finish treatment before she left. At that point, I thought I would eliminate once and for all what, in my mind, was still the main cause of the problem. I proposed putting brackets on her lower incisors, promising her that they would be perfectly aligned in a couple of months. I was fairly confident, because I

was convinced that once I eliminated adherence as a variable, those few teeth would align quickly.

Two months later, her incisors were still crooked.

That was probably the first time I seriously considered a possibility I had all but ruled out: perhaps this patient had never lied to me, and the problem I had attributed for months to her adherence might have been due, at least in part, to her biological response to orthodontic movement.

## Teeth Don't Move at the Same Speed in Every Patient

That experience prompted me to look more closely at individual differences in the rate of orthodontic tooth movement, and I discovered that what I had observed clinically had a much stronger biological basis than I had imagined.

When we apply orthodontic force, we are not mechanically moving a tooth the way we might move an object. The force is the initial stimulus for a complex biological response involving the periodontal ligament, alveolar bone, osteoclasts, osteoblasts, inflammatory mediators, and numerous cellular and molecular mechanisms. The speed of this response varies considerably from person to person, and even when we try to standardize the force applied, different people can show very different rates of tooth movement.

One of the studies that best demonstrates this phenomenon was published by Giannopoulou and colleagues in the *European Journal of Orthodontics*. The authors used a standardized experimental protocol in which two premolars in each participant were moved buccally for eight weeks using a force of 1 N. Although the force was standardized and local or functional obstacles to movement had been ruled out, the amount of movement varied enormously between individuals, ranging from just 0.6 mm to 5.8 mm, with a mean of 2.7 mm. Even more interestingly, teeth belonging to the same individual tended to behave similarly, showing a strong within-person correlation. The authors concluded that this wide variability made it possible to identify *slow movers* and *fast movers* within the sample.

The concept has also been examined in subsequent clinical studies. In a randomized trial on extraction-space closure, for example, the authors explicitly used the terms *fast movers* and *slow movers* to distinguish patients who showed different amounts of tooth movement over the same period. In that particular study, cases with movement of at least 1 mm per month were defined as *fast movers*. Of course, that threshold is not a universal definition we can apply in clinical practice, and it does not allow us to simply sort all our patients into two categories. But it shows that

differences in individual rates of movement are real and that the concept is used in the orthodontic literature itself.

The reasons for this variability are probably numerous and are not yet fully understood. Orthodontic tooth movement depends on the tissues' biological response to force, and researchers have studied individual differences in the molecular systems involved in bone and periodontal ligament remodeling. Some studies have also investigated a possible genetic contribution, particularly polymorphisms in genes involved in the RANK, RANKL, and OPG systems and in other mediators of the orthodontic response. Interestingly, in this area too, the authors explicitly refer to the clinical phenomenon of *slow movers* and *fast movers*, suggesting that at least part of the difference in movement rates may depend on each person's biological characteristics.

This biological variability, of course, also existed when we treated patients with fixed appliances. In that setting, though, a *slow mover* was mainly a matter of time. If a tooth had not yet reached the desired position, in most cases we could simply leave the same archwire in place longer and wait for the biological response to run its course. As my great mentor Davide Mirabella has always told me: *"Let them cook."* Let them cook. Sometimes the best thing to do was simply give the teeth the time they needed to move.

The fact that a patient needed more time did not necessarily mean we had to change the treatment sequence: we could adjust the pace of treatment to the speed at which their teeth were actually moving.

With aligners, the problem takes on a different dimension. It is no longer just a question of how long a tooth takes to move, but of whether its biological rate of movement can stay in sync with the planned sequence. Each aligner is designed on the assumption that the movement planned for the previous aligner has occurred to a sufficient degree. If a patient is a *slow mover* and we keep switching aligners at a pace that is too fast for their biological response, a tooth can gradually fall behind its planned position. The next aligner then tries to fit a tooth that has not yet reached that position, and the discrepancy can build with each change until it becomes clinically apparent as a loss of tracking.

This is probably one reason why the concept of a *slow mover*, though far from new, becomes particularly relevant in aligner treatment. With fixed appliances, we could often simply afford to wait longer; with aligners, we need to match the pace at which we advance through the sequence to the pace at which that patient's biology can actually produce movement.

None of this means, of course, that a widespread loss of tracking should automatically be attributed to biology, or that being a *slow mover* is the most likely explanation whenever aligner treatment does not go as planned. Adherence remains one of the first variables to consider when several teeth begin losing tracking at the same time. But the literature on *slow movers* reminds us of something equally important: from clinical observation alone, we cannot always know for certain how much of what we are seeing is due to the patient's behavior and how much may be influenced by their individual biological response.

In the end, that was exactly what Chantal had taught me. In her case, I had eliminated adherence as a variable by switching to fixed appliances, and her teeth had continued to move slowly: **she really was a *slow mover*.** But that experience also helped me understand something else. If a slower biological response could genuinely explain a loss of tracking, then I could raise that same clinical possibility when I faced a patient whose adherence I strongly suspected was the problem instead, without accusing them or having to prove that what they were telling me differed from what they were actually doing.

Over the years, that insight gave rise to one of the strategies I use most often.

## I Don't Have to Prove the Patient Is Lying

Today, when I see a patient with a widespread loss of tracking, I no longer start with an accusation or immediately try to convince them that they are not wearing their aligners. First, I try to work out with them what is happening by going through their daily habits and looking for times when the aligners stay out of their mouth longer than they realize. This step matters in light of everything we saw in the first part of the book about reconstructive memory and the gap that can exist between what we actually do and how we remember and describe our behavior.

If, after this conversation, the patient continues to assure me that they are wearing their aligners exactly as prescribed, I no longer try to prove otherwise. Instead, I deliberately accept their account and introduce another possible explanation:

> *"Look, we have two pieces of information we need to put together. I can see that your teeth aren't tracking with the aligners at the rate we expected, and you're telling me you're wearing them correctly. If that's the case, we need to consider another possibility."*

At this point, the patient usually pays close attention, because they do not feel accused and want to understand what the other explanation might be. I continue:

*"Teeth don't move at the same speed in everyone. Most patients respond within the time frames we normally use, but there is a small percentage of people whose biological response is considerably slower. The literature even uses the terms fast movers and slow movers. If you're wearing your aligners as prescribed and your teeth keep falling behind, we need to consider the possibility that you're a slow mover."*

At this point, I have not contradicted the patient or asked them to admit anything. On the contrary, I have accepted their account. But precisely because I have accepted it, I need to draw a consistent clinical conclusion:

*"If your teeth move more slowly, one week might not be enough for them to complete the movement planned for each aligner. If we kept changing aligners every seven days anyway, the gap between your teeth's planned and actual positions could gradually grow, making the loss of tracking worse and worse."*

The consequence I propose is simple:

*"For that reason, at least for now, instead of changing your aligner every week, we should change it every three weeks. Then we'll check how your teeth are responding and, if we see that tracking is improving, we can gradually try to speed things up by switching, for example, to a change every two weeks."*

Changing aligners every three weeks is a cautious clinical decision on my part, not an interval the literature has identified as necessary for all patients with a slower biological response. For the patient, though, the shift from seven to twenty-one days makes a consequence that may have seemed abstract immediately concrete: if we needed three times as long as planned for each aligner, and kept to that pace, the overall treatment could even take three times as long.

And this is often when the most interesting thing happens.

The patient thinks about it for a few seconds and spontaneously begins to describe their behavior a little differently:

*"Well, Doc, now that I think about it, maybe it isn't quite twenty-two hours every single day. On weekends I sometimes wear them less, and when I go out, they might stay out longer. If I really make an effort to wear them all the time, can we keep changing them every week?"*

It is important to understand that I have not asked them to confess anything. I have not said I do not believe them, and I have not tried to prove they were lying to me. I have simply taken what they told me seriously and shown them what the consistent clinical consequence would be if, despite wearing their aligners correctly, their teeth truly could not keep up with the planned pace.

Once the patient understands that this could mean going from seven to twenty-one days and therefore greatly prolonging treatment, they also start to see those hours without aligners differently. What, just a few minutes earlier, they had summed up as *"I wear them all the time"* can become *"Actually, a little less on weekends,"* or *"When I go out, they sometimes stay out quite a while."*

And when that happens, in most cases **I do not switch to three-week changes at all**. If the patient spontaneously acknowledges that they have worn their aligners for fewer hours than prescribed and tells me they want to try to be more consistent, I keep the weekly changes and give them a chance to show it through what happens over the following weeks. Three-week changes remain the consistent clinical consequence when the patient continues to assure me that they have worn their aligners correctly.

In other words, I do not use a slower pace as punishment. I use time as a motivator to make the consequence of what I am seeing concrete and allow the patient to choose whether to change their behavior.

## At This Stage, Time Is the Most Powerful Motivator

As I have already described, Time, Outcome, and Investment are three of the motivators I use most often when communicating during aligner treatment, but they do not all carry the same weight at every stage of therapy.

When the patient started treatment only a few months ago, using the investment as a motivator generally makes little sense: treatment has just begun, and the patient has only recently paid for it. The outcome certainly remains important, but at this stage it can still feel relatively far away, and the patient may have trouble grasping how poor adherence today could jeopardize something they will not see for many months. Time, on the other hand, is immediately understandable.

If someone began treatment expecting to change aligners every week and to finish in roughly fifteen or eighteen months, hearing that they might now need to wear each aligner for three weeks makes the consequence extremely concrete. We are no longer talking vaguely about the possibility that *"treatment will take longer"*: the patient immediately sees that the pace of treatment could become three times slower than they had imagined at the start.

So I explain:

> *"We inevitably have to allow for the possibility that treatment could take much longer. If your teeth really need two or three times as long to keep up with each aligner, we can't expect treatment to finish in the time we originally planned. First*

*we need to find out the pace at which your teeth can actually keep up with the aligners, and then we can give you a more reliable estimate."*

I do not need to tell the patient that an eighteen-month treatment will definitely take four and a half years, because I cannot know that, and it would be wrong to turn a possible consequence into a certain prediction. But I can help the patient understand that, **if the pace really remained one aligner every three weeks instead of every seven days, the time required could be as much as three times longer**.

It is precisely the difference between a vague consequence and a concrete one that makes time as a motivator so effective. Saying *"if you don't wear them, treatment will take longer"* is something the patient has probably heard before, and it can remain entirely abstract. Going from seven to twenty-one days, on the other hand, translates that same consequence into something they can see, measure, and relate to their goal.

## A Strategy That Works in Either Scenario

What interests me most about this approach is that I do not need to know with absolute certainty which of the two explanations is correct.

If the patient really is wearing their aligners twenty-two hours a day and has an unusually slow biological response, slowing down the changes and reassessing tracking is a cautious choice consistent with what I am seeing clinically.

If, instead, as is probably the case in most situations where I use this strategy, the patient is wearing their aligners for far fewer hours than they say, I have not accused them of lying or forced them to defend themselves. I have simply shown them a consequence of their current behavior that matters to them right now.

Very often, that is what prompts them to reconsider their account on their own. The patient does not have to jump abruptly from *"I wear them all the time"* to *"I was lying to you."* They can simply pause, think back over their week, and recognize that perhaps *all the time* did not really mean twenty-two hours every day. If they then decide to make a greater effort, I can keep the weekly changes and see whether their words are followed by a real change in behavior.

If, instead, they continue to insist that they are already wearing the aligners for the prescribed number of hours, I have no reason to argue: I accept their explanation, slow down the changes, and assess the response.

Either way, I have achieved what really mattered to me: **a concrete decision about what will happen from that point on.**

## A Confession Doesn't Move Teeth

I think this has been one of the most important changes in how I communicate with patients who struggle with adherence. For a long time, I thought my job was to make the patient understand that they were not doing what I had asked, almost as though admitting the mistake were a necessary condition for getting back on track. Over time, though, I realized that getting that admission was not what I needed at all.

Paradoxically, today that kind of confession comes more often precisely because I have stopped seeking it. When the patient does not feel accused and instead understands the concrete consequence of what we are seeing, they can reconsider their behavior without feeling that changing their story means admitting defeat.

If they tell me, *"Actually, I wore them less over the weekend, but starting today I'm really going to make an effort,"* that information is certainly useful to me, because it allows me to keep the weekly changes and see whether better adherence is enough to restore tracking. But the confession is not what has therapeutic value: what matters is what the patient does next.

If, instead, after ten minutes of discussion, I managed to force them to admit, *"You're right, I was actually wearing them twelve hours a day,"* I might have the satisfaction of knowing I had interpreted the loss of tracking correctly, but clinically I still would not have achieved what really matters to me. **A confession does not move teeth; a change in behavior does.**

The *slow mover* strategy allows the patient to change their behavior without forcing them to admit that what they had told me just minutes earlier was not entirely true. It gives them a way to save face while making it much harder for them to ignore the consequence of continuing as they have been.

If we put someone in a position where they have to choose between admitting they have not followed our instructions and continuing to defend their account, they may choose the latter even when it does nothing to help them achieve the result they want. If, instead, we set that argument aside and focus on the clinical consequences and possible solutions, we allow them to change their behavior without first having to openly acknowledge that they made a mistake.

That is why, when the clinical evidence and the patient's account do not match, I now spend much less time trying to establish who is right and much more time finding a strategy that can work regardless of the explanation. **We do not need to prove to the patient that we are right; we need to create the conditions for them to start doing the right thing.**

**PRACTICAL SCRIPT**

When I observe a generalized loss of tracking, after reviewing the patient's habits with them and hearing them confirm that they are wearing their aligners correctly, I can frame the conversation this way:

> *"I can see that several teeth aren't keeping up with the aligners at the pace we expected. We've gone over how you're wearing them, and you've confirmed that you're wearing them correctly, so we need to consider another possibility. Teeth don't move at the same speed in everyone, and some patients have a slower biological response to orthodontic tooth movement. The literature also uses the terms fast movers and slow movers, and if you're wearing your aligners as prescribed, you may be in that second group.*
>
> *If your teeth move more slowly, one week may not be enough for them to reach the position each aligner is designed to achieve. If we kept changing aligners every seven days, that small lag could gradually grow, and tracking would get worse and worse. That's why, if you're wearing them for the prescribed number of hours, we need to give your teeth more time and switch, at least initially, from one week to three weeks per aligner.*
>
> *That would make the pace of treatment three times slower. I can't tell you today that the entire treatment will take three times as long, because we first need to see how your teeth respond, and we may be able to speed things up again. But if we had to maintain this pace throughout treatment, the total time could increase substantially. So now we need to find out the pace at which your teeth can actually keep up with the aligners."*

At this point, I give the patient room to respond. If they say:

> *"Doc, now that I think about it, maybe it isn't always twenty-two hours. If I really make an effort, can we try to keep the weekly changes?"*

my answer can be just as simple:

> *"Of course. If you think there are hours we can make up, I'd rather give you that chance. Let's keep the weekly changes, but from today on, we need to be very consistent. At your next visit, tracking will tell us whether that change was enough."*

If, instead, the patient continues to assure me that they are already wearing the aligners for the prescribed number of hours, I do not press the issue or try to get them to change their story: I slow down the changes as I explained and assess the response later.

## Why This Protocol Works

- **Reducing Reactance**

Directly contradicting the patient and telling them they are not wearing their aligners as they claim may make them defend their position even more strongly. Initially accepting their account and proposing an alternative explanation avoids a confrontation and keeps the conversation focused on solving the problem.

- **Saving Face**

The *slow mover* strategy allows a patient who has not worn their aligners enough to change their behavior without being forced to admit they lied. They can simply reconsider what they meant by *"all the time"* and acknowledge that they missed some hours. That way, changing their behavior does not first require a personal defeat.

- **Making the Consequence Clear**

Saying *"you need to wear them more"* tells the patient what to do, but does not necessarily make clear why they should do it. Going from seven to twenty-one days, on the other hand, translates the consequence into something they can immediately understand.

- **Time as a Motivator**

In the early stages of treatment, the patient has only recently made the investment, and the final outcome may still feel far away. Time, meanwhile, is an immediate and easily understood consequence. The prospect of moving at one-third the pace can quickly change how the patient values those hours when the aligners are out.

- **Feedback**

The protocol does not end with the conversation. If the patient decides to keep the weekly changes and make a greater effort, tracking at the next visit provides the feedback we need to see whether the change in behavior was enough. If, instead, we slow down the changes and tracking still does not improve, we must also be willing to reconsider our clinical hypothesis.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

**Main motivators: time and control.**

With a Dominance-style patient, I would avoid a direct confrontation, because openly questioning what they are telling us could quickly turn the conversation into a contest. It is more effective to present the two possibilities clearly and let what the patient does next determine which path we can take.

> *"If you're already wearing them for the right number of hours, we need to slow down because your teeth need more time. If, instead, you think you could be more consistent, we can keep the weekly changes and see right away whether tracking improves. We'll have our answer at your next visit."*

**I**

### Influence

**Main motivators: the outcome and motivation.**

With an Influence-style patient, I would avoid an overly long or technical explanation and focus more on the possibility of getting things back on track quickly and keeping the original pace if they make a real change in their behavior.

> *"We can still try to keep the weekly changes. If you know you wore them less on some days, let's get back to the prescribed wear time starting today and see how your teeth respond. If tracking is back where we want it at your next visit, we'll continue this way."*

**S**

### Steadiness

**Main motivators: reassurance and a clear path forward.**

With a Steadiness-style patient, it is important to keep them from interpreting the situation as a sign that treatment is failing. I would explain that we have two possible paths and that we can monitor either one over time.

> *"This doesn't mean the treatment isn't working. We just need to find out whether your teeth need more time or whether we can make your wear time more consistent. If you think you can be more consistent, we'll keep the weekly changes for now and look at how your teeth are responding together at your next visit."*

## Adapting Communication to DISC Profiles

CONTINUED

**C**

### Conscientiousness

**Main motivators: data and explanation.**

With a Conscientiousness-style patient, I would make greater use of what we can observe clinically, showing them how the aligners are tracking and explaining the two possibilities precisely, along with how the next visit will help us distinguish between them.

> *"Clinically, we can see that your teeth have fallen behind the planned positions. If you're already wearing the aligners for the right number of hours, we need to consider a slower biological response and give each aligner more time. If, instead, you think you've missed some hours of wear, we can keep the weekly changes for now, return to the prescribed wear time, and check at your next visit whether the gap has narrowed."*

CHAPTER 6

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | At the first follow-up visit in the aligner sequence, usually after about two months, when a progressive, generalized loss of tracking appears. Also assess the treatment plan and biological response before attributing it to adherence. |
| **GOAL** | Restore tracking by changing the patient's behavior or the aligner change interval, then assessing the response. |
| **WHAT TO DO** | 1. Review the patient's day: meals, work, evenings, weekends, and how long they wait before putting their aligners back in.<br>2. If the patient confirms that they are wearing them as prescribed, also consider a slower biological response.<br>3. Discuss moving from seven to twenty-one days per aligner, explaining that this would significantly slow treatment. Twenty-one days is a choice in the protocol described here, not a universal prescription.<br>4. If the patient acknowledges that they have not worn the aligners enough, agree on the prescribed hours of wear and keep the weekly changes when clinical conditions allow.<br>5. If the patient confirms that they are wearing the aligners as prescribed, slow down the changes and reassess tracking, then decide whether to maintain that pace or speed up.<br>6. If tracking does not improve, reconsider the initial clinical hypothesis as well. |
| **WHAT TO CHECK** | The distribution of the loss of tracking, hours and routine of wear, the aligner change interval, and the response at the next visit. A promise to make a greater effort must be followed by a verifiable change. |

CHAPTER

# 7

# The Patient Who Wears Their Aligners but Not Elastics

*"A chain is only as strong as its weakest link."*

— ENGLISH PROVERB

When I began using intermaxillary elastics with aligners many years ago, relatively few clinicians were systematically incorporating them into their treatment mechanics, especially to correct Class II malocclusions through distalization or other **sagittal mechanics**. To me, they were a fundamental part of treatment, and, as often happened whenever I introduced something new into my practice, I watched every case closely, trying to understand which mechanics worked best and which movements were more predictable.

Some treatments proceeded exactly as I had expected, while others seemed to behave completely differently. I would start a distalization, wait a few months, and

then find an apparently contradictory situation at the follow-up visit: the aligners were tracking the teeth perfectly, tracking was excellent, crowding was resolving, rotations were improving, and everything that depended directly on the aligners seemed to be moving in the right direction. **The Class II malocclusion, however, was not improving.**

Over time, I began to recognize a pattern that recurred fairly regularly: **a patient could wear their aligners correctly while being far less consistent with their orthodontic elastics.** I realized that I had made a simple conceptual mistake: I tended to think of adherence as a general characteristic of the patient, when wearing aligners correctly and wearing orthodontic elastics correctly are two entirely different behaviors. Today, that distinction seems almost obvious to me, but for many years it was anything but.

## When the Aligners Work but the Elastics Don't

In my practice, I use intermaxillary elastics primarily for sagittal and transverse correction. They can be very useful for crossbites, although I generally introduce them at a later stage of treatment, so adherence presents somewhat different challenges there. The situation in which their use becomes particularly important, and in which I have learned the most from observing patient behavior, is sagittal correction.

When I perform a distalization, I prescribe orthodontic elastics from the start and expect sagittal correction to progress alongside the movements planned for the aligners. If, after a few months, I see spaces opening between the posterior teeth and the aligners continuing to track well, but the Class II molar relationship is not improving and unwanted effects are beginning to appear in the anterior teeth, I strongly suspect that the patient is not wearing their elastics consistently enough. This is essentially what we saw happen in Laura's case, described in the first part of the book.

When I use interarch mechanics instead, I introduce orthodontic elastics after aligning, leveling, and coordinating the arches transversely. Here, too, the sign is fairly clear: the aligners continue to fit well and everything proceeds as planned, but the sagittal correction does not become apparent as I had expected.

This is a very different situation from that of a patient who does not wear their aligners correctly. In the previous chapter, a generalized loss of tracking was one of the main warning signs; here, tracking can be perfect, and the patient can look in the mirror every day and see their teeth getting progressively straighter, becoming

convinced that treatment is working exactly as it should. Paradoxically, that is one reason it can be so difficult to help them understand the importance of orthodontic elastics.

Once the aligners are in, they can stay in the mouth for many hours without requiring any further decisions. The patient takes them out to eat, generally has them right in front of them, and, when the meal is over, simply has to remember to put them back in.

Orthodontic elastics, by contrast, require a series of small additional actions repeated several times throughout the day. They must be removed and replaced regularly, the patient has to remember to always have a pack on hand, and, especially at first, putting them on can be harder and more uncomfortable. Every meal, outing, workday, school day, sporting event, or trip can become a point at which that routine breaks down.

The literature specifically on orthodontic elastics confirms how important these small obstacles can be. Veeroo and colleagues interviewed patients who had been prescribed full-time wear of intraoral elastics and identified forgetfulness, meals, pain or discomfort, social situations, sports, lost or broken elastics, and a limited perception of the benefits among the main barriers to adherence.

**It is not enough to know whether a patient is following instructions. We need to understand which part of treatment they are following and, above all, where the behavior we prescribed breaks down.**

The first way to find out is to avoid the seemingly most natural question, which is probably also the one that gives us the least information. If I ask *"Are you wearing your elastics?"*, the answer is almost always predictable: *"Yes, Doctor. Always."*

But if I try to reconstruct a normal day, that *always* often becomes less absolute. So I start asking: *"Do you take them off when you have breakfast? And what do you do as soon as you finish? Do you put them back on immediately after lunch? Do you wear them at work or school? When you go out in the evening? On weekends? Do you always have a pack with you?"*

At that point, it may emerge that *"always"* actually means overnight and for much of the day, but not when the patient is at work; or during the week, but much less on weekends; or that several hours pass after lunch before they remember to put them back on. This does not necessarily mean they were deliberately lying to me before. As we saw in the first part of the book, our perception of our own behavior can differ from what we actually do throughout the day, and we know that patients tend to report more hours of wear than objective measurements show. That is why I have no interest in asking ten times, *"Are you sure you're wearing them?"* I need to

understand not just whether the patient wears them, but **when they wear them, when they take them off, and, above all, when they do not put them back on**.

There are also a few small signs that I have learned to watch for over the years and that can help me get a clearer picture of what the patient actually does. One line I have heard so many times that it has become almost a classic comes from the patient who arrives for a follow-up visit without their elastics and, when I ask why they did not put them on that day, replies: *"I didn't put them on because I was coming here anyway."* At that point, I simply answer: *"What does that have to do with it? If you're supposed to wear them all the time, you need to have them on when you come to see me, too."*

It is not the hour spent without elastics before the visit that particularly worries me, but the reasoning behind that decision. If the patient thinks it is perfectly natural to stop wearing them on the very day they come to the orthodontist, they probably have not yet taken the central point of the prescription to heart. Orthodontic elastics are not something to wear when you remember or when they are not too much trouble: if they are prescribed for continuous wear, they need to become as much a part of treatment as the aligners.

Another small indicator I find extremely useful in practice is watching **how** the patient puts on their elastics. When I have doubts, I hand them the elastics and simply say, *"Show me how you put them on."* A patient who has genuinely been wearing elastics for weeks or months tends to do it automatically. They know where the hooks are, know how to position their fingers, and can generally put the elastics on quickly, often without even using a mirror. When, instead, someone who is supposed to have been wearing them for months asks me for a mirror, spends a long time looking for the hooks, needs the tool, or takes several minutes to get them on, that raises another small red flag for me.

The same principle applies to another behavioral indicator I have used for many years: **how many orthodontic elastics the patient actually wears**. When I prescribe them, I normally give the patient two packs, about two hundred elastics, and ask them to replace them at least three times a day. I do not tell them that this supply will not last until the next visit, which I generally schedule three or four months later, because I want to see what happens without prompting them.

If the patient really follows the prescription, they should run out well before the appointment and will have to contact the office to say something like, *"Doctor, I'm running out of elastics. Can I stop by to pick up some more?"* That call cannot, of course, tell me how many hours a day they wore them, but it is a behavior consistent with

what I prescribed and, above all, it happens spontaneously, without my having told them that I would expect them to ask for more.

The opposite situation is even more interesting. After three or four months, the patient returns for a follow-up visit. I ask how their supply of elastics is holding up, and they might answer: *"Great, doctor. I still have an almost full pack."* At that point, I do not need to start a discussion about how many hours they actually wore them. If I gave them about two hundred elastics and asked them to replace them at least three times a day, that supply simply is not consistent with several months of wear as prescribed.

Interestingly, Veeroo and colleagues also used the number of elastics consumed as an indicator of adherence, asking participants to save and return the ones they had worn. Mine is not intended as a scientific measure of adherence, but the principle is similar: **sometimes what a patient uses up tells us more about their behavior than what they remember doing.**

**Of course, none of these signs, taken alone, proves that a patient is not following instructions.** Difficulty putting on orthodontic elastics can depend on manual dexterity, eyesight, age, or the complexity of the configuration; the number of elastics left can be affected if the patient has lost some, received more, or used a different pack. What matters to me is the **convergence of clues**: sagittal correction that is not progressing as expected, an inconsistent account of the patient's day, little familiarity with putting on elastics, and a rate of use that does not match the prescription. **When several clues begin to tell the same story, the picture becomes much more meaningful.**

## Orthodontic Elastics Work If Worn as Prescribed

Once I understand **where** adherence tends to break down, there is a second step I consider just as important: making sure the patient has truly understood what the prescription I gave them means. If I prescribe orthodontic elastics for all-day wear, the patient cannot decide on their own to wear them only at night or only when it is more convenient. Wearing them only at night does not mean following the same treatment a little less faithfully; it means **following a different prescription from the one I planned**.

For years, I have used a very simple example to make this idea easy to grasp: *"Imagine you're in a car going uphill with no brakes. When you put on your elastics, it's as if you're pressing the gas pedal: the car climbs and climbs and climbs, and little by little we get the correction we want. But when you take off the elastics, the car doesn't stay where it*

*is: it starts rolling backward, and you lose some of the ground you gained. That's why even one day without elastics can undo what we achieved over the previous three or four days. But we shouldn't think only in terms of whole days, because the same problem can occur for several hours every day. Think, for example, about what happens if you wear them properly from the afternoon through the evening and all night, but then you go to school in the morning and don't put them on again until five in the afternoon. Or you put them on in the morning, take them off during a break to eat something, and then, because you don't have a fresh pack with you, go without elastics for the rest of the day. During those hours, we aren't simply pausing treatment: we risk losing some of the effect we achieved in the preceding hours. The same goes for the weekend. If you wear them properly Monday through Friday, but on Saturday and Sunday you go out, play sports, see friends, and decide not to wear them, we risk losing much of what we gained during the week in those two days. So the issue isn't just how many hours of wear you can add up by the end of the day, but how continuously we can keep the elastics working. That's why I insist on this so much: I don't want you to wear elastics only when it's convenient, because they need to support the correction we're trying to achieve without interruption."*

The numerical comparison I use with patients—that one day without orthodontic elastics can undo what was achieved over the previous three or four days— **does not come from an experimental measurement and should not be interpreted as one**. There are no data that would allow us to determine with that degree of precision how much correction is lost after a day, or after a given number of hours, without elastics. Yet I have deliberately used this line for years because it makes the need for continuous force easy to grasp and, above all, because **I want the possibility of interrupting elastic wear to worry the patient**.

In a sense, it is another of the **white lies** we discussed in the first part of the book: I am not inventing a risk that does not exist, because wearing elastics intermittently really can disrupt the continuity of the mechanics I planned, but I deliberately turn that risk into a simpler, more definite, more memorable message than the literature would allow me to quantify. **I do it with a specific goal: to increase the likelihood that the patient will follow the prescription.** Saying that a day without elastics could cost the patient several days of correction makes the consequences of that behavior much more concrete than a general recommendation to wear them more consistently.

There is another factor to consider as well: the force exerted by orthodontic elastics does not remain constant over time. In an *in vivo* study of patients wearing Class II orthodontic elastics, Qodcieh and colleagues observed a progressive decline

in force, particularly noticeable within the first few hours of wear. **That is another reason I ask my patients to replace their elastics at least three times a day.**

## Before Trying to Motivate Them, I Find Out Why They Aren't Wearing Them

When I realize a patient is not wearing their elastics as prescribed, I do not automatically assume the problem is a lack of motivation. First, I try to understand what is actually happening. There is a big difference between someone who has not understood why they matter; someone who wears them only at night because they have decided on their own that eight hours is enough; someone who finds them too uncomfortable to wear at work; and someone who sincerely wants to wear them but finishes lunch each day, returns to their desk, and forgets about them until evening.

In that last case, continuing to explain how important orthodontic elastics are does little good, because the problem is not persuading the patient further but **making the behavior easier**. I might suggest keeping a pack of elastics on their desk, one in their bag, and perhaps one in the car or another place where they regularly spend part of the day. Or I can use the principle of **implementation intentions**, described in Chapter 5, by linking putting the elastics back on to something the patient always does and turning a vague recommendation like *"Remember to put them back on"* into a much more concrete rule: *"When you finish eating, brush your teeth and put your aligners and elastics back on."*

Digital reminders can also help when the problem is simply remembering to repeat a behavior. In a study of 42 Class II patients, Leone and colleagues sent motivational and reminder messages to the experimental group twice a week for three months and observed 3.7 times as much Class II correction as in the control group. More recently, Şenocak and Camcı studied 124 adolescents, 68 of whom were assessed specifically for Class II elastic wear, comparing text reminders, video reminders, and no reminders. **When the problem is remembering a repetitive behavior, increasing motivation is not always enough; sometimes we need to build a better routine.**

The situation is entirely different when a patient does not forget their elastics but **consciously chooses not to wear them at certain times of day**. They wear them at home and overnight but take them off at work, at school, or when they go out because the elastics are visible, interfere with their speech, or make them feel self-conscious. In this case, reminders and spare packs do little good, because the problem is not remembering to put them back on: the patient has decided that, in

certain situations, the social cost or discomfort of wearing elastics matters more than the prescription we gave them.

And this is exactly where our approach needs to change. Making the behavior easier is not enough: we need to make **the consequences of not following the prescription more meaningful**. After all, the problem is not just that treatment might take longer. If the patient wears elastics only while at home, correction may proceed more slowly, remain incomplete, and affect the final result as well. At that point, I need to make clear to the patient that they have a choice: follow the prescription we planned, consider whether a clinically appropriate alternative exists, or accept that treatment may become longer and less predictable, with a result that could differ from what we originally planned.

## The Patient Sees Straight Teeth, Not the Class II

When I find that the problem is primarily motivation, the approach I use most often is, once again, **time as a motivator**. We might think that the outcome as a motivator would be more effective with elastics, because what we risk losing is precisely the sagittal correction. But in practice, there is a very simple problem: **the patient often does not see what we see**.

We see a Class II relationship, assess the molar and canine relationships, and know exactly how much correction remains. The patient, meanwhile, looks in the mirror and sees, above all, their incisors getting straighter, the crowding diminishing, and their smile gradually improving. From their perspective, treatment is working, so it is hard to understand why we keep insisting so much on two little elastics.

That is why I explain: *"Your teeth are getting straighter, and you can see that too, because that part of treatment depends mainly on the aligners. The elastics, though, are doing a different job: they correct the relationship between the upper and lower arches. If that part falls behind, we risk reaching a point where the teeth are perfectly aligned but the bite still needs correcting. Then we will have to keep working on the part that fell behind, and* ***treatment will take longer than we planned****."*

Using **time as a motivator** makes a consequence tangible that patients have trouble seeing in the mirror: they may not recognize a residual Class II relationship, but they understand perfectly well what it means to **have to continue treatment longer**.

At this point, a perfectly understandable question often comes up: *"But how long will I have to wear them?"* If I am emphasizing that elastics must be worn consistently

and that insufficient adherence can prolong treatment, it is natural for the patient to want to know how long they will have to keep wearing them.

This is where, in my practice, **DentalMonitoring**, which we will discuss in a later chapter, offers an enormous advantage. I explain to the patient that they should not decide on their own when to reduce or stop wearing elastics, and that they do not need to wait for their next office visit, because I can use their periodic smartphone scans to track the correction remotely. I tell them: *"Don't worry about figuring out when it's time to wear them less or stop wearing them. Keep wearing them exactly as I prescribed because, even if I won't see you in the office again for three, four, or perhaps six months, I'll track your progress through your smartphone scans. When I see that we've achieved the correction we want, I'll tell you how to change your elastic wear. You don't have to decide when the elastics have finished their job."*

This does more than reassure the patient: it also **removes any ambiguity from the prescription**. Until I tell them otherwise, they must continue wearing the elastics exactly as prescribed. This also reduces the risk that they will start cutting back on their hours on their own simply because they think they have worn them enough.

All these elements help me understand where adherence breaks down and choose the best strategy for restoring it. **As in the previous protocol, my goal is not to get a confession or prove to the patient that they are not wearing the elastics as they claim, but to create the conditions for their behavior to change from that point on.**

### PRACTICAL SCRIPT

When the aligners are working properly but sagittal correction is not progressing as expected, I can frame the conversation this way:

> *"The aligners are working well, and your teeth are gradually getting straighter. The elastics have a different job, though, and right now that part of treatment isn't progressing as I expect. Before changing anything, I want us to look together at how you wear them on a typical day. I'm not just interested in whether you put them on, but especially in when you leave them off: what happens after lunch? Do you put them back on right away? Do you wear them at work or school? What about when you go out in the evening? On the weekend?"*
>
> *"If the problem is that you simply forget them at certain times, we need to make remembering easier: keep a pack at work or school, carry one with you at all times, and make a firm rule. When you finish eating, brush your teeth and put*

> *your aligners and elastics back on. But if you choose not to wear them in certain situations because they're visible, uncomfortable, or interfere with your speech, we need to approach this differently, because wearing them only when it's convenient means following a different prescription from the one we planned."*
>
> *"Your teeth are getting straighter, so when you look in the mirror, you may feel that everything is going perfectly. But the elastics correct the relationship between the upper and lower arches. If that part falls behind, we risk reaching a point where your teeth are perfectly aligned but your bite still needs correcting, and treatment could take longer than we planned. That's why, starting today, I want elastics to be just as much a part of your treatment as the aligners."*

## Why This Protocol Works

- **Reconstructing Behavior**

Simply asking *"Are you wearing your elastics?"* tends to elicit a broad, predictable answer, whereas going through the different parts of the day helps identify when the patient actually stops following the prescription. That shifts the conversation away from whether the patient is being honest and focuses it on the behavior we need to change.

- **Behavioral Indicators**

The progress of sagittal correction, whether the patient is wearing their orthodontic elastics at the appointment, how naturally they put them on, and how many packs they have used can, taken together, provide much more useful information than a verbal statement alone. These indicators are not meant to prove that the patient is lying, but to help us reconstruct what is actually happening.

- **Reducing Friction**

If the problem is practical, simply increasing motivation does little good. Keeping orthodontic elastics in several strategic places and always having a pack available reduces the risk that the patient will stop wearing them because they forgot their pack or do not have one on hand.

- **Implementation Intentions**

A rule such as *"When you finish eating, brush your teeth and put your aligners and elastics back on"* uses the principle of **implementation intentions**, described in Chapter 5, by linking the behavior to a specific event that happens every day and making it easier to turn into a routine.

### Making the Consequence Meaningful

When the problem is not forgetting the orthodontic elastics but consciously choosing not to wear them at certain times, making the behavior easier is not enough. We need to make the consequence of that choice more meaningful, turning something the patient has difficulty seeing into something that has a concrete meaning for them.

### Time as a Motivator

Patients can easily see that their incisors are getting straighter, while they may not notice at all that sagittal correction has fallen behind. That is why **time as a motivator** is often more effective than the outcome as a motivator: explaining that they could reach a point where their teeth are aligned but part of treatment remains to be completed makes it immediately clear why adherence needs to improve now rather than at the end of treatment.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

**Main motivators: time and control.**

With a Dominance-style patient, I would avoid lengthy discussions about how many hours they have actually worn their orthodontic elastics. Instead, I would show them what is not happening clinically and connect it to the possibility of needlessly prolonging treatment.

> *"The aligners are doing their job, but the Class II relationship isn't correcting as it should. If this part falls behind, we'll end up with straight teeth but still have part of treatment to complete, and we'll lose time. Starting today, we need to wear the elastics as consistently as we agreed."*

**I**

### Influence

**Main motivators: simplicity and the outcome.**

With an Influence-style patient, I would focus above all on making the routine simple and compatible with an active social life, using a few concrete rules rather than lengthy technical explanations.

> *"Let's keep it simple: one pack with you at all times and one at work or school, and one firm rule. When you finish eating, brush your teeth and put your aligners and elastics back on. That way you don't have to think about it every time, and both parts of treatment can work together."*

**S**

### Steadiness

**Main motivators: routine and reassurance.**

With a Steadiness-style patient, I would avoid letting the lack of correction feel like a treatment failure and focus on building a predictable routine that is easy to repeat.

> *"The aligners are working well, and we don't need to overhaul anything. We simply need to make orthodontic elastic wear more consistent. Let's look together at when you tend to forget them, build a routine that becomes automatic, and then check how the correction changes."*

## Adapting Communication to DISC Profiles

CONTINUED

**C**

### Conscientiousness

**Main motivators: clinical evidence and precision.**

With a Conscientiousness-style patient, I would show them the sagittal relationship directly and clearly distinguish what the aligners are accomplishing from what depends more heavily on the orthodontic elastics.

> *"Look here: alignment is progressing well and the aligners fit properly, but the sagittal relationship hasn't changed as much as we expected. These are two different parts of treatment, which is why we need you to wear the elastics more consistently as prescribed. We'll check your next scans to see whether the correction resumes as expected."*

CHAPTER 7

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When the aligners continue to fit well and tooth movements are progressing, but the sagittal correction that depends on orthodontic elastics is falling behind. |
| **GOAL** | Identify where adherence to the prescription breaks down and achieve more consistent orthodontic elastic wear. |
| **WHAT TO DO** | 1. Go through how the patient wears them during meals, at work, at school, on evenings out, and on weekends. Ask when they take them off and how much time passes before they put them back on.<br>2. Note whether the patient is wearing their orthodontic elastics, how naturally they put them on, and how many packs they have used, interpreting these observations alongside the clinical findings.<br>3. Distinguish forgetfulness and practical obstacles from a conscious choice not to wear them.<br>4. If the problem is practical, keep packs in the places the patient regularly spends time, link putting the orthodontic elastics back on to a routine, and use reminders when needed.<br>5. If the problem is motivational, explain that the teeth can become aligned while sagittal correction remains incomplete, making it necessary to extend treatment.<br>6. Reinforce the prescribed wear and replacement schedule; check the response at subsequent appointments or in subsequent scans. |
| **WHAT TO CHECK** | Progress of sagittal correction, consistency of wear, the patient's routine, and their response to the agreed-upon intervention. No indirect indicator alone proves a lack of adherence. |

CHAPTER

# 8

# The Routine Refinement

*"All models are wrong, but some are useful."*

— GEORGE BOX

Whenever I plan the first set of aligners, I always add **four passive aligners** at the end of the sequence. We call them passive because, at least on paper, they contain no newly programmed movements, but I have always considered that definition a little simplistic. If the movements programmed into the preceding aligners have not been fully achieved, even an aligner labeled passive in the digital treatment setup may not be entirely passive clinically. It can continue to hold what we have achieved and, in some cases, give the preceding movements more time to finish.

But that is not the main reason I include them.

The most important reason concerns something we have discussed many times in this book: **behavior**. I do not want to reach the end of the active phase of the

first set, take a new scan, and tell the patient that they now have to wear the same aligner for two or three weeks while we wait for the refinement to be planned and manufactured. For months, I have asked them to build a specific routine: each week, they take a new aligner, replace the previous one, and continue treatment. If I suddenly interrupt that routine and tell them to keep wearing the same aligner for weeks, I am changing the very habit that has helped me maintain their adherence up to that point.

This is not just my clinical impression. Research on habits shows how repeated behaviors gradually become associated with the cues and context in which they occur. Wood and colleagues observed that, when those elements remain stable, a habit is more likely to continue guiding behavior, whereas a change in context can interrupt its automatic expression. Verplanken and Wood later developed this concept, showing how a disruption in the cues that support a habit can be one of the moments when the behavior becomes more dependent on deliberate thought.

That does not mean, of course, that if a patient goes a few weeks without changing aligners, they suddenly forget everything they have done over the preceding months. It means something much simpler and, to me, clinically relevant: **if a routine works, I prefer not to interrupt it without a reason.**

That is why I plan for four passive aligners from the outset and schedule the appointment for when the patient reaches the first passive aligner. At that point, I check the bite, assess which movements have been achieved as planned and which need to be fine-tuned, and, if we are looking at what I consider a normal routine refinement, I take a new scan. While I plan the next phase and we wait for the new aligners to arrive, the patient simply continues their routine: they move on to the second passive aligner, then the third, and finally the fourth, changing aligners at the same frequency as before.

It may sound like an organizational detail, but to me it is anything but. **I am not using passive aligners only to maintain the position of the teeth: I am also using them to maintain the patient's behavior.**

And it is at this appointment, when the patient has reached the first passive aligner and the active phase of the first set is over, that I have been asked the same question countless times over the years. The patient sits down in the chair knowing this is an important appointment. They have gone through thirty, thirty-five, perhaps forty aligners, one after another. They look at their teeth and, in most cases, see a better smile, very different from the one they started with. I check the bite, assess which movements have been achieved as planned and which need to

be fine-tuned, then explain that we will take a new scan. That is when the question comes:

*"Wait, we're not done?"*

At first, it surprised me. At the initial consultation, I had explained that treatment would take, for example, fifteen to eighteen months. That was only an estimate, of course: some treatments may take twelve months, others twenty-four or more. But we had discussed the overall duration, and the patient had accepted it.

Then the aligners had arrived. A long sequence of aligners, each with its own number printed on it. And from that moment, a new countdown began, almost without the patient realizing it: 12 of 40, 18 of 40, 25 of 40, 32 of 40... Each week, the patient finished another one and moved a little closer to the final number they could see right in front of them.

Meanwhile, the fifteen to eighteen months we had discussed at the initial consultation faded further and further into the background. Not necessarily because the patient had forgotten what I told them, but because that estimate in months had gradually been replaced by something much more concrete, something they could see, touch, and count every week. Treatment was no longer a span of time; it had become a sequence with a beginning and, above all, a clearly visible end.

As they approached the last few aligners, in their mind they were not simply finishing the first set of aligners: they were finishing treatment. So when I explained that we would take a new scan to order a refinement, their surprise was entirely understandable. At that point, I could have simply reminded them of what I had said months earlier:

*"Remember? We talked about fifteen to eighteen months."*

And for a while, that is exactly what I did. Then I realized I was trying to correct, at the end, an expectation I had allowed to build for months. If that reaction kept coming up, the problem was not finding a better answer when the patient reached the end of the set. It was preventing them from reaching that point convinced that the last aligner meant the end of treatment.

So I needed to address it when I gave the patient the first set, as they were beginning to form an idea of how long their treatment would last. I needed a simple way to help them understand that those aligners represented only the first major phase. That is how I started talking about 80%.

## 80%

So I began changing what I said when I handed over the aligners. I no longer wanted the patient to equate the number of aligners they received with the total duration of treatment, and I needed a concept simple enough for them to remember many months later. I decided to use a number because, over the years, I have noticed that numbers make information that might otherwise remain vague feel more concrete. They also give it a greater sense of precision.

Saying *"we'll probably still have a few things to do after this first set"* is very different from saying *"we'll do about 80% of the work with this first set."* The second sentence is more immediate and, above all, easier to remember.

I deliberately chose a simple number: 80%. It is not a scientific finding, nor is it meant to predict mathematically how much we will achieve with the first set. It is a communication tool I use to convey a very simple idea: **the first box is not the entire treatment**. When I hand over the aligners, I always say:

> *"These are the first aligners of your treatment. We'll do the most important part of the work with this first set, and our goal is to get roughly 80% of the way to the final result. When we reach the end, we'll take a new scan, see which movements have been achieved exactly as we wanted, and use a second set to work on the details and on any movements that need greater precision."*

Since I started introducing this idea early, the conversation at the end of the first set has become much easier. When the patient reaches the last part of the sequence, the refinement is no longer a surprise: they have known from the beginning that the first set represents the most important part of treatment, but not necessarily the last.

Bringing up the refinement from the outset also lets me clarify another important point: **a refinement does not, in itself, mean that treatment has failed**. No biological system necessarily carries out every movement with the same precision with which we programmed it on the screen. Some teeth follow the sequence perfectly; others may need further correction. Some movements are fully achieved; others less so. That is exactly what the refinement is for: starting with what we have actually achieved and using a new plan to complete what remains and fine-tune the result.

The problem is that, just as we reach this phase, something happens that can put adherence at risk.

## The Refinement Paradox

The patient looks at their teeth and, very often, is already happy. What they see is dramatically different from where they started: their incisors are aligned, their smile has changed, and friends and family may already have told them their teeth look beautiful. We, on the other hand, still see what is missing: a rotation that has not been fully achieved, torque that needs fine-tuning, a vertical movement that needs to be completed, some contacts that need refinement.

This is where an important difference emerges between what we see and what the patient sees. We know we are entering the phase when we need to be even more precise; the patient may feel that the work is practically done.

And this is often where the patient's behavior changes, too. If I feel that the most important part of the work has already been done and only a few details remain, I will naturally place less value on every hour of wear. The twenty-two hours that seemed essential during treatment start to become twenty, then eighteen. Maybe one evening I leave the aligner out a little longer because, after all, my teeth are already straight.

The problem is that, clinically, we should think almost the opposite way. If some movements remained incomplete after the first set, those are precisely the ones we need to focus on during the refinement. I cannot expect a movement that has already proved less predictable to be achieved more successfully while the patient is also cutting back on the hours I am applying the necessary force.

This is the idea I want to turn around in the patient's mind: **the refinement is not the time when they can be less consistent because so little remains; it is the time when they need to be more consistent precisely because what remains requires greater precision.** And it is this paradox that gives rise to two of the objections I have heard most often over the years. The first almost always comes when the patient sees the new set:

*"But why are there still so many?"*

The patient received, say, forty aligners in the first set, and now we hand them a refinement with fifteen, eighteen, or twenty. They look at them and make an entirely understandable calculation:

*"But if we did almost everything with forty aligners, why do I need so many now to do so little?"*

Over time, this question, too, has forced me to make my explanation more precise.

The first reason is that, in the refinement, I am working on the very movements that were hardest to achieve. If a movement did not follow the first digital treatment

setup as I expected, it would make little sense simply to try it again at the same rate. I may decide to slow it down, spread it over more steps, and apply the force more gradually. As a result, a seemingly small movement may require more aligners than the patient imagines just by looking at the position of their teeth.

The second reason is that I am not thinking only about the teeth that still need to move. While I work on the remaining details, I also want to maintain and consolidate everything I have already achieved. So the refinement period is not exclusively about producing new movements; it is also about preserving those that have already been achieved as planned.

Then there is a third reason, much simpler and almost a matter of arithmetic, that can significantly affect how the patient perceives the total number of aligners. It relates directly to the four passive aligners we discussed at the beginning of the chapter. If the first set contains forty aligners and four are passive, that means thirty-six are intended for the active phase: the four passive aligners account for 10% of the entire sequence, so their effect on the total number is relatively small. In a short refinement, however, that proportion changes completely. If, for example, I had twelve active aligners and added four passive ones, the sequence would grow to sixteen, and a quarter of the aligners I gave the patient would be passive. The patient sees sixteen aligners and thinks all of them will be needed to move the few teeth that still need work, when part of the sequence serves a completely different purpose. That is why I explain the number, too:

> *"Don't let the number on the box scare you. It doesn't necessarily mean you'll have to wear every one of these aligners. There are only a few movements left to make, but we've slowed them down because they require greater precision. I've also added four passive aligners at the end. They have no new movements programmed into them and are mainly there to keep your routine going while we prepare the next phase, if one is needed. So the number you see isn't the number of active aligners between you and the end of treatment."*

This explanation has another advantage: it takes away another possible reason for the patient to be less consistent. Fifteen or twenty new aligners do not mean that treatment has suddenly gone backward; they mean we have entered a different phase, in which we are working more slowly and toward more specific goals.

But there is another interpretation that can lead to exactly the same result. This time, it does not come from the number of aligners, but from what the patient feels when wearing them:

> *"Doctor, these aligners don't feel tight."*

Or:

*"Doctor, these aligners feel much looser."*

At first, my answer was purely technical. I explained to the patient that this was normal because the new set had been made from the scan we had just taken. The first aligner therefore started from a position extremely close to the actual position of their teeth and would not necessarily produce the same sensation they remembered from other phases of treatment.

The explanation was correct, but at some point I realized it did not fully solve the problem.

Over the months, the feeling of pressure had become a kind of physical proof to the patient that the aligner was working. When they felt it pulling, they thought the aligner was doing its job; when they felt almost nothing, they could easily reach the opposite conclusion. And if something does not seem to be working, it becomes much easier to give yourself permission to wear it less.

So I began being much more explicit about this point, too:

*"At first, these aligners may feel looser, and that's normal. They were made from the scan we just took, so the beginning of the sequence starts from exactly where your teeth are today. But the movements we need to achieve are the very ones that needed more help in the first set. That's why, in this phase, wearing them for twenty-two hours is even more important than it was before."*

This explanation lets me head off the wrong interpretation before the patient can turn it into a reason to cut back on wear time. Over the years, I have also looked for a solution that goes beyond communication.

Since I began using a system that lets me choose among materials with different properties, I have used the stiffer material for refinements. For me, the choice also has a biomechanical rationale, especially when I need to work on root control. But over time, I noticed another interesting effect: the patient was more aware of the aligner in their mouth and continued to feel that it still had a job to do.

Of course, this does not mean that an aligner that *"pulls harder"* is necessarily more effective, or that the sensation the patient reports can be used as a measure of orthodontic force. The advantage that interests me here is primarily behavioral: being able to feel the aligner reinforces the patient's sense that treatment is still active and, consequently, that they need to keep wearing it just as carefully.

It is one of those observations that came more from the treatment chair than from books, and over time I have come to sum it up in a very simple sentence: if the

patient cannot feel the aligners working, they may stop seeing them as important and become less consistent about wearing them.

And it is precisely during a refinement, when the patient can already see the result and may naturally be inclined to be less consistent, that I do not want this to happen.

## PRACTICAL SCRIPT

All these observations, accumulated over the years, have gradually changed the way I hand over a refinement. Today I do not wait for the patient to tell me they thought they were finished, that there are too many aligners, or that they do not feel them pulling. If I know these objections come up repeatedly, I would rather address them first.

The conversation might go something like this:

> *"We've reached exactly the phase we talked about at the beginning. With the first set, we did the biggest part of the work and got to roughly 80% of the result we wanted. Now we need to work on the details, especially the movements that were not achieved with the same precision as the others.*
>
> *That is exactly why this phase is no less important than the first. In fact, some of the movements we need to make now are the very ones that have already shown they need more attention, and we have deliberately slowed them down to make them more gradual. To achieve them, we still need the same thing we have needed all along: consistent force, which means twenty-two hours of wear.*
>
> *You'll also see that the number of aligners may seem high compared with the few things we still need to fix, but that doesn't mean we have to redo half the treatment. We've spread some movements over more aligners precisely because we want to make them more slowly. I've also added four passive aligners at the end so you can keep up your routine while we prepare the next phase, if one is needed.*
>
> *At first, you may also feel that these aligners aren't as tight. That's normal, because they were made from a scan of your teeth as they are today. But don't take that to mean you need to wear them any less: the movements we're working toward now are precisely the ones for which I want you to be as consistent as possible.*
>
> *We've already completed the biggest part of the journey, and you can see the result. It would be a shame to ease up now, on the very movements that will take us from a good result to the result we planned."*

This conversation did not come from any single theory of communication. It is the result of many small objections I have heard over the years and the realization that I needed to stop waiting until they came up to respond to them.

Over time, I have learned that when an objection comes up often enough, getting better at answering it is not necessarily the best solution. I would rather make it less likely to come up in the first place.

If I know the patient might think they are finished, I set that expectation when I hand over the first set. If I know they might think fifteen or twenty aligners are too many, I explain where that number comes from. If I know they might take a weaker feeling of pressure to mean they do not need to wear them as much, I clear up that ambiguity before it becomes a reason to change their behavior.

Ultimately, that is the real goal of the protocol: **to leave as little room as possible for interpretations that could justify wearing the aligners less consistently, while making clear how much of the result still depends on the patient's behavior.**

A refinement is not a failure, nor is it a phase when we can finally relax. It is when we move from major changes to the details, and because those details often involve the movements that have been hardest to achieve, the patient's consistency remains crucial.

The patient sees that only a little remains. Our job is to help them understand how much that little matters.

## Why This Protocol Works

### ■ Framing and Managing Expectations

The refinement is not presented as *"another set because we aren't finished,"* but as an expected phase of treatment in which we move from major movements to precision. The clinical facts do not change, but the frame through which the patient interprets them does: we are not starting over; we are completing what we planned.

The approximate 80% figure reinforces that expectation. Saying *"we'll probably still have something to do after this first set"* leaves the patient with a vague piece of information that is easy to forget or reinterpret as the months pass. Saying instead *"this first set will do about 80% of the work"* gives them a concrete reference point that is easier to remember. The 80% figure is not used as a scientific finding, but as a communication anchor: when the patient approaches the end of the first set months later and the number printed on the aligners suggests that treatment is finished, we can bring them back to the expectation we established from the start.

### Preventing Ambiguity

The main objections are addressed before they arise. The patient already knows that the end of the first set may not be the end of treatment, why a refinement may contain a seemingly large number of aligners, and why the new aligners may feel different.

Explaining these things in advance leaves less room for interpretations that could become a reason to wear the aligners less consistently. If I know an objection comes up often enough, I would rather prevent it than keep getting better at answering it.

### Goal Gradient

The literature on the *goal gradient* shows how getting closer to a goal can increase the effort people put into reaching it. In my clinical experience, however, a different problem can arise during a refinement: if the patient thinks the result is already good enough, they may begin to ease up just as they approach the end.

The protocol therefore aims to prevent the patient from seeing how close they are to the result as a reason to relax. Having only a little left to do does not mean that little matters less. On the contrary, because we are working on the movements that have been harder to achieve, the final details require the same consistency as the previous phase, if not greater care.

### Commitment and Consistency

During a refinement, we are not asking the patient to adopt a new behavior, but to continue one they have maintained for months. The twenty-two hours of wear remain the same and are presented as a natural continuation of the effort that allowed them to achieve the result they see in the mirror.

The same principle explains why I add four passive aligners rather than leave the patient wearing the same aligner for weeks. If a routine is working, I would rather protect it than interrupt it unnecessarily. When the patient starts the first passive aligner, I perform the checkup and take the scan; while we wait, the patient moves on to the second, third, and finally fourth. The weekly changes continue and, from the patient's perspective, their behavior stays essentially the same.

### Making the Outcome Salient

At this point, the patient can already see much of the result, and what they have achieved can itself become the protocol's main motivator. The message is not simply that a few details remain, but that those details matter and that we need to finish what we have already begun.

Simply saying *"you need to keep wearing your aligners for twenty-two hours"* states a rule. Explaining instead that the remaining movements are the very ones that

have been harder to achieve makes clear why that rule still matters. The perspective changes: we are not asking the patient to start over, but to complete the result they have already spent months working toward.

A refinement, then, must not become the point at which the patient feels treatment is slowly winding down. In how we communicate it, we need to achieve exactly the opposite: help them understand that we have entered the phase where the result is close, but where every detail matters precisely for that reason.

When a refinement is not routine, but has become necessary because the patient has not followed the treatment plan, the problem is entirely different. In that case, we will need to do more than explain a new series of aligners: we will need to manage it without teaching the patient that failing to follow the plan has no consequences.

But that deserves a protocol of its own.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

**Main motivators: time and outcome.**

With a Dominance-style patient, I would avoid dwelling on the clinical reasons for refinement and focus instead on the fact that the main part is already done and that maintaining the same consistency will let us complete the final movements without wasting time.

> *"The main part is done. Now we need to finish the last movements properly without wasting time: twenty-two hours of wear, and we'll reach the result we planned."*

**I**

### Influence

**Main motivators: outcome and motivation.**

With an Influence-style patient, I would make more use of what they can already see in the mirror, turning how close they are to the result into a reason to stay consistent rather than an excuse to ease up.

> *"Look how much your smile has changed. We're really close, but these last details are what will make the difference. Let's not let up now."*

**S**

### Steadiness

**Main motivators: continuity and reassurance.**

With a Steadiness-style patient, I would emphasize above all that I am not asking them to change their habits again. The routine that has worked so far simply needs to continue as the clinical goals of this phase change.

> *"You don't need to change anything about the routine you've followed up to now. We'll continue with the same twenty-two hours of wear and the same schedule; the only difference is that now we're working on the final details."*

**C**

### Conscientiousness

**Main motivators: precision and clinical explanation.**

With a Conscientiousness-style patient, I would show them what we have achieved and what still needs to be fine-tuned, explaining why the remaining movements have been planned more gradually.

> *"The new scan let us measure which movements were fully achieved and which were not. We've planned the remaining movements more gradually, so consistently wearing the aligners for twenty-two hours remains essential."*

CHAPTER 8

# Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | Start when planning the first set, preparing the patient for a possible refinement; put the protocol into practice when the patient starts the first passive aligner. |
| **GOAL** | Complete the remaining movements while maintaining the hours of wear and the routine established during the first sequence. |
| **WHAT TO DO** | 1. Plan four passive aligners at the end of the sequence.<br>2. When handing over the aligners, explain that the first set represents the main part of treatment, but not necessarily all of it. The 80% figure is a communication tool, not a scientific measurement.<br>3. When the patient starts the first passive aligner, check the occlusion and the movements achieved, take a scan, and plan a refinement if needed.<br>4. While waiting, maintain the change schedule by moving to the second, third, and fourth passive aligners.<br>5. Explain which movements remain and why they are being made more gradually in the new sequence.<br>6. Clarify that the number of aligners alone does not indicate how much work remains, and that feeling less pressure is not a reason to reduce the hours of wear. |
| **WHAT TO CHECK** | Occlusion, movements achieved and still needed, expectations, consistency of wear, and tracking of the new sequence. Being close to the result should sustain the patient's effort, not become a reason to ease up. |

CHAPTER

# 9

# When Refinements Teach the Wrong Lesson

*"The consequences of an action influence the probability that it will be repeated."*

— B. F. SKINNER

A refinement does more than correct teeth; unfortunately, it can also teach the patient that following the treatment plan is not essential.

When widespread loss of tracking appears for the first time, I accept that there may be different explanations and try to get the patient back on track without demanding a confession, as I explained in Chapter 6. But if the patient has already stopped wearing the aligners in several sequences, received new sets, and keeps returning with the same problem, I am no longer dealing only with poor adherence.

I am also dealing with what the patient learned from previous refinements: that a new scan will erase the consequences and treatment can start over under the same terms. This is where a clinically useful option risks teaching the wrong lesson.

## Alessandro's Case

About ten years ago, I began treating Alessandro, a thirteen-year-old boy who was missing both upper lateral incisors and had a full Class II malocclusion. The goal of treatment was to correct the Class II malocclusion and create the space needed to replace the missing teeth when the time was right. This plan required consistent adherence because the result depended not only on wearing the aligners, but also on wearing orthodontic elastics.

Alessandro, however, almost never wore his aligners. I saw him every four months, and we always found ourselves in the same situation: tracking had broken down, the planned movements had not occurred, and we could not continue with the set we had given him. We stopped, took another scan, and prepared a new treatment plan, each time starting over with the same explanation and his promise that this time he would follow the plan seriously.

**During the first two years of treatment, we had already prepared five refinements, or at least what I still thought of as refinements at the time.** Each new scan seemed to give us a chance to start over, but in reality it did not change Alessandro's behavior. He had gradually learned that if he did not wear his aligners, we could simply stop that set and make another.

After nearly two years, something finally changed. Alessandro began wearing his aligners more regularly, and over about six months, his teeth became visibly straighter. For the first time, we had achieved a significant clinical result, although we still needed to complete the Class II correction because he had not worn his orthodontic elastics as consistently. I therefore decided to do a refinement, explaining that the new set would need to consolidate what we had achieved and complete the remaining movements.

When I saw him again four months later, I was surprised by the clinical picture. Not only had the teeth failed to stay in the positions we had achieved with the previous refinement, they had actually become more crooked. After finally managing to straighten them, we were practically back where we had started because Alessandro had not worn any of the new aligners.

As we looked at the photographs together and compared the scan with the previous one, his father made an observation that would change the way I handled these situations forever:

> *"Wow, this time his teeth are even more crooked than before. I imagine there will be a cost now, because the company will see that Alessandro didn't follow the instructions. I don't think it can keep making aligners for free if the patient doesn't wear them."*

That was when a lightbulb went on for me. Until then, I had continued to present each new scan as an option included in the treatment, without realizing that this apparent lack of limits had helped erase the consequences of failing to follow the plan. Alessandro's father, on the other hand, had spontaneously identified something I had not yet managed to turn into a communication strategy: if a new set produces no significant improvement, or if the situation has actually worsened, it cannot continue to be treated as a normal refinement included in the treatment plan. I replied:

> *"That's exactly right. The company will compare this scan with the previous one and see that the last set produced no significant improvement. If we need to make more aligners, there will be a cost for the new phase."*

That answer came to me spontaneously during the conversation, but it helped me understand something much more important. Up to that point, Alessandro had been given five new chances, each accompanied by advice and promises, but none had been tied to a concrete consequence. For the first time, the possibility of a cost made it clear that a new scan could not once again erase what had happened over the preceding months.

So it was not an economic theory that suggested this strategy to me, but the spontaneous observation of a father who, seeing that his son's teeth had gotten worse, thought it entirely natural that poor adherence should finally have a consequence.

## When a Second Chance Becomes the Rule

Some patients admit they have not followed the plan very well and accompany each request for another chance with a credible explanation and a seemingly sincere promise. Others continue to insist they have worn their aligners correctly even when a comparison of the scans tells a completely different story. After repeated fresh starts, however, determining which account is more sincere becomes less

important than breaking the pattern by which both automatically lead to the same outcome: another set of aligners.

The honest patient may learn that all they need to do is acknowledge their difficulty and promise to change, while the less honest one may realize that continuing to deny it leads to no different consequence. The problem, then, is not just what the patient says, but how our practice responds. **If every explanation, whether true or false, is followed by a new scan and another fresh start under the same terms, we are the ones making the patient's adherence matter less and less.**

The first time, we may have offered a second chance; by the second or third time, though, we may have unwittingly begun to teach a rule: **following the treatment plan is certainly preferable, but it is not really essential, because starting over will always be an option.**

This is where the problem also becomes interesting from a behavioral perspective. Every clinical decision we make has a consequence, and that consequence can influence what happens next. When we say *"Don't worry, we'll start over,"* we want to reassure the patient and preserve the therapeutic relationship, but the message they may take away is different: what happened over the preceding months has no concrete consequence, because a new set will let them start over anyway.

This is one of the principles underlying **operant conditioning**: the consequences of a behavior help determine how likely it is to be repeated. Of course, we are not intentionally rewarding poor adherence, but without realizing it, we can create a system in which its consequences are erased each time by a new scan. The same general principle underlies *contingency management* interventions, used in other health care settings to promote adherence through consequences established in advance and linked to verifiable behaviors. This does not prove that introducing a cost will automatically improve aligner wear, but it supports a much more cautious principle that is useful for our protocol: **consequences work best when they are clear, communicated in advance, and tied to something verifiable.**

When treatments include additional sets without a new production charge, a second problem arises. Clinically, having those sets available is a tremendous advantage, but from the patient's perspective, it can gradually make the difference between completing a set and having to repeat it because they did not actually wear it less clear. Research on the *zero-price effect* shows that something priced at zero is not simply perceived as costing very little; it is evaluated differently. This does not mean that something free is automatically considered worthless and, above all, it does not mean that a patient wears their aligners less because they know refinements are included. The problem is subtler: **if every fresh start seems free,**

**automatic, and potentially unlimited, the concrete consequence of not completing the previous set becomes less apparent.**

A concept used in health economics that can help us understand this mechanism is moral hazard. The classic example is insurance: when someone knows they are protected from the financial consequences of a possible loss, they may have less incentive to prevent it than when they know they will bear those consequences directly. This does not mean they intentionally cause the loss or act in bad faith; it simply means that having protection can change behavior. In our case, the mechanism may be similar. A patient does not necessarily decide to wear their aligners less because they know they can receive more, but they may gradually learn that poor adherence does not really change what happens: if they do not wear one set, another will simply be made. **The problem, then, is not that patients knowingly take advantage of the system, but that the system ends up making poor adherence almost consequence-free.**

And it would be too easy to place all the responsibility on the patient, because we clinicians are often the ones who unwittingly reinforce this pattern. We schedule a ten-minute appointment to check tracking and discover that the patient has not followed the plan, the teeth have not improved, and they cannot continue with the set. At that point, we should stop, document the situation, show the comparison with the previous scan, and have a potentially uncomfortable conversation. But to avoid it, the easiest solution is to take another scan. It takes only a few minutes, seems to resolve the problem, and lets us put everything off until the new set arrives. **Taking a scan is easy; changing behavior is much harder.** And so what seems like the quickest solution during the appointment may become the least efficient one over the following months: the patient avoids facing the consequences of not following the plan, we put off the problem, and a few months later we risk finding ourselves in exactly the same situation.

This does not mean it is wrong to offer a second chance. Orthodontic treatment can last many months, and during that time, personal, family, or work problems can temporarily make it harder to follow the plan. Being understanding also means recognizing these circumstances and accepting that one difficult period need not jeopardize the entire treatment.

**The problem is not offering a second chance. The problem is offering one without making it clear that it is a second chance.**

If all we do is take another scan and ask for another promise, nothing distinguishes that fresh start from any that may follow. Instead, we can accept the patient's explanation while introducing the rule that will guide the rest of treatment:

> *"I understand you've been through a difficult time, and this time we can start over. But I need to explain that this new scan became necessary mainly because you didn't wear the aligners as planned. When you finish the next set, we'll compare the new position of your teeth with where they are today, and we'll need to see a significant improvement consistent with following the treatment plan. If we don't see that improvement, we won't be able to automatically treat another set as a normal refinement that's included in your treatment."*

This wording does not blame the patient or turn the refinement into a punishment. **It gives them another chance while establishing that the chance has a limit and, above all, defining in advance how we will determine whether they made use of it.**

## From the Patient's Account to a Verifiable Result

Alessandro's case, and especially his father's words, helped me turn an insight into a very simple rule that I began using in these situations and have used ever since: when I offer another chance, I no longer want to assess only the patient's promise, but the result that promise will have produced. That is why I explain from the outset that we will compare the next scan with the starting scan, and that we will be looking not for every individual movement to be perfect, but for clinically significant overall progress consistent with wearing the aligners as prescribed.

After all, we know that even a patient who follows the treatment plan perfectly can reach the end of a set with an incomplete rotation, a movement still to be completed, or a tooth that has lost tracking. Likewise, a patient who previously almost never wore their aligners may genuinely have increased their wear time and feel that they are now following the plan adequately. That improvement deserves recognition, but the measure cannot be only how much the patient has improved compared with before. It must also be whether their behavior now meets the demands of treatment. Someone can have come a long way from where they started and still not be wearing their aligners enough for treatment to progress as it should.

This led me to a distinction that has become essential. If the next scan shows clear overall progress and only a few movements remain incomplete, I am looking at a normal refinement: the patient has done their part, and now it is up to me to complete the treatment. If, instead, I see only marginal movement, no substantial progress, or even a worsening, the situation is entirely different. I am no longer refining treatment the patient has followed; I am trying to repeat a phase they did not really carry through. **The criterion, then, is not whether every movement**

**was perfect, but whether the patient actually wore the set and achieved a result consistent with following the treatment plan adequately.**

Comparing the scans does not, of course, tell me how many hours the patient actually wore their aligners, nor is it meant to. But it lets me shift the conversation to something else: I no longer have to debate whether they wore them for eighteen, twenty, or twenty-two hours, or try to determine how truthful their account is, because I can compare where they started with where they ended up. In other words, **I stop debating the promise and start looking at the result.**

Once we have established that the new scan must show significant improvement, we also need to make clear what will happen if it does not. **We cannot continue indefinitely to treat every new set as a normal refinement included in treatment**, and this rule can be tied to the aligner manufacturer or presented as a clear practice policy. In the first case, we reduce the chance that the patient will interpret the decision as an arbitrary punishment imposed by the clinician: the new scan is presented not simply as a picture we can use to start over, but as objective documentation that can be compared with the previous scan to assess the result achieved.

The rule can be explained this way:

> *"When we send in the next scan, the company will be able to compare it with today's scan and assess the overall result. Every tooth will not need to have completed every planned movement perfectly, but there will need to be significant improvement consistent with wearing the aligners as prescribed. If that improvement is clear, we can address any incomplete movements through a normal refinement. But if progress is only marginal, there is no substantial improvement, or your teeth have actually gotten worse, the company may not consider the new set a normal refinement included in treatment, so additional aligners could carry an extra cost."*

Alternatively, the clinician can present the rule as a practice policy:

> *"This time, we can take a new scan and start over, but at the end of the next set, we'll need to see significant improvement consistent with wearing the aligners as prescribed. If treatment has progressed and only a few movements remain incomplete, we can complete them through a normal refinement. But if there is no substantial progress, we cannot keep repeating the same phase. We'll have to consider either stopping treatment or continuing with a new plan and an additional cost."*

Charging an additional fee is not a punishment. It makes a concrete distinction between two situations that are already different clinically: completing a treatment

the patient has actually followed, and repeating a phase that produced no results because the aligners were not worn as prescribed.

As we saw in Chapter 6, when we first encounter a widespread loss of tracking, we focus primarily on making the impact on **time** tangible: if weekly changes are not enough, the time needed for each aligner may increase, along with the overall length of treatment. But when a patient has already started over two, three, or four times, time as a motivator may gradually lose its effectiveness, because waiting longer and starting over have become routine. This is where the **investment** as a motivator comes into play—not to threaten the patient or make them pay for not following instructions, but to make it clear that another fresh start is not automatic and that repeating a phase they did not complete is not the same as finishing a few remaining movements.

For this consequence to be fair and effective, however, it must be communicated **before** the new set, not introduced at the next appointment. If the patient knows about it before starting over, it is a clear condition of the new chance we are offering them; if they learn about it only after once again failing to follow the plan, they may see it as a rule invented after the fact. The crucial step, then, is to replace a promise with a verifiable condition. We no longer ask, *"Do you promise you'll follow the plan this time?"*—because we have probably heard that promise before. Instead, we explain that continuing under the same terms will depend on the result we can see in the next scan.

A new scan captures the current position of the teeth and allows us to prepare more aligners, but it does not restore the time lost and, above all, it does not, by itself, change the behavior that caused the problem. If we keep starting over without changing the conditions, we repeat the same treatment while relying once again on the same promise. A refinement retains its full value when it completes a course of treatment the patient has actually followed. When it is used to repeatedly redo phases the patient never completed, it ceases to be only a tool for clinical precision and risks becoming a way to compensate for the patient's failure to follow the plan.

**Included refinements are meant to complete a treatment the patient has followed, not to indefinitely replace one they never really followed.**

Being understanding does not mean eliminating every consequence. We can offer a second chance, and in many cases we should. But for it to remain a second chance, it cannot become an unlimited number of chances.

**PRACTICAL SCRIPT**

At this point, the conversation with the patient must turn what we have just explained into a simple, understandable rule established before the new set begins. We should not ask for another promise or try to get the patient to admit they have not followed the plan. We need to make clear what result we expect and what will happen if they do not achieve it.

> *"This time, we can take a new scan and start over, but I need to explain an important rule. When you finish this set, we'll compare the new scan with today's scan, and it will need to show significant improvement consistent with wearing the aligners as prescribed. That doesn't mean every tooth has to reach its planned position perfectly, because some movements can be less predictable. But the treatment as a whole will need to have made real progress.*
>
> *If you've followed the plan and only a few movements remain incomplete, we can proceed with a normal refinement. But if there's no substantial improvement, or your teeth have actually gotten worse, the company may not consider the new set a normal refinement included in treatment, so additional aligners could carry an extra cost. This isn't a punishment. It simply means we wouldn't be completing a treatment you'd followed; we'd be repeating, yet again, a phase you hadn't carried through."*

## Why This Protocol Works

- **It Breaks the Cycle of Unintended Reinforcement**

If every time a patient fails to follow the plan, they automatically receive a new set, **the chance to start over without any concrete consequence may encourage the same pattern to continue**. Establishing a limit in advance breaks that pattern without turning the appointment into a punishment.

- **It Makes the Consequence Clear**

A general recommendation to follow the plan more closely remains abstract, whereas **the possibility that another set may no longer qualify as a normal refinement included in treatment** makes the difference between completing a treatment and having to repeat a phase immediately clear.

- **It Shifts the Focus from Promises to Results**

Comparing the scans keeps us from basing the decision solely on what the patient says and introduces an observable clinical criterion, without claiming to measure wear time directly.

- **It Uses the Investment as a Motivator**

When time as a motivator is no longer enough to change a repeated behavior, the possibility of an additional cost gives the new chance weight and makes it harder to take for granted.

- **It Preserves the Therapeutic Relationship**

The consequence is communicated in advance, tied **to criteria established before the new set**, and presented as a condition for continuing treatment, so the patient does not interpret it as a personal decision or an emotional reaction on the clinician's part.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

**Main motivators: control and a concrete consequence.**

With a Dominance-style patient, it is best to communicate directly and briefly, focusing on the decision and avoiding lengthy explanations they might see as a waste of time.

> *"We can start over, but from now on we'll use a clear criterion: the next scan must show significant improvement. If treatment has made real progress, we'll complete the remaining movements as usual. If there's no substantial progress, another set cannot be considered a normal refinement included in treatment and will carry an additional cost."*

**I**

### Influence

**Main motivators: motivation and the outcome.**

With an Influence-style patient, it helps to keep the tone positive and engaging without letting enthusiasm for a new promise take the place of a commitment we can verify.

> *"This really could be the set that gets your treatment back on track. At the next scan, though, we'll need to see significant improvement, because the result will tell us whether you followed the plan adequately this time. If treatment has made real progress, we can complete it as usual. If we don't see substantial progress, another set cannot be considered a normal refinement included in treatment and will carry an additional cost."*

**S**

### Steadiness

**Main motivators: reassurance and clarity.**

With a Steadiness-style patient, we should avoid making the rule feel like a threat. Explain it calmly and emphasize that its purpose is to help bring treatment to an orderly conclusion.

> *"I understand these past few months have been difficult, and we can start over without turning this appointment into a reprimand. But I want to explain clearly what will happen: we'll compare the next scan with today's, and we'll need to see significant improvement. If treatment is making real progress, we can complete it as usual. If the improvement isn't sufficient, we'll need to decide together how to proceed and discuss any cost for the new phase before starting over again."*

## Adapting Communication to DISC Profiles

CONTINUED

**C**

### Conscientiousness

**Main motivators: an objective criterion and a clinical distinction.**

With a Conscientiousness-style patient, it helps to explain precisely that the comparison concerns the overall result and does not require every movement to match the digital treatment setup perfectly.

> *"We won't make the comparison by evaluating a single tooth. We'll look at overall progress relative to the planned movements. We don't expect a perfect match with the digital treatment setup, because some movements may not be fully achieved even when the aligners are worn as prescribed. There must, however, be clinically significant improvement consistent with adequate adherence. Without that progress, the next phase cannot be considered a normal refinement included in treatment."*

CHAPTER 9

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When a phase needs to be repeated primarily because of insufficient adherence and the patient has already been given a chance to start over. |
| **GOAL** | Make the new chance contingent on verifiable progress, breaking the automatic cycle of starting over without consequences. |
| **WHAT TO DO** | 1. Reassess the reasons treatment has not progressed, distinguishing insufficient adherence from other clinical difficulties.<br>2. Acknowledge the difficulty and offer another chance when appropriate.<br>3. Before the new set, explain that the next scan will be compared with the starting scan.<br>4. Make clear that we expect clinically significant improvement, not just another promise to follow the plan.<br>5. If treatment progresses and some movements remain incomplete, handle them as a normal refinement.<br>6. If there is no substantial progress or the situation worsens, reassess whether to continue. Another set may carry an additional cost under the terms communicated in advance. |
| **WHAT TO CHECK** | Aligner fit, comparison of the scans, overall progress, remaining movements, any worsening, and the history of starting over. Confirm that the patient understands the difference between completing treatment and repeating a phase they did not complete. |

CHAPTER

# 10

# The Patient Who Disappears

*"Lost time is never found again."*

— BENJAMIN FRANKLIN

Some patients decide to stop or suspend treatment and tell us so openly: they come to the office, explain why they do not want to or cannot continue at that point, and we decide together how to handle the situation. **In my office, when a patient decides to suspend treatment, we have them sign a treatment suspension consent form that makes it explicit that treatment is being interrupted and defines the conditions under which it may be resumed.** That leaves no ambiguity: we know treatment is suspended, and the patient knows they are not simply missing a few appointments. Then there is another category of patients, probably much more common than we would like, who never actually stop treatment, at least not in words, but simply stop coming in at some point and disappear.

At first, it does not even seem like a particularly serious problem. The patient misses an appointment, the front desk calls them, and often gets a perfectly reassuring response: *"Yes, I know. I'm sorry, things are just crazy for me right now. I'll call you as soon as I can."* Of course, they do not call, and after a few weeks the office tries to reach them again. This time they might say: *"Look, I really can't right now. Maybe next month."* Another month passes, then two. At first we keep trying to reach them, but eventually the attempts inevitably become less frequent, because we have other patients to care for, other calls to make, and other problems to solve. Without anyone ever formally deciding to stop treatment, that name simply disappears from the schedule and, as time goes by, from our day-to-day work as well.

A year may pass, or two, three, or even four, until one morning the receptionist comes into my office and says: *"Remember Mario Rossi? He called because he'd like to make an appointment to continue treatment."* And that word, *continue*, contains almost the entire problem this chapter addresses, in my opinion.

From his point of view, treatment may never really have ended. It was simply suspended, almost like a movie he paused and now thinks he can resume from exactly the same frame. He knows he has not come in for three years, of course, but that does not necessarily mean he saw that long absence as the end of treatment. He may believe that what he bought and paid for is still there, waiting for him.

The problem is that his mouth did not stay on pause during those three years. His teeth may have moved, his bite may have changed, he may have worn his last aligners or retainer unpredictably, restorations, cavities, or periodontal problems may have appeared, and even what he wants may differ from what we agreed on at the start. The clinical, logistical, and financial conditions under which we planned that treatment may also have changed.

**Time does not freeze orthodontic treatment.**

Beckwith and, later, Skidmore showed that the duration of orthodontic treatment is also influenced by factors related to patient adherence and continuity of care; Melo and colleagues highlighted the impact of missed appointments among adult patients, while Bichara and colleagues observed treatment times nearly a year longer in patients who had missed more than two appointments. The literature thus confirms something we know well from clinical experience: losing continuity is not simply a scheduling problem.

But one question remains that the literature cannot answer for us, and it is the real problem this chapter addresses: **how long can a treatment remain open when the patient has simply disappeared?** When that patient reappears after three years saying he wants to "continue," before we even decide which aligners to order,

we need to understand something more important: **are we really continuing the old treatment, or has what we are dealing with become a new treatment in the meantime?**

## The Five-Year Misunderstanding

In the world of aligners, there is a very concrete reason this expectation may have developed, and if we are being fair, at least some of the responsibility is probably ours. Many commercial packages allow the practice to request additional sets of aligners at no extra cost within a specified manufacturer's ordering window, under terms that vary from company to company. In some cases, that window may extend to five years, for example.

The problem arises when we decide to explain this to the patient and, as inevitably happens in everyday communication, try to simplify it. Many of us have probably said at least once something like, *"With this treatment, your aligners are included for five years,"* or, even more simply, *"You have five years of treatment."* We know what we mean, but a statement that, to us, describes a commercial term can easily turn into something very different in the patient's mind: *"I paid for a treatment that lasts five years, so until then, whatever happens is included."*

It is understandable, then, that a patient might think they can stop for a few months, then a year, perhaps disappear altogether, and finally reappear before the deadline simply asking to resume. This is where the misunderstanding lies, because **the manufacturer's ordering window available to the practice and the clinical duration of treatment are not the same thing**. The first determines how long the practice can request additional sets of aligners at no extra cost; the second is the time needed to complete treatment.

Imagine a patient who reappears after four years with a package that has a five-year window. We might still be able to order a new set, but we need to reassess the patient, perform a new intraoral scan, replan the case, and, above all, determine how much treatment remains. If another eighteen months are needed, that window could close while the patient is still in treatment.

That is why I think the real mistake is telling patients they have "five years of treatment," because **we are not selling them five years of time**. Those five years represent a manufacturer's ordering window, not the duration of treatment.

## When They Return, We Don't Yet Know Whether We're Continuing Treatment

When a patient reappears after three or four years, the most natural impulse is to reopen the old treatment plan immediately and figure out which aligner they can start with. After all, we already know that patient, and we still have photos, scans, and the digital treatment setup. The problem is that everything we see on the screen belongs to the patient as they were three or four years ago, while the patient in the chair is the patient as they are today.

During that time, their teeth may have shifted, their bite may have changed, dental or periodontal problems may have developed, and often we do not know exactly what they did with their last aligners. They may have stopped wearing them after a few weeks, kept wearing the same one for years, worn them only at night as a sort of retainer, or continued switching on their own to the aligners they still had at home. What they want may have changed, too: three years ago, we might have been close to finishing and they were satisfied; today, they may return asking us to change something that was not even part of the goals agreed on at the start.

So the first question should not be *"Which aligner can we start with?"* but rather, **"Where are we today, and what do we need to do from here?"** Before deciding whether we are continuing the old treatment, we need to examine the patient again, piece together what happened during the absence, assess the current clinical situation, and compare it with the situation when we last saw them.

The old scan, the old digital treatment setup, and all the files remain extremely useful because they allow us to reconstruct where we started and how far we got. **It is not the file that has "expired"; it is the clinical situation that may have changed.** That is why I consider an appointment after a long absence a full reassessment, not simply the first checkup after a break: **first we reassess the patient, and only then do we decide whether what we are about to do can still be called "continuing."**

At this point, we might be tempted to establish a rule that applies to everyone: after twelve months, treatment is considered interrupted; after two years, resuming it carries a new charge; after three, we start from scratch. It would be simple, but I don't think the time elapsed can be the only criterion, because three years away can tell completely different stories. One patient may have struggled with adherence from the start, repeatedly missed checkups, and gradually abandoned a treatment that was already progressing with difficulty; another may have followed the plan perfectly until an illness, a serious family problem, or a move made it genuinely

impossible to continue. The same three years may have passed on the calendar, but that does not necessarily mean we should make the same decision.

This is not about deciding who deserves to be rewarded or punished, but about understanding **how much of the work I need to do today can still reasonably fall within the commitment I made at the start, and how much requires a new commitment from the practice**. This is also how we need to approach the financial question: **any new charge is not meant to make the patient pay for what happened in the past, but for the work needed from today onward**.

In my practice, I group these returning patients into three broad scenarios, not as rigid rules, but as criteria that help me stay reasonably consistent in my decisions.

In the **first scenario**, I still consider it reasonable to resume treatment under the original financial terms. This might be a patient who followed the plan well before the interruption, had to stop for a serious and unforeseen reason, and returns with a clinical situation that is still fairly easy to get back on track. I may decide to take on the work required because I consider what I need to do still close enough to the commitment I made at the start.

In the **second scenario**, the original treatment can still be brought back on track, but the situation has become more complex. Several years have passed, I need to replan a substantial part of the case, or I am approaching the end of the original manufacturer's ordering window. I may decide to restart with very clear limits or ask the patient to contribute an amount proportionate to the new work required. I don't think it makes sense to set a universal amount: what I consider essential is that the patient know the terms **before** starting again.

In the **third scenario**, finally, the connection to the original treatment has become too tenuous to call it simply a continuation. Many years may have passed, the clinical situation may be significantly different, much of what we achieved may have been lost, or I may need to redo the diagnosis, treatment planning, and treatment to a substantial degree. In that case, I think it is more appropriate to propose a new course of treatment and establish a new financial agreement. Even in this third scenario, the message remains the same: I am not asking the patient to pay because he disappeared, but because **what I need to do today is no longer simply a continuation of what we left unfinished back then**.

## Warn the Patient as They're Drifting Away

So far, we have focused mainly on what to do when the patient reappears, but an important part of the problem begins much earlier, while they are gradually

dropping out of follow-up care. That is precisely when we can prevent many future misunderstandings.

Of course, our first goal must be to get the patient back into a regular treatment routine. A systematic review and meta-analysis by Mohammed and colleagues showed that reminder systems can improve attendance at orthodontic appointments and reduce the risk of missed appointments. But when absences start to recur, what we say on the phone needs to change: it is no longer enough to ask when the patient wants to schedule the next checkup. We need to explain that a long interruption has consequences. The front desk staff might say:

> *"I understand that it's hard for you to come in right now, but we can't leave your treatment on hold without checkups for too long. If the interruption continues, the doctor will need to reassess your situation when you're ready to resume, and the financial terms of your treatment may change as well. That's why it's best to work out together now how to handle this period."*

I would rather the patient hear this while the problem is developing than have to explain it three years later. It is not a threat, but advance notice of a real consequence that lets them decide whether to make arrangements to continue or agree with us on a formal suspension. **If they decide to suspend treatment, at my practice we formalize that decision by having them sign a treatment suspension consent form**, so it is clear to both of us that treatment is not simply being left open until the patient decides on their own when to return. In fact, we should explain this from the very beginning:

> *"If something comes up during treatment and you need to stop for a while, that's no problem, but don't disappear. Call us, and we'll decide together how to handle it, because leaving treatment on hold for a long time without checkups can change both the clinical situation and the financial terms under which we can resume."*

The problem, then, is not that a patient may need to stop, but that they do so without telling us. If they let us know, we can assess the situation, protect what we have achieved, and agree on a suspension. If, instead, they keep missing appointments and gradually stop responding as well, their silence cannot become an implicit way to keep treatment open indefinitely.

After making reasonable attempts to bring the patient back, I think it is useful to send a written notice making clear that, without the necessary checkups, we are no longer able to actively supervise treatment, and that returning after a long interruption will require a new clinical evaluation and a review of the terms under which treatment can resume. In the same notice, we can tell the patient what to do

with any aligners and other appliances they still have and remind them not to continue treatment on their own. All attempts to contact the patient and all important communications should be documented in the patient's chart. If the patient returns after three years, we need to be able to reconstruct when continuity of care was lost, what attempts were made to restore it, and what information the patient was given as treatment was being interrupted.

**The interruption needs to be made clear while it is happening, not reconstructed three years later.**

## When the Patient Reappears

When, perhaps after three years, the patient calls saying they want to *"continue treatment,"* the front desk staff should not decide whether that treatment is still open, whether it is still covered, or how much it will cost to resume, because at that point we do not yet have the information to know. Likewise, I would avoid getting into a discussion on the phone about what was said years earlier or why the patient disappeared. The response can be very simple:

> *"Of course, let's schedule an appointment with the doctor. It's been quite a while since your last checkup, so first we need to reassess the situation. Then the doctor will explain how we can resume and under what terms."*

When the patient comes in, I piece together what has happened since I last saw them, assess the current clinical situation, compare it with the situation when we left off, and determine how much of the original treatment can still be brought back on track. **Only then do I decide what solution to propose and under what terms we can restart.** That way, any new charge is presented not as a consequence of the patient's disappearance, but as a charge for **the work needed today to complete treatment.**

### PRACTICAL SCRIPT

Once I have decided which of the three scenarios applies, I try to follow the same sequence every time: I start with the situation today, explain what needs to be done to finish treatment, and only then clarify the terms under which we can restart.

If we can resume under the original terms:

> *"I've reassessed everything, and we can still resume treatment under the terms we agreed on at the start. From this point on, though, we need to keep treatment on track. If something happens that prevents you from continuing, call us and we'll decide together how to handle it, but we can't leave treatment on hold for years again."*

If the original treatment can be brought back on track but we are approaching the limits of the original terms:

> *"We can still resume, but we're now close to the end of the manufacturer's ordering window under your original package. If we can complete treatment within that window, nothing changes. If you need more aligners after that date, though, there may be additional costs. I'd rather tell you now, before we start again."*

If what we need to do has become a new treatment:

> *"Several years have passed, and the situation today is different from where we left off. To achieve the outcome, I can't simply order more aligners and pick up where we stopped. I need to replan the case and begin a new course of treatment. That's why what I'm proposing today is a new treatment, with a new financial agreement."*

In none of these three cases do I feel the need to begin by reminding the patient how many appointments they missed, how many times the front desk staff called, or how many times they promised to return. That information needs to be documented in the patient's chart, but using it to open the conversation only risks turning it into a debate over who is right and who is wrong. **What matters to me is bringing the conversation back to the present: this is the situation we have today, this is what we need to do, and these are the terms under which we can do it.**

After all, one of the principles we have already encountered applies here, too: **a confession doesn't move teeth**. I don't need the patient to admit they were wrong; I need them to understand where we stand today and, if we decide to restart, know exactly under what terms we are doing so.

## Why This Protocol Works

- **It Shifts the Discussion from the Past to the Present**

We do not try to determine who was right about what happened during the years the patient was away. We start with the current clinical situation and what needs to be done from this point forward, so the conversation does not turn into an argument about who was responsible for what happened in the past.

- **It Removes Ambiguity About Restarting**

The fact that treatment was never formally concluded does not mean it can be restarted at any time under the original terms. The reassessment establishes how much of the original treatment can still be brought back on track and under what terms we can restart.

- **It Ties the Cost to Future Work**

Any new charge is presented not as a penalty for disappearing, but as a consequence of the clinical and administrative work, and the expense, required from that point forward. This lets us use the investment as a motivator without turning it into a punishment.

- **It Gives Advance Notice of the Consequences**

Warning the patient while they are falling out of a regular treatment routine lets them know that a prolonged absence may change both the clinical situation and the terms under which they can resume. A consequence communicated in advance is very different from a condition imposed when the patient returns years later.

So when I see that name on my schedule again after years have passed, I do not automatically consider the treatment we left unfinished to be reopened. Instead, I consider a possibility reopened: the possibility of seeing the patient again, understanding where we stand today, and deciding together whether and how we can restart.

**The patient can disappear. Treatment cannot remain open forever.**

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## Adapting Communication to DISC Profiles

**D**

### Dominance

**Main motivator: decisions and terms.**

With a Dominance-style patient, I would avoid a lengthy review of the past and get to the point quickly:

> *"I've reassessed the situation, and these are our options today. We can bring part of the original treatment back on track under these terms or, if necessary, replan the case entirely. I'll explain the differences, and we can decide how to proceed."*

If there is a new charge: *"The charge isn't because you missed appointments. It's for the work we need to do from today onward to achieve the outcome."*

**I**

### Influence

**Main motivator: the future and the outcome.**

With an Influence-style patient, I would avoid letting the conversation become a long review of everything they failed to do. I can use their willingness to start again to turn their attention immediately to the future:

> *"Great, then let's look ahead. Let's see what we can build on, what we need to redo, and how best to organize this new phase."*

The substance of the decision does not change, including any financial terms, but the communication focuses primarily on the outcome.

**S**

### Steadiness

**Main motivator: reassurance and clarity.**

A Steadiness-style patient may arrive already worried about being scolded or having to start over from scratch:

> *"First, let's take a careful look at where we stand today and how much of what we did can still be brought back on track. Then I'll explain our options and the terms under which we can restart."*

The goal is to reassure the patient about the process without making promises we cannot yet keep.

**C**

### Conscientiousness

**Main motivator: data and criteria.**

With a Conscientiousness-style patient, I would make greater use of the available records:

> *"This was the situation when we stopped, this is where we stand today, and this is the work we still need to do. That is why I can bring this part of the original treatment back on track, while this other part needs to be replanned."*

CHAPTER 10

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When a patient stops coming in for checkups or returns after a long absence asking to continue treatment. |
| **GOAL** | Restore continuity of care when possible and make any resumption of treatment contingent on the current clinical situation. |
| **WHAT TO DO** | 1. Contact the patient when they begin missing appointments and try to understand what is making it difficult for them to come in.<br>2. Explain in advance that a prolonged absence may change the clinical situation and the terms under which treatment can resume.<br>3. If the patient decides to stop, formalize the suspension with a treatment suspension consent form.<br>4. If the patient disappears, document attempts to contact them and communicate that treatment has been interrupted.<br>5. When the patient returns, reassess them: the previous records help reconstruct the course of treatment but do not replace a current evaluation.<br>6. Distinguish between continuing the original treatment, continuing under new terms, and starting a new treatment.<br>7. Clarify the treatment plan and any costs before restarting. The cost reflects the work needed; it is not a penalty for the absence. |
| **WHAT TO CHECK** | Time elapsed, reason for the interruption, clinical situation, previous adherence, how much of the treatment can be brought back on track, remaining terms of the original package, and work still needed. |

CHAPTER

# 11

# The Patient Who's Never Satisfied

*"Nothing is enough for the man to whom enough is too little."*

— EPICURUS

Over the years, I have learned to pay particular attention to a certain kind of patient. They are not necessarily difficult patients. They are not the ones who fail to wear their aligners, nor are they the ones who miss appointments or routinely forget their orthodontic elastics. In fact, they are often extremely attentive to their treatment: they constantly examine their teeth, compare photos taken weeks apart, and arrive at appointments knowing exactly which detail they want to show me. The problem is that they are never completely satisfied.

Once we have corrected what brought them to us, their attention tends to shift to increasingly small details that they probably had not even noticed at first. Almost

without realizing it, we risk turning treatment into a pursuit of perfection that becomes harder and harder to achieve and, sometimes, impossible.

In my experience, aligners have made this risk even more apparent. Digital simulations allow patients to examine the outcome in detail, photos are compared constantly, and the ability to plan further tooth movements can easily give the impression that there is always something more to improve. There is also a very simple difference from fixed braces: after many months of treatment, a patient with braces generally cannot wait to get them off, while continuing with aligners for a few more months is often seen as a much smaller sacrifice. The desire to finish, then, can weaken just as attention to ever smaller details grows.

For a long time, I thought this way too. If a patient showed me a detail that could be improved and I thought I could improve it, I tried to accommodate them. It seemed like the most logical choice: improve the outcome further and, at the same time, increase their satisfaction. Over time, I came to understand that those two things do not necessarily go together and that, with some patients, each additional correction may simply shift their attention to the next detail.

The question, then, is not just **how much more we can improve an outcome**, but **when continuing to pursue improvements stops being genuinely useful**.

## When Improvement Isn't Enough

Stefano was about forty when he came to my office, after consulting several other orthodontists. His malocclusion was far from simple: a Class II subdivision malocclusion, with Class I on the right and full Class II on the left, an almost 100% deep bite, and significant crowding in the lower arch.

I planned treatment with aligners and orthodontic elastics and explained that we would need approximately twelve to fifteen months. Stefano followed the plan, and by the end of the first set of aligners, the result was very good: we had achieved a perfect Class I relationship on the left as well, opened the bite, and obtained excellent alignment. At the time, I was still early in my experience treating complex cases with aligners, and that result meant something special to me. I was proud to have corrected a malocclusion that many colleagues would not even have considered treating with aligners.

Stefano, however, pointed out a small detail involving his lower left lateral incisor. He was right: that tooth could be improved, so we decided to do a refinement. I wanted to complete treatment as well as possible, but I also wanted Stefano to

be fully satisfied. I thought the two went hand in hand: **the more I improved the outcome, the more satisfied he would be.**

By the end of the first refinement, that detail had improved, but Stefano had begun focusing on another one. So we did another scan, then another refinement, and then yet another. Each time, we managed to improve a detail, but his attention shifted to something new. In the end, we did eight refinements, and treatment, which was supposed to take approximately twelve to fifteen months, stretched to twenty-eight.

The paradox was clear: **the clinical result kept improving, but Stefano's satisfaction did not increase along with it**. Every time we corrected an imperfection, his attention shifted to an even smaller detail. Even after the eighth refinement, he was not completely satisfied. He said something like:

> *"We're almost there now. There's still one little thing we could improve, but treatment is taking too long, and I don't want to keep going like this."*

So treatment did not end because Stefano was finally satisfied. It ended because, after twenty-eight months, **he was no longer willing to invest any more time in it**. If continuing had not come at that cost for him, we probably would have found something else to change.

By then, I was tired and frustrated too. We had achieved the clinical result we were aiming for long before, but every new detail had become reason enough to continue. I realized I had made a mistake: **I had kept asking myself what else we could improve, without asking whether it really made sense to do so.** Stefano taught me a lesson I have never forgotten:

**Listening to the patient does not mean handing them the steering wheel for their treatment.**

## When the Point of Reference Changes

At the start of treatment, a patient may be concerned about significant crowding, a deep bite, or an incisor that is noticeably out of position. Once those problems are corrected, though, their point of reference changes. As the months pass, they also tend to forget how different things looked at the start and end up comparing the result not with the smile they had when they first came to us, but with what could still be improved. A detail they probably would not even have noticed at first can become the new problem to correct.

The literature on smile perception also shows us that orthodontists and patients do not necessarily place the same importance on the same details, and that their

thresholds for detecting small dental and gingival asymmetries differ. Studies using smile photographs, and more recently eye-tracking systems, have shown just how selective our attention can become when we focus on increasingly small details.

In clinical practice, this has led me over the years to distinguish between two very different ways of looking at the same result, which I call **social esthetics** and **inspection esthetics**. These are not terms from the literature, but two expressions I use to describe something I see constantly in my patients.

**Social esthetics** is how a smile looks in real life: when a person talks, laughs, moves, and interacts with others. No one examines a single incisor from a few inches away; we perceive the smile as a whole.

**Inspection esthetics**, by contrast, comes into play when we stop looking at a smile and start scrutinizing it. The patient moves closer to the mirror, tilts their head, changes the lighting, enlarges a photo, or compares two selfies, looking for a difference. The longer they examine a detail, the more important that detail may become.

**The problem is that correcting that detail does not necessarily put an end to this kind of scrutiny.** Attention may simply shift to the next detail, exactly as it did with Stefano. The result keeps improving, but at the same time, it becomes increasingly difficult to reach a point where the patient can find nothing else to change.

That is why **improving the result further does not necessarily increase satisfaction** as well.

One situation in which this gap between the clinical result and the patient's perception becomes especially clear is one I encounter fairly often at the end of treatment. I show the patient the final photos or digital scan, and as we look at them together, they too acknowledge that their teeth are properly aligned. Then they look in the mirror and tell me:

*"They look straight on the screen, but when I look at myself, I still don't like them."*

Simply telling them that their teeth are straight does not help. Their dissatisfaction is real, but that does not necessarily mean it still comes from the position of their teeth. What continues to bother them may have to do with the shape of the teeth, wear along their edges, anatomical differences between the two sides, how the teeth look in relation to the lips, or simply how they perceive their smile as a whole.

This is where it becomes essential to distinguish between two statements that sound similar: *"My teeth aren't straight"* and *"I don't like my smile."* In the first case, we should be able to identify a tooth, its position, and the movement needed to change it. In the second, we are describing a broader esthetic perception that may

not be improved by moving the teeth further. That is why, before planning another refinement, I ask the patient:

> *"Can you show me exactly which tooth bothers you and what you'd like to change? What would we need to see differently at the end of another refinement to say it worked?"*

The answers to these questions change the situation completely. If we can identify a tooth, a movement, and an outcome we can assess, we can consider whether it makes sense to try to achieve it. But if the request remains *"I'd like them to look better," "I'd like them to be even straighter,"* or *"I'd like to like the way I look more,"* **we still have not identified an orthodontic goal on which to base a new phase of treatment**.

We can plan a tooth movement, but we cannot plan how the patient will see themselves in the mirror. Continuing to move teeth in the hope of changing a feeling we have not been able to translate into a precise goal may simply lead us to another refinement and, a few months later, the same conversation. So in these cases, I try not to dismiss what the patient is feeling, but to clarify the limits of what another set of aligners can actually do:

> *"I'm not questioning what you see when you look in the mirror. But we need to understand exactly what bothers you and whether it has to do with the position of your teeth. If we can identify a specific movement that might improve it, we can consider it. But if we can't define what we would need to change, another set of aligners might move your teeth further without changing what you don't like right now."*

Listening to the patient's request and trying to understand it is essential. In recent years, we have rightly worked to involve patients more and more in treatment decisions: they need to understand the goals of treatment, be able to express their expectations, and discuss with us the options for achieving those goals. Involving them, though, does not mean that every request must automatically become a new phase of treatment. We remain responsible for assessing whether what they want to correct is a clinically appropriate goal and whether the expected benefit justifies continuing treatment.

For a long time, at the end of a refinement, I too would ask almost automatically: *"Is there anything else you'd like to change?"* It seems like an innocent question, but it already contains a suggestion: **that there is still something to look for**. For a patient who pays particularly close attention to details, it can almost become an invitation to examine their smile even more closely until they find a new imperfection.

Over time, I have changed how I handle the final checkups. First, I perform my clinical evaluation and determine whether we have achieved the agreed-upon goals, checking the occlusion, alignment, overjet, overbite, contacts, and all the other elements needed to consider the active phase of treatment complete. **Only then do I involve the patient in the final assessment**, show them what we have achieved, and listen to anything they still want to discuss.

The difference may seem small, but it completely changes the meaning of the conversation. The patient remains involved and still has all the space they need to express what they see and how they feel, **but we are not asking them to look for something wrong**. Our responsibility remains to assess whether what they are showing us is still a clinical goal that justifies continuing treatment.

One of the things I now do most often with these patients is **take them back to the starting point**. With Stefano, I should have done it much sooner: instead of examining every new detail with him, I should have stepped back and shown him how much the result had changed from where we started.

So I bring out the initial photos and compare them with the current ones. I do not do this to prove to the patient that they should be satisfied, because satisfaction cannot be imposed, but **to put the detail they are now examining from a few inches away alongside the overall problem we had decided to treat at the start.**

> *"Let's look at the result as a whole. This is where we started, and these were the goals we set. We've achieved them. I see the detail you're showing me too, but we need to ask whether changing it would provide a real benefit or whether we're trying to improve something that is already clinically sound."*

That way, I do not dismiss what the patient sees or tell them the detail does not exist. **I help them see the treatment as a whole again**, after months in which their attention has gradually narrowed to smaller and smaller details.

At this point, though, we need to understand what the patient's request to continue really means, because not all requests mean the same thing.

If one of the goals we agreed on has not been achieved, there is no question in my mind: treatment needs to be completed, and if another refinement is necessary, I am the first to propose it.

Then there is the opposite situation, in which the patient's request **does not correspond to a specific orthodontic goal or would require a movement I consider unnecessary, unpredictable, or clinically inadvisable**. In that case, I do not proceed, regardless of their willingness to continue.

But there is a third possibility, and it is the one that most often creates difficulties with a patient who is never satisfied. **The treatment goals have been achieved, and I do not consider another refinement necessary, but the patient points out another detail—one that is real, can be changed, and is clinically acceptable—that they still want to address.**

This is the hardest situation to manage, because I cannot tell the patient that the detail they are showing me does not exist or that it cannot be changed. I can, however, explain that **completing the treatment goals is one thing; deciding to continue beyond those goals in pursuit of further improvement is another**.

## When Explanations Are Not Enough

For a long time, I tried to handle these situations primarily through communication. I showed the initial photos, reminded the patient of the goals we had agreed on, compared where we started with the result we had achieved, and explained why another refinement would not necessarily produce a significant improvement. All of this remains essential and is still the first part of my approach today, but I realized it was not enough with some patients. I could spend a great deal of time explaining that we had achieved the goals and that the detail we were looking at did not warrant another phase of treatment, but in the end, the question was almost always the same:

*"I understand, Doctor, but if we can still try, why wouldn't we?"*

From the patient's perspective, it was a perfectly logical question. If continuing meant no additional payment, if wearing aligners for a few more months was no longer a particularly significant sacrifice, and if we could produce another set anyway, **why stop?** Even if the expected benefit was small and uncertain, the perceived cost of trying was practically zero.

That was the part I had failed to understand for a long time. I kept looking for better arguments to explain why stopping made sense, while the patient continued to see a chance for improvement that, from their perspective, had almost no downside. As long as the two options were *stop* or *try again for free*, even a very small detail could seem like reason enough to choose the latter. I realized, then, that in this specific situation, explanations alone were not enough. **I had to make the decision to continue concrete.**

In my practice, I chose to make that decision concrete by introducing a fee for the additional phase of treatment, generally **1,000 euros.** The amount itself is not the point, of course, and I do not mean to suggest a rule that applies to everyone:

each colleague can set an amount that fits their own practice. What matters is that the amount be significant enough to turn what had, until then, seemed like a simple opportunity to *"try again"* into a real decision. I want to be very clear about this, because **I do not introduce the fee to sell another refinement; on the contrary, when I introduce it, I hope the patient will decide not to proceed. Its purpose is deliberately to discourage them**.

Until then, the patient had been asking themselves, *"If we can still try, why should we stop?"* Once continuing requires a new investment, the question becomes, *"Is this detail really important enough to justify another 1,000 euros and a few more months of treatment?"* That is exactly the shift I am looking for: the detail is the same, and the movement we might try to achieve is the same, but **the value the patient places on that detail changes when continuing is no longer without consequences**. In fact, very often, after I explain the fee, the patient looks at their smile again and asks me, *"But do you really think it's worth it?"* To me, that is the moment the strategy has achieved its goal, because the patient is finally asking not just whether they can continue, but how important continuing really is to them.

Of course, a willingness to pay this fee does not make every request acceptable. This option applies only to the situation we have just described: the treatment goals have been achieved, and the requested change is not necessary, but it is specific, clinically acceptable, and reasonably achievable. If I consider the movement unnecessary, unpredictable, or potentially harmful, I do not proceed, regardless of the patient's willingness to pay. Likewise, if the patient continues to say only *"I don't like them"* without being able to identify what they want to change, we still do not have a goal on which to base a new phase of treatment. A fee does not make an inadvisable movement advisable: it comes into play only after we have established that the change can be made but is not needed to complete treatment.

### PRACTICAL SCRIPT

When I consider the treatment goals achieved but the patient points out another detail they would like to change, the conversation might go something like this:

> *"Let's look together at where we started and the goals we set. I see the detail you're showing me too, and we can try to change it, but from my point of view, another refinement is not needed to complete treatment, because we've achieved the goals we agreed on.*

> *If I thought there was still something we needed to correct, I would be the first to propose another set of aligners, and of course there would be no additional charge. In this case, though, we're talking about a further improvement you'd like beyond the result we've already achieved.*
>
> *If this detail is important enough to you that you want to continue, we can consider it, provided we can define exactly what we want to change and the movement is clinically acceptable and reasonably predictable. In my practice, this additional phase costs 1,000 euros.*
>
> *I'm not recommending that you do it; on the contrary, my recommendation is that we stop. I'm simply giving you the option to decide whether this detail is important enough to you to justify a new investment and a few more months of treatment."*

If, at this point, the patient asks me, *"But do you really think it's worth it?"* my answer must be consistent with what I have just explained. If I consider the goals achieved and the expected benefit marginal, I tell them exactly that. **The fee is not meant to sell the refinement, but to make the choice to proceed with it real.**

## Why This Protocol Works

- **It Brings the Comparison Back to the Starting Point**

Over time, patients tend to get used to the result they have achieved and gradually focus more on what could still be improved. Bringing out the initial photos is not meant to convince them that they should be satisfied, but to keep the last small detail from becoming the only measure by which they judge the entire treatment.

- **It Avoids Suggesting That There Must Be Something to Correct**

The question "Is there anything else you want to change?" implicitly invites the patient to look for something. That is why I prefer to perform my clinical evaluation first and only then listen to any concerns the patient may have. I do not reduce their involvement, but I avoid turning the final checkup into a systematic search for the next imperfection.

- **It Turns an Option into a Decision**

When another refinement seems available with no consequences, even a very small benefit can seem like reason enough to try. Introducing a fee changes that assessment because it brings back into view certain factors that the option to continue had gradually pushed into the background: time, effort, and above all, how much the patient truly values the detail they want to change.

The point is not to persuade the patient with money, but to interrupt the automatic "we can try again" response. That is why the amount must be significant enough to require a deliberate choice, while remaining, of course, a practice decision that each professional must adapt to their own circumstances.

### It Keeps Desire Separate from Clinical Indication

The patient's willingness to pay the fee does not change our clinical judgment. First, we must establish that the request is specific, clinically acceptable, and reasonably achievable; only then can we discuss a possible additional phase. **The availability of more aligners is not, by itself, an indication to use them.**

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## Adapting Communication to DISC Profiles

**D**

### Dominance

With a **D** patient, I tend to be direct and focused on the decision: "We've achieved the goals we set. We can try to change this detail further, but I don't consider another refinement necessary. If you still want to continue, we can consider it as an additional phase."

**I**

### Influence

With an **I** patient, I focus more on the overall change and social esthetics: "Look how much your smile has changed since we started. The detail you're showing me is real, but before we continue, I'd like us to consider how important it really is compared with the overall result we've achieved."

**S**

### Steadiness

With an **S** patient, I try above all to avoid making the end of treatment feel like abandoning something unfinished: "We've achieved the goals we set, and I think it's right to end treatment here. If this detail remains very important to you, we can still consider it, but it's not something I think needs correcting."

**C**

### Conscientiousness

With a **C** patient, I use more analytical, verifiable criteria: "We've reviewed the initial goals and the clinical parameters we established. The detail you're seeing is real, and we can define it precisely, but the expected benefit of another refinement is limited. We can therefore consider together whether continuing really makes sense."

CHAPTER 11

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When the patient keeps asking to continue after the goals have been achieved or expresses dissatisfaction without identifying a specific change. |
| **GOAL** | Distinguish incomplete treatment from an additional request and proceed only when there is a clinically appropriate, verifiable goal. |
| **WHAT TO DO** | 1. Assess whether the agreed-upon goals have been achieved.<br>2. Compare the initial photos with the current situation, showing the change achieved.<br>3. Identify the request: which tooth, which movement, what result should be different?<br>4. If an agreed-upon goal has not been achieved, complete the treatment.<br>5. If the request is vague or clinically inappropriate, do not begin a new phase.<br>6. If there is a real detail that can be changed safely but does not need to be changed to complete treatment, consider an additional phase.<br>7. Define the goal, expected benefit, limitations, and any fee before proceeding. A willingness to pay does not make an inappropriate request appropriate. |
| **WHAT TO CHECK** | Initial goals, current result, how specific the request is, expected benefit, and the patient's understanding of the criteria we will use to consider any additional phase complete. |

CHAPTER

# 12

# The 95% Patient

*"The best is the enemy of the good."*

— VOLTAIRE

Some patients make it clear as soon as they sit down in the chair that we have a complex treatment ahead of us. Then there are others who give us exactly the opposite impression. They smile, and we almost wonder why they came in: their teeth are already straight, their bite is good, their smile is attractive, and they may even have had orthodontic treatment before. At first glance, there seems to be very little to correct. Then the patient points to a slightly rotated incisor, a small space, or a tooth just out of position and says:

*"Doctor, I just want to fix this one little thing."*

Over time, I've learned that the word **just** should immediately make us pay closer attention.

We see a small movement and naturally think of a small treatment: a few aligners, a short time, little difficulty. But that is not necessarily how it works. In fact, some treatments that seem simplest from a biomechanical standpoint can be among the most delicate to manage, because when the starting point is already very good, even the smallest detail takes on an entirely different importance.

Over time, I've come to call them **"95% patients"**: people who are already very close to their desired result and ask us to address only the small gap that still separates them from their goal.

Imagine two patients. The first comes to us with an obvious problem: visibly crooked teeth, significant crowding, or another situation they clearly want to change. At the end of treatment, the result will never be perfect, because perfection is not natural and, in fact, does not exist. But the improvement will be so clear that the patient will not have to look for it: they need only compare the before and after to see how much their smile has changed. The second patient, by contrast, comes into our practice with an already beautiful smile and teeth that are practically straight. They are not looking for a major transformation: they came in for that one small detail they keep noticing and want to correct.

And that is the difference that completely changes the picture. For the first patient, the improvement is substantial, obvious, and spread across many aspects of the smile. For the 95% patient, by contrast, the entire value of treatment is concentrated in one very small detail. What may be a minor correction to us is the whole reason they decided to begin treatment: **that 5%, to them, is 100% of the problem**. That is why I consider the 95% patient potentially more delicate to treat than many apparently complex cases: **we have little room for improvement and a great deal to preserve**.

And it is precisely that word, *just*, that can influence us too. If the patient considers the problem small, we can easily start to consider the treatment small as well. *"It'll be easy," "it'll only take a few aligners,"* and *"it's nothing"* are phrases that sound reassuring but that I have stopped using. Once we say something will be easy, we are no longer just describing the biomechanics: we are creating an expectation. The patient may take that statement as a guarantee that the result they want will be simple to achieve and perfectly predictable, when we know very well that a biological movement does not become perfectly predictable simply because it is small.

There is another point worth clarifying from the start: the fact that the patient sees only one tooth out of position does not necessarily mean treatment will consist simply of moving that tooth into the right place. To rotate an incisor, we might need to create space; to close a small diastema, we might need to control the inclination

of the neighboring teeth; and to maintain a good bite, we might need to involve other teeth or even both arches. A tooth may also fail to respond exactly as expected, a refinement may be needed, and torque, contacts with neighboring teeth, periodontal tissues, or anatomical limitations may come into play—factors the patient cannot know about when they ask us to *"move this tooth just a little."* Explaining this beforehand matters, because our goal remains to correct that detail, but doing it properly may require us to move teeth the patient considers perfectly fine as they are. If we have not made that clear from the start, any movement beyond that one tooth may feel like something they never asked for. **How small a problem looks does not, by itself, tell us how easy it will be to achieve exactly the result we have in mind.**

The same misunderstanding can arise over cost. The patient sees only one tooth that needs moving and may think that, because it is a small change, treatment must necessarily cost very little. Of course, each colleague is free to set their own fee structure, but a few aligners do not automatically mean little value or little responsibility: even a small treatment requires diagnosis, planning, and clinical responsibility. **The patient is not buying four, six, or ten aligners: they are undergoing orthodontic treatment.**

## Define the Goal Before Starting

With these patients, I think it is essential to slow down precisely when everything seems to suggest moving quickly. Before starting, I need to understand exactly what the patient wants to change—not in general terms, but very specifically. I look at the photos, the scan, and the clinical situation with them, ask them to point out exactly what they do not like, and try to understand the result they imagine. Then I ask a question I consider essential: **"If we could correct exactly this one thing and everything else stayed just as it is today, would you be satisfied?"**

It is a very simple question, but it often tells us immediately whether the request really is limited to that detail. If the answer is an unequivocal yes, we have identified a specific, shared goal. But if the answer is *"Yes... but maybe, while we're at it, we could also...,"* that probably means the small correction is not the only thing the patient would like to change, and it is much better to find that out before starting. In more delicate cases, I think it is useful for what we have agreed on to be **put in writing and signed by the patient**—not to turn the doctor-patient relationship into something legal or defensive, but simply so that, as time passes, the reason we began treatment and the result we set out to achieve remain clear.

If the patient wants to rotate an incisor slightly, that rotation will be the goal; if they want to close a small space, closing that space will be the goal. It may seem like an obvious distinction, but defining the endpoint precisely also means defining when treatment can be considered complete. **If we do not know what has to happen before we can say "we're done," every improvement risks becoming merely the starting point for a new request.**

Once the initial goal has been reached, the patient's attention may shift to something they had not noticed before. That does not necessarily mean we are dealing with someone who is chronically dissatisfied: it may simply be a consequence of the change we achieved. The point is to distinguish what remains to be done to reach the agreed-upon goal from a new request. If the patient came in to close a space and that space is not completely closed, we are still completing treatment; if, on the other hand, we have closed it as agreed and now another detail comes up, we are talking about something different. Patients rarely put it in those terms. More often, they say: *"While we're at it..."* And it is precisely that *while we're at it* that can turn a small touch-up into a process with no real end.

One case probably taught me more than any other how dangerous it can be to confuse the desire to accommodate a patient with the clinically sounder choice. I still consider it one of the most stressful episodes of my professional career. Nadia was the mother of a young patient I had treated for a Class III malocclusion with a maxillary expander. She was very pleased with her son's result and, during one of his checkups, said to me, almost jokingly: *"Doc, sooner or later we need to do something about my teeth too."* I asked her what she didn't like, and she explained that, overall, she was happy with her smile, but she felt her upper arch was a little narrow. Above all, she didn't like the position of her upper right lateral incisor (FDI 12), which sat slightly lingual to the other teeth. In photos, that tooth fell more into shadow and looked darker, and that was the detail that bothered her.

Since she was already in the office, I suggested doing an intraoral scan to get a better sense of the situation. That scan showed us that the case was less straightforward than it appeared if we looked only at the lateral incisor: Nadia had a tendency toward a Class III malocclusion, a tendency toward a crossbite on the right side around the premolars and first molar, and some crowding in the lower arch. I explained that if we wanted to plan comprehensive orthodontic treatment, we would need to consider both arches and correct the overall occlusal relationship. But she was very clear: she didn't care about the lower crowding because *"you can't see the lower teeth anyway,"* while what she really wanted was to move that upper lateral incisor farther labially so it would no longer fall into shadow in photos. So I

agreed to treat only the upper arch with a treatment package limited to 20 aligners. The goal was specific and relatively simple: correct the position of the upper right lateral incisor (FDI 12) without turning her request into comprehensive orthodontic treatment, which she did not want to undergo at that point.

After about ten to twelve aligners, in just over three months, we achieved exactly what Nadia had asked me to do. The lateral incisor was in the right position, her smile had improved, and she was very pleased. I asked her explicitly whether she liked the result, and she confirmed that she did. With respect to the goal that had brought her in, treatment could have been considered complete. But it was precisely then that Nadia began to look at her smile differently: she started noticing the right posterior area more and told me she wanted the arch to look a little wider around the premolars, reducing what she saw as an overly noticeable buccal corridor. I explained why I did not think expanding only the upper arch was a good idea: for one thing, we would need to maintain a proper relationship with the lower arch, or we risked creating premature contacts; for another, expanding the premolars would inevitably create spaces that would be difficult to manage. But Nadia persisted. She didn't want a lot of expansion, she told me, *"just a little, whatever you can do."*

And that is where I made my mistake. I felt I knew her, I had treated her son, our relationship had been excellent up to that point, and Nadia had always seemed reasonable to me. So I decided to accommodate her and proposed a new scan, explaining that I would prepare an extremely conservative simulation, with the least expansion I thought compatible with her bite. When she came back to see the digital treatment setup, something happened that I now recognize immediately as one of the most important warning signs with a 95% patient: she looked at the simulation and told me she wanted a little more expansion. I adjusted the digital treatment setup slightly and showed her that increasing the movement could create spaces and change the relationship with the lower arch, but she asked for another small increase. I showed her again what the consequences might be, emphasizing the risk of creating premature contacts and the need to find a compromise, until she told me that amount of expansion seemed sufficient, and I approved the refinement.

Three months later, Nadia returned extremely dissatisfied. Expanding the premolars had created small spaces where food tended to get stuck, and in trying to manage the arch form, the upper right canine had moved slightly inward and developed a premature contact with the lower canine. At that point, I tried to correct the problem with a second refinement, moving the canine farther labially and changing the position of the premolars again. But each time I corrected one aspect, another issue emerged: the canine improved, but the posterior contacts changed; we fixed

the contacts, and some other detail appeared that Nadia found bothersome. What had begun as a small touch-up was gradually turning into a treatment with no real end. In a further attempt to satisfy her, I even opened a new case at my own expense so I would have additional sets of aligners available and could keep looking for a solution that would finally make her happy. What had initially been a treatment lasting about three months, completed with a patient pleased with the result for her lateral incisor, thus became a process lasting fifteen to eighteen months.

Each appointment grew progressively more difficult, and a visit that would normally have taken ten minutes could turn into an hour or even an hour and a half of discussion. Nadia was increasingly angry, I was increasingly stressed, and the relationship that had been excellent at the start deteriorated visit by visit. I still remember exactly how I felt when I saw her name on the schedule: I knew I was in for a long, difficult appointment, and even before she arrived at the office, I could feel the tension building. It stayed with me all day. I kept reminding her that, before we began that second phase, I had explained several times why I did not think it was advisable to expand only the upper arch. But her response was probably one of the most important lessons I have ever learned from a patient:

*"Yes, Doc, but you're the doctor. You should have told me no."*

Over time, I had to admit that Nadia was right. Not because I had failed to understand the biomechanics of the case—I understood them perfectly and had explained them to her—but because, once I had determined that her request was not clinically advisable, the responsibility for holding that line remained mine. Patients can insist, tell us they accept the risk, ask us for *"just a little,"* but we are still the ones who decide whether to make that movement. The situation came very close to becoming a malpractice dispute: Nadia sought another opinion and was told that properly correcting the situation would require treating both arches and planning the case comprehensively—in essence, the very treatment I had proposed at the start and she had declined.

In the end, to keep the situation from deteriorating further, I decided to give her a full refund. The toll was heavy in every respect: I had personally covered the cost of two cases, refunded the entire fee, used an enormous amount of clinical time and staff energy, and turned a treatment that could have ended successfully after three months into one that lasted more than a year. Nadia was dissatisfied, and so was I. A patient who had initially placed enormous trust in our practice was unlikely to keep speaking about us with the same enthusiasm she had shown after her son's treatment. But the greatest cost, at least for me, was not financial: it was the stress

of a situation I had helped create by trying to accommodate a patient, even though I had known from the start that expansion limited to the upper arch was not the solution I would have chosen clinically.

In hindsight, it is easy to pinpoint when I should have stopped: at the appointment after the first set of aligners, when Nadia was satisfied and the lateral incisor she had come in about was in the right position. We had achieved the agreed-upon goal, and everything that followed was not the completion of that treatment but the start of a new one—a treatment I myself did not consider clinically sound but accepted anyway in an attempt to please her. That is precisely the danger of the 95% patient: not so much the small movement they initially come to us for, but the possibility that, once we have achieved that goal, the target shifts and we keep chasing it, until we lose sight of the line between listening to what the patient wants and going along with a request we no longer consider clinically sound.

Today, faced with the same situation, my answer would be very different: *"I completely understand what you'd like to improve, and I understand why, now that you're looking at your smile differently, this bothers you more. The problem is that, to try to make that change, I would have to alter a balance that works well today, and I'm not sure that would be the best choice for you. If you'd like, we can discuss a more comprehensive treatment that also addresses this concern; otherwise, given the goal we started with, I think we can consider treatment complete."* This experience taught me a rule I now consider essential when managing 95% patients: **listening to patients means taking what they want seriously, but it does not necessarily mean doing what they ask us to do.**

## Preserving the 95%

Nadia's case profoundly changed how I approach these patients. Before asking myself how to achieve the movement they are requesting, I now try to understand what I risk changing in the process. A very good starting point presents a paradox: we have little to gain, because the flaw we want to correct is small, but we may have a great deal to lose, because the teeth, contacts, and occlusal relationships around that flaw already function properly. Every movement must therefore offer a benefit that justifies what we have to change to achieve it.

The first thing to check is whether orthodontic treatment is indicated, because what looks like a problem with tooth position does not always stem from the tooth's position. A difference between two incisors may be due to the shape of their crowns, their incisal edges, the gingiva, wear, color, or simply the way light falls on two different surfaces. In these cases, continuing to move a tooth may not resolve what

bothers the patient at all, while a small restorative correction, reshaping the incisal edge, or, in selected cases, treatment of the gingival tissues may produce a result much more proportionate to the problem. The question we need to ask ourselves, then, is: **am I trying to correct something with orthodontics that is actually an orthodontic issue?**

Even when the answer is yes, however, a small movement is still a biological movement. With aligners, we can plan rotations of a few degrees, extrusions of a few tenths of a millimeter, changes in torque, or extremely precise space closures on a screen, but what we see in the software remains a plan, and the clinical response does not become mathematically predictable simply because the movement is small. This matters even more when that movement is the entire reason the patient decided to start treatment.

We also need to think about stability from the outset. In a patient who has corrected a significant problem, a tiny change in a tooth's position may even go unnoticed; in a patient who came in solely to close half a millimeter of space or correct a slight rotation, the same relapse may mean the exact problem that led them to seek treatment has returned. **Retention, then, cannot be a detail we discuss at the end; it must be part of the initial conversation.**

**The patient is asking us to correct the 5%. We are responsible for not damaging the 95% they already have.**

### PRACTICAL SCRIPT

With this type of patient, I prefer to address the issue before we even begin, because being clear at the outset almost always prevents much more difficult conversations at the end:

> *"Your starting point is already very good, and that's a positive thing. Precisely because of that, we need to be very clear about what you want to change: you're asking me to address a small detail in a situation that already works well. First, we need to find out whether what bothers you really comes down to the tooth's position. If it does, that will become our goal. Keep in mind, though, that even if you want to correct just one tooth, we may need to move other teeth, create space, or work on both arches to do it properly without compromising what already works well.*
>
> *If we decide to start, let's agree on the result we want and on when we can consider treatment complete. If we reach that goal, we'll have achieved what we set out to do. If you notice another detail along the way, of course we'll assess it together, but*

> *it will be a new request, and we'll need to decide whether intervening really makes sense. And precisely because we're correcting something very small, we also need to think from the outset about how to maintain the result over time."*

The tone should remain positive. We are not trying to scare the patient or put together a series of defensive disclaimers; we are simply making clear that we are agreeing to pursue **a defined goal**, not to begin a process of improvement with no end.

## Why This Protocol Works

- **It Defines the Target at the Outset**

For the 95% patient, the detail they want to correct makes up a tiny part of their smile but accounts for almost the entire reason they decided to start treatment. Defining it precisely allows the clinician and the patient to know from the outset what they are trying to achieve and, above all, to recognize when they have achieved it.

- **It Puts the Word "Just" in Perspective**

A small problem does not necessarily mean simple treatment. Explaining from the outset that we may need to involve other teeth, create space, or treat both arches keeps each necessary movement from coming as a surprise. **The size of the problem as the patient perceives it does not necessarily determine the complexity of the treatment needed to correct it.**

- **It Protects What Already Works**

Starting from a very good situation changes the balance between the expected benefit and what we need to preserve. Before agreeing to the request, then, we must determine not only whether the movement is possible, but also whether it is orthodontically indicated and whether the expected benefit justifies the changes required to achieve it.

- **It Defines Both the Result and Its Stability**

For the 95% patient, a small relapse may amount to the return of the entire problem as they perceived it. Discussing retention before treatment therefore makes clear that correcting the detail is only one part of the process; the other is maintaining the result.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

With a **D** patient, I get straight to the result and the boundaries of treatment: *"Your starting point is already very good. Let's define exactly what detail you want to correct and what result we want to achieve. We may need to involve other teeth to do it properly, but our goal will remain the one we agreed on. Once we reach it, treatment will be complete."*

**I**

### Influence

With an **I** patient, I focus more on how their smile looks and on agreeing on the result together: *"Your smile already looks very good, and that's exactly why I want to protect everything that's working well. Show me exactly what detail you'd like to improve. We'll define it together and document it, so that at the end we can compare the result with what you wanted to achieve. Once we've reached that goal, treatment will be complete, and then maintaining the result will be essential."*

**S**

### Steadiness

With an **S** patient, I emphasize above all that we don't want to disrupt a situation that already works well: *"We're starting from a very favorable situation, and the goal isn't to change everything. It's to address this specific detail carefully while keeping what works stable. Before we begin, we'll define together what you want to achieve, and I'll explain which teeth we may need to involve, so there won't be any surprises along the way."*

**C**

### Conscientiousness

With a **C** patient, I use a more analytical approach: *"We first need to analyze the detail you've pointed out to understand what's really causing it. If it's a problem with tooth position, we'll assess the movement needed and its effect on space, contacts, and occlusion. If it's due instead to the shape of the tooth or the surrounding tissues, we may consider a different solution. So let's define the goal, the criteria we'll use to consider treatment complete, and how we'll maintain the result."*

CHAPTER 12

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When the patient is starting from an already favorable situation and asks to correct a detail, presenting it as a minor change. |
| **GOAL** | Correct the detail when indicated, while protecting what already works and defining the endpoint in advance. |
| **WHAT TO DO** | 1. Identify the exact tooth, detail, and desired result.<br>2. Check that the problem stems from tooth position and that the movement is orthodontically indicated.<br>3. Assess which teeth, contacts, and occlusal relationships will need to be involved and which must be preserved.<br>4. Explain that a limited request may require more extensive movements than the patient expects.<br>5. Define and document the goal and the criteria for completing treatment, distinguishing them from any new requests.<br>6. Make clear that a small movement is still a biological movement and does not allow promises of mathematical precision.<br>7. Plan retention from the outset: even a slight relapse can bring back the entire problem as the patient perceived it. |
| **WHAT TO CHECK** | The source of the flaw, clinical indication, effects on occlusion, expectations, whether the goal has been reached, and whether the result is maintained. Don’t underestimate “just”: that detail may account for 100% of the reason the patient wants treatment. |

CHAPTER

# 13

# When Finishing Doesn't Mean You're Done

*"The end crowns the work."*

— MEDIEVAL LATIN PROVERB

For years, I noticed something in my patients that I couldn't fully explain. I had the sense that some results achieved with aligners tended to relapse more readily than results I had achieved with fixed appliances. It didn't happen in every case, and it didn't affect every type of movement in the same way, but in my experience, rotations and certain vertical movements, such as intrusion and extrusion, were what caught my attention most.

At first, it was only a clinical impression, and for that very reason, I couldn't automatically turn it into a conclusion. The first explanation I considered was the most obvious one: perhaps I wasn't seeing a stability problem, but simply movements that

were harder to achieve fully with aligners. The literature reminds us that treatment is not equally effective for every type of movement, and that controlling certain rotations, torque, and vertical movements can be more challenging than other types of tooth movement.

That explanation, however, didn't fully account for what I was seeing, because I wasn't just looking at teeth that had failed to reach their planned positions. In many cases, they had reached those positions correctly; the difficulty came later, when it was time to maintain them.

So I began to wonder whether I had been looking at the problem from the wrong angle. Perhaps the question wasn't just **how I had moved that tooth**, but **how much time I had given it to stabilize after moving it**.

When we move a tooth orthodontically, we aren't simply changing the coordinates of its crown within the arch. We are altering a biological system made up of bone, the periodontal ligament, and gingival tissues, which must gradually adapt to the new position. The end of orthodontic movement and the end of biological reorganization, then, do not necessarily coincide.

Classic studies of post-treatment stability have shown that different tissues do not all reorganize at the same rate. The periodontal ligament tends to reorganize over approximately three to four months, while gingival collagen fibers may take four to six months. Some components of the supracrestal fibers may take even longer; classic histologic studies described them as not yet fully reorganized **232 days after orthodontic movement ended**, especially following the correction of rotations.

That last finding in particular caught my attention: **232 days is almost eight months**. That does not mean every tooth needs almost eight months to become stable, nor does it mean the literature recommends the protocol I later developed. It did, however, help me bring a fundamental point into focus: **a tooth may have finished moving long before the tissues around it have completed their adaptation to its new position.**

At that point, I also began to think about a very simple difference between fixed appliances and aligners. With fixed appliances, a rotation may be fully corrected in the early stages of treatment, just as a curve of Spee may be leveled or a vertical movement completed long before the appliances are removed. But treatment continues: we move on to stiffer archwires, sagittal mechanics, space management, and refinement of the occlusion, while that tooth remains for months in a system that limits its movement.

By the time we finally remove the appliances, a long time may have passed since that particular movement was completed. This is where I began to form a possible

explanation for what I was seeing: **with fixed appliances, some stabilization may take place while treatment is still underway.**

With aligners, by contrast, that interval can be much shorter. A rotation, intrusion, or extrusion may be programmed through the last aligners and, if necessary, corrected again during a refinement. We may therefore reach the end of active treatment at virtually the same time that we are completing one of the very movements we consider most delicate.

This led me to look at treatment duration differently. **Eighteen months of treatment does not necessarily mean eighteen months of stabilization for every movement we have made.** Two patients may both have undergone eighteen months of treatment, but an incisor may have been in its final position for nine months in the first patient and only a few weeks in the second.

**This remains my clinical hypothesis, not a mechanism demonstrated by the literature.** Together with what we know about how long tissues take to reorganize, however, it led me to change the way I concluded my treatments.

**My stabilization protocol grew out of that reflection.**

## Nine Months That Weren't There Before

In the past, when I was satisfied with the result, the transition was very simple: I ended active treatment, took a scan for the retainers, and asked the patient to wear them at night, following the protocol I'll describe in the next chapter. In essence, I went almost directly from about twenty-two hours of aligner wear to about ten hours of nighttime retainer wear.

At some point, I stopped doing that. If the tissues needed time to reorganize, it seemed inconsistent to complete the final movements and then immediately, drastically reduce the hours those teeth were held in their newly achieved positions. So I added a phase between active treatment and definitive retention that had not existed before: **the stabilization phase**.

In my practice, it lasts a total of nine months. That duration does not come from a study showing that nine months is the ideal period. It is a clinical decision I reached over time, drawing on what I observed in my patients and the long biological window described for the reorganization of some supracrestal tissues—up to the **232 days** we discussed.

But I still had to decide how to organize those nine months. I did not want them to be merely a long waiting period. I wanted them to guide the patient gradually from about twenty-two hours of wear during active treatment to about ten hours

of nighttime retainer wear. So I divided them into **three consecutive three-month phases**, each with a specific goal.

**Three months to stabilize. Three months to taper. Three months to allow the occlusion to settle.**

That is how I developed the sequence I still use today, the **Pasciuti 3-3-3 Stabilization Protocol**, which calls for a total of 24 stabilization aligners. I reserve it, however, for patients who have suitable clinical conditions and are sufficiently adherent. If a lack of adherence makes another nine months of aligners unrealistic, I end the active phase and move directly to retention.

### The First Three Months: Stabilize

The first twelve aligners are worn exactly like the active aligners that preceded them: one per week, about twenty-two hours a day. To the patient, very little appears to change, because they continue the routine they became accustomed to during treatment. From a therapeutic standpoint, however, the goal changes completely. I am no longer trying to modify the occlusion or complete new, clinically significant movements. I am maintaining what we have achieved and giving the tissues an initial period to adapt to the position reached.

### The Second Three Months: Taper

After the first twelve aligners, I begin to reduce wear time gradually. Aligner number 13 is worn twenty-one hours a day, number 14 for twenty hours, number 15 for nineteen, and so on, decreasing by one hour with each weekly change until aligner number 24 is worn for ten hours a day.

This way, the transition from nearly continuous wear to nighttime wear does not happen abruptly. Each week, we add one more hour without an aligner, gradually guiding the patient toward the next phase.

### The Last Three Months: Allow the Occlusion to Settle

When the patient reaches aligner number 24, the plan changes once again. They no longer replace it, but continue to wear **that same aligner for the next three months**, only at night, for about ten hours.

At this point, six months have passed since active movements ended, and I no longer want only to hold the teeth in the positions they have reached. I also want to give function more room to play a role, so the occlusion can gradually find its balance.

I have always thought that an orthodontist's job is not to build an occlusion on a screen and then try to hold it indefinitely in that position. Our goal is to put the teeth

in the best possible position so that, once orthodontic movements are complete, chewing, contacts, and function can also contribute to their settling.

That is why, during the last three months, there is no longer any material between the arches during the day. The teeth come back into contact during function, promoting the gradual process of **occlusal settling**.

The decision to keep wearing the same aligner for all three months is intentional. A thermoplastic material worn repeatedly does not necessarily retain the same mechanical properties over time that it had when it was new. Viscoelastic phenomena such as *stress relaxation* cause the forces exerted by the material to change progressively.

I do not consider this gradual change in the aligner's mechanical properties to be the cause of *occlusal settling*, nor has my protocol been studied experimentally to demonstrate such a relationship. **The decision to keep wearing the same aligner is instead part of a strategy in which the aligner gradually stops trying to rigidly control every detail of the occlusion**, while still protecting the position achieved at night.

By the end of this phase, **nine months have passed since active movements ended**. Only then do I consider the stabilization period complete and take a new scan to make the definitive retainers.

#### In Practice: How I Obtain the 24 Stabilization Aligners

There is, however, a very practical problem: deciding to use twenty-four stabilization aligners does not necessarily mean I can order them as twenty-four passive copies of the same final position. Commercial systems may limit the ability to request such long sequences without programmed movements.

Over time, I have developed a planning strategy that allows me to obtain the sequence the protocol requires. At the end of treatment, I take a new scan and request a refinement. When I receive the new digital treatment setup, I remove all the proposed tooth movements and essentially return the plan to the position achieved at the end of the previous treatment.

That, however, would produce only one aligner per arch. To obtain twenty-four aligners, I select two posterior-most teeth, usually second or third molars, one in each arch, and intentionally program a clinically insignificant movement for each, such as **0.2 mm of intrusion** or a minimal root movement involving torque or tip.

The software thus recognizes that a movement is present and generates a sequence of aligners. I then adjust the staging, intentionally slowing that minimal activation so it is distributed across the **twenty-four aligners** the protocol requires.

From a digital standpoint, these aligners are therefore not completely passive, because the digital treatment setup retains a programmed micro-activation on one posterior-most tooth in each arch. **Clinically, however, that micro-activation is irrelevant to the treatment outcome:** I include it solely as a planning workaround to obtain the number of aligners the protocol requires.

This procedure is a technical solution I use in my practice with the system I work with. It is not a procedure recommended by the manufacturer or a protocol validated by the literature, and it must, of course, be adapted to the characteristics of the system being used and the individual case.

But one problem remains, and it is behavioral rather than biomechanical. The patient is sitting across from us, looking at their smile and happy with it, and we are happy with it too. After months of treatment, we finally say the words they have been waiting to hear: *"We're done."* A few minutes later, though, we explain that they will need to keep wearing aligners for another nine months.

Their reaction is understandable: *"What do you mean, another nine months? I thought we were done?"*

The problem lies in the word *done*. **We are done moving the teeth, but we are not done with treatment.** We have achieved the result, but we still need to protect it while the tissues adapt to the new position.

If we present stabilization as something that comes after treatment, almost an add-on that the patient can approach more casually, we risk undermining their adherence ourselves. That is why I prefer to present it from the start as a phase that is still fully part of treatment: **we no longer need to achieve the result; we need to avoid losing what we have just achieved.**

### PRACTICAL SCRIPT

When I explain the stabilization phase to a patient, my first priority is to prevent a misunderstanding: I am not telling them that treatment has failed or that we need another nine months to achieve the result. On the contrary, I begin with the fact that we have already achieved it.

> *"Today we achieved the result we wanted. Your teeth have reached the positions we wanted them to reach, so from this point on, we do not need to move them anymore. But there is something important to understand: just because a tooth has reached the right position does not mean it is already stable there. The tissues around it need*

> *time to reorganize, and especially at first, if we suddenly stopped protecting it, we might allow it to start moving again.*
>
> *It would be a shame to come this far, after all the time and effort we have invested, achieve the result we wanted, and then risk losing part of it right now. That is why I consider these nine months part of treatment: we no longer need to move the teeth, but we do need to preserve what we have achieved while it becomes stable.*
>
> *We'll do this gradually. For the first three months, you'll keep wearing your aligners twenty-two hours a day, just as you have until now. Over the next three months, you'll reduce your wear time by one hour each week until you reach ten hours. For the last three months, you'll wear the final aligner only at night.*
>
> *So every three months, we'll take another step: first we stabilize, then we gradually reduce wear time, and finally we give the occlusion more room to settle. After nine months, we'll take a new scan and, if everything has stayed in the positions we achieved, we'll move on to definitive retention. We've done the hard work. Now we need to make sure we don't lose what we've gained."*

In this way, I am not simply asking the patient to keep wearing aligners; I am giving a clear purpose to the behavior I am asking of them. Simply saying *"you'll need to continue for another nine months"* could make stabilization seem, to the patient, like an inexplicable extension of a treatment they believe is already over. The message I want to convey is different: **we have achieved the result, but it still needs to be stabilized**.

As with the other protocols, I can tailor the message here using the three motivators of **outcome, time, and investment**, without changing what I am asking the patient to do. At this stage, the outcome is probably the most natural motivator: *"We've achieved the result. Now we need to protect it while it becomes stable."* For some patients, I can emphasize the time they have already devoted to treatment: *"We've worked for months to bring your teeth into these positions. It wouldn't make sense to risk losing some of what we've achieved in just a few weeks."* For others, I can bring up their overall investment—not just the money, but also their consistency and effort: *"You've invested a lot to get this far. These nine months are about protecting that investment."* The clinical implications do not change. What changes is what I choose to emphasize for the person in front of me.

## Why This Protocol Works

### It Turns the Result into Something to Protect

During stabilization, the result is no longer something to achieve; it is something the patient already has. That brings *loss aversion* and the *endowment effect* into play: **we are no longer asking patients to cooperate to gain something, but to avoid losing what they have gained.**

### It Reframes the Nine Months

The *framing* changes the meaning of the same request. **"Another nine months of treatment"** can sound like an extension of therapy; **"nine months to protect the result"** instead makes that period a phase with a clear purpose.

### It Maintains Continuity in the Patient's Routine

For the first three months, patients continue the same routine they followed during the active phase. Only after that do they reduce wear time, by one hour a week. The change is therefore not sudden or left to vague instructions, but **gradual, predictable, and measurable**.

### It Breaks a Long Period into Understandable Goals

Nine months can seem long. Three consecutive three-month phases, on the other hand, each have an immediately recognizable goal: **stabilize, taper, allow the occlusion to settle**. Patients always know where they are in the process and what comes next.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

With a Dominance-style patient, I prefer to be direct, focusing on the result and on control:

> *"We've achieved the result. Now we need to make sure we don't lose any of it. For nine months, we'll follow the 3-3-3 protocol, and then we'll move on to definitive retention."*

**I**

### Influence

With an Influence-style patient, I can place more emphasis on what they see and the result they have just achieved:

> *"Look how far we've come. You have the smile we wanted. These nine months will protect it while it becomes stable. It would be a real shame to stop now."*

**S**

### Steadiness

With a Steadiness-style patient, I emphasize continuity and gradual change:

> *"Nothing changes all at once. For the first three months, you'll continue exactly as you have until now. Then we'll start to reduce your wear time gradually. Each step will ease you toward wearing the aligner only at night."*

**C**

### Conscientiousness

With a Conscientiousness-style patient, I can offer a more rational, biological explanation:

> *"The tooth movements are complete, but the tissues need more time to reorganize. That's why we have a nine-month phase during which we maintain full-time wear at first, then gradually reduce the hours before moving to definitive retention."*

CHAPTER 13

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | After the active treatment goals have been achieved, in patients who are clinically suitable and sufficiently adherent. This is the author's protocol, not a universally validated regimen. |
| **GOAL** | Protect the result and make the transition from active-phase wear time to nighttime retention gradual. |
| **WHAT TO DO** | 1. **Stabilize — first three months.** Aligners 1–12, changed weekly, twenty-two hours a day: maintain the position achieved.<br>2. **Taper — second three months.** Aligners 13–24, changed weekly: reduce wear time from twenty-one to ten hours a day, by one hour each week.<br>3. **Allow the occlusion to settle — final three months.** Wear the same aligner, number 24, for ten hours each night, maintaining nighttime protection while allowing more room for normal function.<br>4. **Move on to retention.** After nine months, take a new scan and prepare the definitive retainers.<br>The plan includes twenty-four aligners and three three-month phases. If another nine months of aligner wear is not realistically sustainable, conclude the active phase and move directly to definitive retention. |
| **WHAT TO CHECK** | Clinical suitability, adherence, appliance fit, maintenance of tooth positions, and the course of occlusal settling. The reduction in wear time should follow the agreed-upon schedule. |

CHAPTER

# 14

# Delivering the Retainers

*"An ounce of prevention is worth a pound of cure."*

— BENJAMIN FRANKLIN

Delivering the definitive retainers marks the beginning of long-term maintenance, and it can happen at two different points. For suitable, adherent patients, it follows the nine months of the stabilization protocol; for patients for whom that phase is not indicated, it comes directly at the end of active treatment. In either case, it is a less dramatic milestone than removing the attachments or handing over the final aligner, but it is just as important. For months, sometimes years, we have worked together to achieve a result. We have asked the patient to wear their aligners, change them on schedule, wear orthodontic elastics, come to appointments, and follow our instructions, until at last the day they have been waiting for since the beginning arrives—the day we say something that sounds harmless:

*"We're done."*

To the patient, those two words may mean much more than we realize. They may hear them as the end of rules and schedules, as the moment aligners become a thing of the past for good, and, above all, as confirmation that the result is now secure and can no longer be lost.

That is why I try never to present the delivery of retainers as the end of treatment. Instead, I prefer to explain clearly that only one phase of the process is ending:

> *"Today, the phase before definitive retention comes to an end: for you, that was the nine-month stabilization phase"—or, when the protocol was not indicated, "the active phase of treatment." "From this point on, long-term maintenance begins. Preserving the result will depend mainly on you and how consistently you wear your retainers."*

The difference may seem small, but it completely changes the meaning of what is happening, because from that day forward, the day-to-day responsibility for preserving the result gradually passes into the patient's hands.

Precisely because I consider this transfer of responsibility a fundamental part of treatment, when I deliver the retainers, I have the patient read and sign a retention-specific informed consent form. This is not simply an instruction included in the initial consent form for orthodontic treatment, or a recommendation given hastily at the end of the final appointment. It is a document specific to this new phase, which we give the patient and have them sign when we deliver the definitive retainers.

The form explains why retention is necessary, how the retainers should be worn, what to do if they are lost or broken, and the possible consequences of not wearing them enough. Above all, it formalizes the transfer of responsibility I have just explained to the patient, which I sum up in a simple sentence:

"Getting your teeth into the right positions was our job. Keeping them there will be yours."

For me, then, the signature is not just a formality or a means of legal protection. It is, above all, the conclusion of a conversation in which I first explain what changes for the patient from that day forward, then give them the information they need to manage this new phase properly, and finally ask them to confirm that they understand it. This is not a way to assign blame in advance if something goes wrong. It makes their responsibility clear while we are still in the best position to do so: **before a problem arises.**

Before the appointment ends, I check that the retainer seats fully and take several intraoral photographs with it in place. I keep them with the final scan to establish a documented starting point. If the retainer no longer fits properly in the future, we will not have to rely on conflicting recollections. We can see how it fit when it was delivered, determine what has changed, and decide what solution is clinically appropriate. The photographs are not meant to prove that the patient failed to wear the retainer, but to make our assessment of the current situation more objective.

All of this becomes even more important when we consider that orthodontic stability cannot be taken for granted once treatment is complete. There is a tendency toward relapse after treatment, and, independently of that, the dentition continues to change throughout life. Retention must therefore be considered an integral part of treatment, and the retention protocol must be tailored to the clinical situation, the tooth movements performed, and the individual patient's characteristics.

The literature offers no single retention protocol that is universally valid for all patients, but there is growing recognition of the need for very long-term maintenance. In one survey of specialist orthodontists, for example, 67–78% of respondents advised their patients to continue retention for life.

My protocol follows the same philosophy, so I consider retention a permanent phase of treatment. That does not mean, of course, that the same retainer must last a lifetime. It may wear out, become distorted, break, or get lost, and it will need to be replaced when necessary. What does not change is the need to keep protecting the result over time.

**The retainer has a lifespan. Retention does not.**

## Ten Hours, Every Night

When I give patients instructions, especially ones I am asking them to follow for life, I prefer to explain why. In my protocol, thermoplastic retainers must be worn **every night for at least ten hours**, roughly from 9 p.m., after dinner, to 7 a.m., before breakfast. Here, too, the number is not based simply on clinical habit; it has some interesting support in the literature.

İshakoğlu and Çokakoğlu studied 42 patients wearing thermoplastic retainers equipped with microsensors. This allowed them to measure actual wear time objectively rather than relying solely on what patients reported. After twelve months, the patients were divided into two groups based on their average daily wear time: less than nine hours, or nine hours or more. Greater loss of alignment was observed in patients who wore their retainers for less than nine hours a day. The authors there-

fore concluded that an average wear time of less than nine hours was not enough to keep irregularity within clinically acceptable limits, and they recommended at least nine hours a day to maintain anterior alignment.

It is important to interpret this finding correctly. It does not mean there is an absolute biological threshold at which teeth move after eight hours and fifty-nine minutes of wear but stay put after nine hours. The study also involved a specific sample of patients treated with four premolar extractions, and the statistically significant difference between the groups concerned lower anterior alignment. Still, it gives us clinically useful information, suggesting that **an average wear time of less than nine hours may not be sufficient to maintain adequate lower anterior alignment**.

It is on the basis of this finding that I prescribe ten hours in my protocol—not because any study has shown it to be a universal biological threshold, but because I think it is reasonable to add a small margin to the nine hours indicated by the study while giving patients a simple, concrete instruction that is easy to remember: **ten hours, every night, for life**.

That does not mean that, if ten hours is sufficient, wearing a retainer for twenty or twenty-two hours must necessarily provide even greater stability. The available evidence does not show a significant advantage for full-time wear over adequate part-time wear. In a randomized clinical trial, Thickett and Power compared full-time and part-time wear of thermoplastic retainers and, after twelve months, found no statistically significant differences between the two protocols for most of the measures they examined. Reviews of the literature point in the same direction, indicating that part-time wear of removable retainers may be sufficient, while emphasizing that the overall quality of the evidence remains limited.

The message, then, is not that ten hours is the perfect dose for every patient and every clinical situation. Retention remains an area in which there is no single universally accepted protocol. What I want patients to understand is much simpler: once active treatment is over, **I am not asking them to keep wearing an appliance twenty-two hours a day for the rest of their lives, as they did with their aligners**. I am asking them to make retention a sustainable nighttime routine.

But telling a patient *"wear it at night"* still leaves too much room for interpretation. For some people, a night lasts nine hours; for others, six. Some put their retainers in when they go to bed, others only just before they fall asleep, while still others may think nothing of skipping a night because, after all, they were simply told to wear them *at night*. That is why I prefer to turn a general instruction into a precise routine: **9 p.m.–7 a.m.** The retainers go in at nine in the evening, after dinner, and

come out at seven in the morning, before breakfast, completing the prescribed ten hours. Of course, I do not attach any biological significance to nine in the evening or seven in the morning. The benefit is mainly behavioral: tying an action to familiar, recurring points in the day makes it easier to turn into a habit.

This becomes particularly important when we consider that adherence to retention tends to decline over time. Kacer and colleagues examined retainer wear during the first two years after active treatment. They found that, while 69% of patients reported wearing their retainers every night during the first three months, between months 19 and 24 the percentage classified as adherent according to the study's criteria had fallen to 45%, and 19% reported no longer wearing them at all. These were self-reported data collected through a questionnaire, not an objective measurement of behavior. But the trend they show is one we readily recognize in clinical practice: **adherence tends to decline over time**.

The real challenge, then, is not convincing patients to wear their retainers the night we deliver them, when they can see the result, the memory of treatment is still fresh, and the value of what they have achieved is unmistakable. We need to build a habit that will last when that smile has become normal, when treatment is a distant memory, and when we are no longer there every few weeks to remind them what to do. **That is why I prefer a precise routine to a general recommendation.**

### You Can't Make Up for Missed Hours

There is another message I think it is essential to give patients on the day we deliver their retainers. They must not come away thinking of the ten-hour prescription as a kind of bank account: if they skip a night, they can simply wear their retainers longer the next day and make up for the missed hours. The problem is not the hours themselves, but what may happen to their teeth while the retainers are out.

If one or more teeth start to shift, the position of the teeth no longer matches the shape the retainer was made to fit. When the patient tries to put it back in, it may feel tighter, or they may notice that it no longer fits the same way. That is precisely when an apparently logical but potentially dangerous thought can arise: *"It feels tight, so I'll wear it a little longer and it'll move my teeth back into place."*

I do not want patients to wear their retainers on their own as if they were new aligners, because those retainers were made **to hold the teeth in position**, not to carry out a new, planned orthodontic movement. If the teeth no longer match the positions the retainer was made for, it may begin to exert forces that were not planned as part of treatment. That is why I introduce a very simple rule when I deliver it:

> *"If you don't wear it for a few days and it feels very tight when you put it back in, don't think you can make up for the missed time simply by wearing it longer. This retainer was made to keep your teeth in the positions they're in today, not to make new orthodontic movements. If it no longer fits properly or you have to force it in, call us."*

There is no need to explain all the biomechanics that can come into play when a retainer no longer matches the positions of a patient's teeth. It is much more important to keep patients from forming the wrong idea and to help them understand, from the day we deliver it, that **missed hours cannot be made up: what we need to prevent is a loss of tooth position**.

## When a Retainer Is Lost or Broken

Then there is a seemingly minor problem I have faced many times over the years: a patient loses their retainer, calls the office, and simply asks, *"Can you make me another one?"* Nothing unusual so far; the problem can arise when they learn that making a new retainer costs money.

To understand their reaction, we need to look back at how we educated them during active treatment. A patient treated with aligners may have received dozens of trays, lost or broken a few, and gone through one or more refinements without directly seeing the cost of each appliance, especially when the package they purchased included additional aligners. Over time, they may have formed a very simple association: **tray = included in treatment**.

When we give them a clear retainer, to them it looks extremely similar to what they have been wearing up to that point. To us, its function has changed completely, but to the patient it is still a tray. If we do not explicitly change that mental model, we cannot be surprised when they lose or break it and assume it, too, should be replaced for free.

Once again, the best time to address the issue is not when the patient calls to say they have lost it, but **before they lose it**. The retention-specific informed consent form I have patients sign clearly states that replacing a lost or broken retainer will cost money. But I do not want that cost to feel like a punishment or a penalty for being careless: the patient is simply paying for the production of **a new appliance**.

During treatment, they purchased a course of care that included certain services and appliances, but that phase is over. It cannot become an implicit promise to produce replacement retainers for free for the rest of their lives. If patients are to wear retainers over the long term, we need to explain from the outset that individual

appliances may wear out, break, or get lost, and that producing a new retainer when one needs to be replaced will cost money. This, too, is part of the transfer of responsibility we began when we delivered the retainer.

**PRACTICAL SCRIPT**

All these points must be scientifically sound, but they must not turn the final appointment into a college lecture. Patients should leave the office remembering a few things that are perfectly clear, so the conversation can be much simpler than the reasoning behind it.

> *"Today, the phase before definitive retention comes to an end: for you, that was the nine-month stabilization phase"—or, when that protocol was not indicated, "the active phase of treatment." "From this point on, long-term maintenance begins, and keeping the result will depend mainly on you and how consistently you wear your retainer. This phase lasts for life; I always joke that retention is like diamonds: it's forever."*

At that point, I give the patient a concrete instruction, trying to leave them with a rule they can remember without any room for interpretation:

> *"The rule is very simple: ten hours, every night. The easiest thing is to put your retainer in at nine in the evening, after dinner, and take it out at seven in the morning, before breakfast. Nine to seven: that's ten hours. You don't need to wear it all day; you need to wear it consistently every night."*

Right afterward, I explain what might happen if they do not stick to that routine, because I want patients to know what to do before they find themselves in that situation:

> *"And one more thing: you can't make up missed hours. If you skip a few nights and your retainer feels very tight when you put it back in, or it no longer fits the way it used to, don't think, 'I'll wear it longer and everything will go back to normal.' This retainer was made to keep your teeth where they are today, not to carry out a new course of treatment. If it doesn't fit properly or you have to force it in, call us right away."*

At the same time, I explain what patients should do if a retainer is lost or broken and, above all, prevent any future misunderstanding about the cost of replacing it:

> *"If you lose your retainer or it breaks, call us right away. Don't wait a few days thinking nothing will happen, because from that point on, your teeth are no longer being held in the positions we achieved. Remember, too, that if we have to make a new one, it will cost money. Active treatment is over, and from today on, managing your retention is part of your responsibility for maintaining the result."*

Finally, I return to why I am giving them all these instructions. I do not want patients to leave remembering only a set of rules; more importantly, I want them to understand what those rules are protecting:

> *"The most important thing is this: today we're giving you a result we achieved together. From this point on, your job is to preserve it. If you stop wearing your retainers and your teeth shift, simply making a new retainer may not be enough. If you want to regain the result we achieved, you may need another course of orthodontic treatment, with a new cost."*

Only after this explanation do I have the patient read and sign the retention-specific informed consent form. At that point, the signature is not the beginning of the conversation but its conclusion. The patient has seen the result and understands that the clinician-guided phase has ended but maintenance continues. They know their responsibilities, how many hours a day and how long they need to wear their retainer, what to do if it is lost, breaks, or no longer fits properly, and the possible consequences of not following the instructions. We have addressed all these possibilities when we were still in the best position to do so: **before they became a problem.**

## Why This Protocol Works

**It changes what retention means.** Telling patients that treatment is not over but a new phase is beginning changes how they see retention. It is not presented as an accessory to wear whenever they remember, but as the means by which they will protect, over time, a result they already have and have invested time, energy, and money in achieving. This brings together **a shift in framing, the salience of the result, and loss aversion**.

**It turns a recommendation into a specific behavior.** *"Ten hours, every night, 9 p.m.–7 a.m."* leaves much less room for interpretation than *"wear it at night."* Tying putting the retainer in and taking it out to two recognizable points in the day also helps turn the instruction into a consistent routine. This brings together **reduced ambiguity and implementation intention**, concepts we have already explained in earlier chapters and do not need to revisit here.

**It transfers responsibility before a problem arises.** From the day the retainer is delivered, patients know what to do if they skip a few nights, if the retainer feels tight, no longer fits properly, is lost, or breaks. They also know that any treatment needed to correct a relapse would be a new course of treatment. Establishing these rules while the result is stable and there is no conflict yet allows us to address future problems on the basis of responsibilities we have already agreed on.

**It makes the commitment explicit.** The retention-specific informed consent form is signed only after the instructions, responsibilities, possible problems, and costs have been explained. The signature does not replace the conversation; it concludes and formalizes what we have discussed, including the fact that replacing a lost or broken retainer means paying for a new appliance, not a penalty.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

With a Dominance-style patient, the message can be direct, brief, and focused on control over the result: *"We achieved the result we wanted. From today on, you're in control of keeping it that way: ten hours, every night."*

**I**

### Influence

With an Influence-style patient, it may be more effective to emphasize their smile and the esthetic result achieved: *"Look at the smile we've achieved. Now we need to keep it that way: ten hours, every night."*

**S**

### Steadiness

With a Steadiness-style patient, it may help to emphasize how simple the routine is: *"You don't have to do anything complicated: put your retainer in after dinner and take it out in the morning before breakfast. The important thing is to make it something you do every evening."*

**C**

### Conscientiousness

With a Conscientiousness-style patient, a precise instruction and the scientific explanation behind it work particularly well: *"The instruction is to wear it for ten hours every night. That number wasn't chosen arbitrarily: it comes from the data we have on maintaining alignment after treatment."*

CHAPTER 14

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When delivering the definitive retainer, either after the stabilization phase or directly after the active phase ends. |
| **GOAL** | Establish a precise maintenance routine, clarifying responsibilities, how to handle problems, and replacement costs. |
| **WHAT TO DO** | 1. Check the fit, photograph the retainer in place, and save the final intraoral scan.<br>2. Explain that the active phase is ending and maintenance of the result is beginning.<br>3. Under the protocol described here, prescribe ten hours every night, for life. As a practical guide: 9 p.m.–7 a.m.<br>4. Make clear that missed hours cannot be made up simply by wearing the retainer longer later on without consulting us.<br>5. Ask the patient to contact the office if the retainer feels very tight, does not fit properly, or has to be forced into place.<br>6. Ask the patient to report a lost or broken retainer promptly, and explain the cost of a new appliance.<br>7. Explain in advance that, if the teeth shift, a new retainer may not be enough and another course of treatment may be needed.<br>8. After the explanation, have the patient read and sign the retention-specific informed consent form. |
| **WHAT TO CHECK** | Fit of the appliance and the patient's understanding of the instructions, duration, responsibilities, how to handle problems, and costs. |

CHAPTER

# 15

# A Virtual Appointment Every Week

*"When you can measure what you are speaking about, and express it in numbers, you know something about it."*

— LORD KELVIN

During clear-aligner treatment, a lot can happen between office visits. Patients change trays on their own, wear them according to their own habits, and often have to judge for themselves whether they still fit their teeth properly. If an aligner begins to lose tracking—that is, it no longer fits the tooth surfaces precisely—the patient may not notice, or may decide to move on to the next aligner anyway, simply because the prescribed number of days has passed. Even a reduction in daily wear time may seem to have no consequences at first, especially when the patient is convinced they can make up for it by wearing their aligners more in the days just before their appointment.

But if we do not see the patient again for four or six months, that small loss of tracking may have carried through the entire sequence. At that point, we no longer have to correct only the first tooth movement that failed to occur, but also the effects of the subsequent aligners, which were designed on the assumption that all the teeth had reached their planned positions.

It is precisely to shorten this period without observation that I use **DentalMonitoring** in my practice, a system that makes it possible to follow the progress of orthodontic treatment remotely. Using an app installed on their smartphone and a special device that helps hold their lips and cheeks away from their teeth, patients capture a guided series of images of their mouths.

In everyday language, we call this a *scan*, although it is not the same as the three-dimensional intraoral scan we perform in the office to capture the arches and plan the aligners. The DentalMonitoring scan is, more precisely, a guided series of photographs captured by patients with their own phones. It allows the system and the clinical team to assess aligner fit and treatment progress remotely.

Under my protocol, patients take these smartphone scans once a week, before moving on to the next aligner. The images let us assess how well the aligners fit and whether the teeth are maintaining tracking—that is, whether their actual positions remain consistent with those planned in the treatment sequence. **Based on what we see, patients are instructed either to move on to the next aligner or to keep wearing the current one for a while longer.**

Changing aligners weekly is therefore no longer just a date on the calendar. Patients do not automatically move on to the next aligner simply because seven days have passed; they wait for the remote check to confirm that the conditions for moving forward are in place.

DentalMonitoring does not replace an orthodontist's assessment, and it cannot tell us everything happening in a patient's mouth. It does, however, substantially shorten the time between the onset of a problem and our ability to detect it. An early loss of tracking can be addressed while it is still limited, before it affects more teeth and compromises a larger part of the sequence.

DentalMonitoring is, first and foremost, a clinical tool that lets us follow treatment more consistently even when in-person appointments are spaced farther apart. As I used it, though, I realized that it also serves a second purpose, one particularly important in the context of this book: **it makes some aspects of patient adherence more visible.**

When I give patients their aligners, I explain that remote monitoring will allow us, in many cases, to schedule office visits every four to six months, adjusting the

frequency to the clinical needs of each case, of course. That does not mean treatment goes unchecked during that time: between appointments, we receive weekly images and can follow its progress consistently.

Each smartphone scan thus becomes a kind of brief virtual appointment. Patients have to remember to take it, follow the procedure correctly, and wait for instructions through the system before changing aligners. Traditional appointments are therefore supplemented by shorter but much more frequent checks, so fewer office visits do not mean less responsibility.

The weekly scans let me assess aligner fit, but also how reliably patients follow the protocol. How regularly they submit images, whether they follow the instructions they receive, and whether they respond to messages become behaviors we can document; likewise, their absence tells us something too.

More than once, I have given a patient the system and watched their scans, after an initial period of regular use, grow progressively less frequent until they stopped altogether. At the next appointment, the aligners no longer fit properly, yet the explanation was almost always the same:

*"I wore them all the time."*

In the past, a statement like that could have started an argument that went nowhere. The patient would have continued to insist they had followed the treatment plan, and all I would have had to cast doubt on their account was the poor fit of the aligners. With remote monitoring, I can take a completely different approach, because **I do not have to choose immediately between believing the patient and accusing them of being dishonest.**

## Pinpointing When Tracking Was Lost

When a patient says they have worn their aligners as instructed, I do not open their DentalMonitoring scan history by saying, *"Now let's see if that's true."* A remark like that would immediately turn the conversation into a trial, making the patient feel accused and pushing them to defend their version of events. I prefer to present the review of the scans as a clinical investigation we undertake together, looking for the period when treatment began to diverge from the plan:

*"Let's look together at all the scans you've taken over the past few weeks. That way, we can identify the week when tracking started to worsen and try to understand what might have happened."*

I often suggest a specific possibility, too, because I want patients to understand that I am not looking for a confession, but for an actual event that might help us interpret what we are seeing:

> *"Maybe we'll find it happened during the week of your best friend's wedding. Maybe that day you went without your aligners for many hours, and from then on, one tooth stopped tracking perfectly. Let's see if we can piece together what happened."*

The wedding example gives the patient a way out of the standoff between *"I wore them all the time"* and *"I don't believe you."* A wedding, a party, a day of travel, a particularly long dinner, an illness, or a very difficult week may be easier to remember than the total number of hours the aligners were worn. A general question such as *"Did you wear your aligners?"* almost inevitably produces an equally general answer: *"Yes, all the time."* But when I tie the question to a specific moment, the patient may realize, for example, that they took their aligners out during the wedding lunch and did not put them back in until late that evening. We are no longer discussing adherence in the abstract; we are reconstructing what they did at a recognizable moment.

I then open the DentalMonitoring history and review the preceding weeks with the patient. In some cases, we can pinpoint the period when tracking began to worsen and link it to a particular event, a reduction in wear time, insufficient use of aligner chewies, or a tooth movement that failed to occur despite apparently adequate adherence. At other times, though, we discover that the smartphone scans needed to reconstruct what happened simply do not exist: the history is empty or has obvious gaps. When the patient responds, *"That's impossible. I did every scan,"* I do not need to accuse them directly, because the missing data show that the agreed-upon protocol was not followed.

That does not necessarily mean the patient failed to wear their aligners properly. It does mean, however, that they interrupted the agreed-upon channel of supervision. Even if they wore the aligners as instructed, without up-to-date data we could not check their fit, assess tracking, or confirm that the conditions for moving forward with the sequence were in place. At that point, I explain:

> *"That's exactly the problem. If we had a scan for every week, we could pinpoint when tracking started to worsen and try to understand what happened. Because some scans weren't taken, we lost the chance to spot the problem and intervene promptly. Now we need to assess how much tracking has been lost and determine whether we can still continue with these aligners or whether we'll need to change*

*the treatment plan. Either way, from this point on, you'll need to resume both wearing your aligners as instructed and taking weekly scans. Otherwise, we risk ending up in the same situation again."*

I cannot know for certain whether the patient is lying intentionally or has reconstructed their behavior in a way they find more acceptable. They may remember the weeks when they used the system properly and forget the weeks when they did not, or they may confuse an intention to take a scan with a memory of actually having taken one. For clinical management, however, the result is the same: the agreed-upon protocol was not followed, and we could not promptly identify when tracking began to worsen.

Remote monitoring, then, does not prevent a patient from lying, nor does it eliminate the ways memory reconstructs past behavior. But it reduces the room for the patient's account to become the only version available.

There are also small details that, while not actual measures of adherence, can tell us something about a patient's habits. One of the most interesting involves orthodontic elastics. When I prescribe them, I do not ask patients to keep them in during smartphone scans at home; in fact, removing them would be much easier, because they limit how wide patients can open their mouths and can make the process more uncomfortable. The most logical thing to do, then, would be to remove the orthodontic elastics, take the scan, and put them back in immediately afterward.

Some patients, however, take the scan with their orthodontic elastics still in place —not because they want to prove to me that they are following instructions, or because they were told to do so, but simply because they forget to remove them. The orthodontic elastics have become such a habitual part of their day that they start the process without even thinking about taking them out.

To me, this behavior is a very meaningful sign, although, of course, a scan taken with orthodontic elastics in place does not prove that the patient wore them for all the prescribed hours: at least in theory, they could have put them in just before the scan. But when I see someone spontaneously accept the added difficulty of taking a scan with orthodontic elastics in place, I consider that behavior a credible indication that wearing them really is part of their daily routine. This is not proof on its own, but a spontaneous sign that carries weight precisely because it does not appear to have been staged for the doctor.

DentalMonitoring lets me observe not only tooth movement, but also glimpses of the patient's day-to-day experience with treatment. A scan taken on time, a prompt response to an instruction, or an orthodontic elastic left in place during a scan

can all be small indicators of patient adherence. **They are indicators, though, not measurements**, and it is important to be very clear about that limitation: DentalMonitoring does not have a timer inside the aligner and cannot tell us exactly how many hours a day it was worn. It cannot, therefore, tell us that a patient wore an aligner for eighteen hours rather than twenty-two.

What we can consistently observe is aligner fit, treatment progress, and any emerging loss of tracking. If the aligners are worn less than necessary, some effects of that inconsistent wear may become visible over time, even though what we see does not allow us to calculate wear time mathematically.

When talking with patients, I use this capability to encourage them to take responsibility, because I want them to know from the start that what happens between appointments will not remain completely invisible. At the same time, I avoid claiming the system is more precise than it is, and I use wording that clearly distinguishes between directly measuring wear time and observing its possible clinical consequences:

> *"The system doesn't have a timer, so it can't tell us exactly how many hours you've worn your aligner. But the weekly smartphone scans let us keep track of fit and treatment progress. If the aligners are worn less than necessary, the effects of that inconsistent wear often become visible over time."*

This distinction also matters because a tooth's failure to move cannot automatically be attributed to poor adherence. There may be biomechanical difficulties, a problem with an attachment, anatomical factors, or a biological response different from the one expected. Remote monitoring should therefore help us start a conversation and develop clinical hypotheses, not deliver an automatic verdict against the patient. **Its real value lies not in proving that the patient did something wrong, but in helping us recognize sooner when something is not working as planned and, where possible, understand why.**

## The Protocol During Active Treatment

At my practice, the protocol calls for patients to take a smartphone scan once a week, before changing aligners, and to wait for instructions before moving on to the next one. When fit is inadequate, they may need to keep wearing the same aligner for a few more days, increase their wear time, or use aligner chewies more often.

If the problem does not resolve, we can move up the office visit and assess whether it is still appropriate to continue with the sequence or whether we need to stop, take a new three-dimensional intraoral scan, and plan a refinement. In-

person appointments are generally scheduled every four to six months, although the frequency is always adjusted to the needs of each case.

The same principle applies when a patient submits a smartphone scan but the images are not clear enough for a reliable assessment. If it happens occasionally, we go through the process together again and check for any technical difficulties. If the problem persists, the patient needs to be seen in the office. When remote monitoring is not workable, we schedule more frequent clinical visits and may give the patient fewer aligners at a time: **patients cannot continue changing aligners on their own without a reliable channel of supervision.**

That way, the weekly aligner change is no longer just a date on the calendar. The patient's instinctive thought, *"A week has passed, so I can change aligners,"* is replaced by a more accurate principle: *"A week has passed, and I can change aligners if the clinical conditions are right for moving forward."*

## Monitoring During Retention

The same principle can continue after active treatment ends. As we saw in the previous chapter, giving a patient a retainer means gradually entrusting them with the responsibility of protecting the result they have achieved, but that does not necessarily mean they must be left entirely on their own from then on. At my practice, I offer the option of continuing monitoring during retention through an optional monitoring and guarantee program that combines remote checks with a form of guarantee for maintaining the result.

The program costs **500 euros a year** and can be continued for up to three years. The treatment packages I use allow additional aligners to be produced within five years of the start of treatment, and active treatment often lasts a year and a half or two years. That frequently leaves a period of about three years during which we can continue to follow the patient and, when the specified conditions are met, intervene if relapse occurs without a new aligner production cost, provided that production is still covered by the package.

The word *guarantee*, however, must be used carefully, because it could suggest that the practice will automatically correct any future tooth movement, regardless of what the patient does. That is not what the program means. The guarantee exists within an agreement in which the patient also makes specific commitments: they must wear their retainers as prescribed, regularly submit the required smartphone scans through DentalMonitoring, attend scheduled checkups, and promptly tell us if a retainer is lost, breaks, or develops any fit problem.

During this phase, I require a DentalMonitoring smartphone scan once a month and include a professional dental cleaning every six months in the program. At that visit, the hygienists also take a three-dimensional intraoral scan, which we can compare with the one taken at the end of treatment to check the stability of tooth positions over time. The patient is not simply buying a promise that, if something moves, it will be corrected for free. They are entering a system of supervision in which prevention, checkups, and personal responsibility continue to work together.

If the patient has met the agreed-upon conditions and relapse nevertheless occurs that requires further orthodontic treatment, we can intervene without charging a new treatment fee, within the time and clinical limits set by the program, of course. If, instead, the retainers have not been worn as prescribed, the monthly scans have stopped, checkups have been missed, or a problem has been ignored for months before being reported, the basis on which that guarantee was built no longer exists.

This matters because even a guarantee program can become counterproductive if patients see it as shifting all responsibility to the practice. The message should not be *"Whatever happens, we'll take care of it,"* but rather *"If you keep doing your part, we'll keep doing ours."* The guarantee thus becomes a shared responsibility: the patient protects the result by wearing their retainers, submitting scans, and attending the agreed-upon checkups, while we maintain a system that allows us to spot any change early and intervene when necessary.

I also prefer to present the program this way when discussing its cost. The **500 euros a year** are not simply the price of a possible new treatment; they cover ongoing prevention and supervision during retention, as well as dental cleanings.

DentalMonitoring thus plays a role in two very different phases of treatment. During active treatment, it allows us to shorten the time a loss of tracking can go unnoticed; during retention, it can help us recognize a change in the result early. In both cases, though, the principle remains the same: **monitoring works only if someone is watching and someone continues to provide the information needed for that monitoring.**

## PRACTICAL SCRIPT

DentalMonitoring comes up in conversations with patients at different points in treatment, so the script needs to change with the situation. When we give patients their aligners, its main purpose is to establish the rules of the system; when tracking is lost, it should help us reconstruct what happened without turning the

conversation into an accusation; if scans have not been submitted, it should make the consequences of the lack of supervision clear; and during retention, it should clarify that the monitoring and guarantee program works only when responsibility is shared.

#### When Giving Patients Their Aligners

> *"Before each aligner change, you'll need to take a smartphone scan and wait for instructions. If everything is progressing as it should, you can move on to the next aligner. If a tooth needs more time, we may ask you to keep wearing the current one and give you specific instructions.*
>
> *The system doesn't have a timer, so it can't tell us exactly how many hours you've worn your aligners. But the weekly smartphone scans let us keep track of fit and treatment progress. If the aligners are worn less than necessary, the effects of that inconsistent wear often become visible over time.*
>
> *You'll have fewer appointments at the office, but you'll have a very specific responsibility: the weekly smartphone scan is just as much a part of treatment as wearing your aligners."*

#### When Tracking Has Been Lost

> *"Let's look at the scans from the past few weeks together. That way, we can pinpoint when tracking started to worsen and try to understand what might have happened. We may find that there was a wedding, a trip, or something else that week that kept your aligners out of your mouth longer than planned."*

#### When Scans Have Not Been Submitted

> *"If we had a scan for every week, we could pinpoint when fit started to worsen and try to understand what happened. Because some scans were not submitted, we lost the chance to notice the problem and intervene promptly.*
>
> *Now we need to assess how much tracking has been lost and determine whether we can continue with these aligners or need to change the treatment plan. Either way, from this point on, we'll need to get back to both wearing the aligners correctly and submitting weekly scans. Otherwise, we could find ourselves in the same situation again."*

**When Offering the Program During Retention**

> *"From today on, maintaining your result will depend primarily on wearing your retainers. You can manage this phase on your own or enroll in our monitoring and guarantee program for the next three years.*
>
> *At my practice, the program costs 500 euros a year and includes two dental cleanings, one every six months, monitoring through a monthly smartphone scan taken at home, and periodic comparisons between your current tooth positions and those at the end of treatment.*
>
> *If you follow the protocol and we nevertheless detect tooth movement, we'll promptly assess how to address it. When the program's conditions are met, you won't be charged for a new course of orthodontic treatment.*
>
> *To keep the program active, you'll need to wear your retainers as directed, submit the monthly smartphone scan, and attend the two scheduled dental cleanings. It isn't just financial coverage; it's a system that lets us spot quickly if something is changing."*

## Why This Protocol Works

- **It Makes What Happens Between Checkups Visible**

The weekly smartphone scan turns the long interval between checkups into a series of brief digital check-ins, so patients know their treatment progress can be observed regularly. At the same time, timely feedback creates a clearer link between behavior and its possible clinical consequences: if fit starts to worsen, we can intervene while the problem is still limited, rather than discovering it many months later.

- **It Lets Us Reconstruct Events Instead of Assigning Blame**

When tracking is lost or scheduled scans have not been submitted, there is no need to turn the conversation into a contest between the patient's account and the doctor's. The available scan history lets us reconstruct what happened together. Connecting the change to specific events, such as a wedding, a trip, or an unusual week, also helps bring back more specific memories than a general question about wear time would. This way, patients can recognize that they may not have followed the plan as closely as intended without immediately feeling accused of lying.

- **It Keeps the System's Limitations Clear**

DentalMonitoring does not directly measure wear time, and a tooth's failure to move does not automatically prove poor adherence. Being upfront about these limitations makes monitoring more credible and prevents it from becoming a tool for proving

that the patient did something wrong. Instead, it preserves the system's most useful function: spotting a deviation early and helping us understand its possible causes.

### It Makes Monitoring a Shared Responsibility

Monitoring works only if both sides fulfill their roles: the practice must review the information it receives and intervene when necessary, while the patient must submit scans, follow instructions, and keep the channel of supervision open. This principle becomes even clearer during retention, when the guarantee does not eliminate the patient's responsibility but makes it part of a mutual, verifiable agreement.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

With a Dominance-style patient, it helps to present DentalMonitoring as a tool for quickly identifying a problem, making a decision, and reducing the risk of delays:

> *"With weekly smartphone scans, we can quickly tell whether treatment is progressing as it should and intervene before a small problem delays your result."*

**I**

### Influence

With an Influence-style patient, it helps to emphasize how easy the system is to use, the contact with the team, and the opportunity to receive frequent reassurance without having to keep returning to the office:

> *"You'll send us smartphone scans every week, and we'll stay in touch through the app, so you'll know whether everything is going well without coming into the office each time."*

**S**

### Steadiness

With a Steadiness-style patient, it is important to emphasize the continuity of monitoring and the team's ongoing presence, so that longer intervals between appointments do not feel like abandonment:

> *"Even though your appointments will be farther apart, you won't be left on your own. We'll review your smartphone scans regularly and guide you through every step."*

**C**

### Conscientiousness

With a Conscientiousness-style patient, it is best to provide a precise protocol, clearly defining the frequency, procedure, criteria for changing aligners, and limitations of the system:

> *"You'll need to take a smartphone scan once a week, before changing aligners. Based on the fit, you'll be told either to move on or to keep wearing your current aligner. The system doesn't calculate wear time, but it lets us document your clinical progress consistently."*

CHAPTER 15

# Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | During active treatment and, for those who enroll, in the optional retention program. |
| **GOAL** | Spot deviations early and maintain ongoing supervision between office visits. |
| **WHAT TO DO** | ***Active Treatment***<br>1. Require a weekly smartphone scan before each aligner change and have the patient wait for instructions before moving on.<br>2. If fit is inadequate, consider having the patient wear the current aligner longer, reiterate the prescribed wear time and the use of aligner chewies when indicated, then monitor.<br>3. If the problem persists, move up the checkup and assess whether to continue, take a new scan, or proceed with a refinement.<br>4. If smartphone scans are missing or inadequate, get the patient back on schedule; if necessary, increase the frequency of checkups and reduce the number of aligners dispensed.<br>5. Generally schedule checkups every four to six months, adjusting the interval to the case.<br>***Retention — Optional Program***<br>1. Agree on the duration and cost: in the protocol described here, up to three years at 500 euros a year.<br>2. Require monthly smartphone scans taken at home; schedule a checkup, dental cleaning, and intraoral scan every six months.<br>3. Make the guarantee contingent on active enrollment in the program, proper retainer wear, scans, checkups, and prompt reporting of problems. |
| **WHAT TO CHECK** | Scan quality and regularity, fit, progress, and response to instructions. Monitoring does not measure wear time and does not replace all in-person checkups. During retention, check that the conditions of the guarantee are being met. |

CHAPTER

# 16

# When It's Really Time to End Treatment

*"For everything there is a season, and a time for every matter under heaven."*

— ECCLESIASTES 3:1

For a long time, ending treatment before achieving every goal felt like admitting defeat. If something still needed correcting, I felt it was almost my duty to find another way to do it. My name was on the treatment, and I continued to see the incomplete result as my personal responsibility, even when the patient's adherence had remained insufficient.

The greatest difficulty came when I knew what I wanted to achieve, but the patient continued not to wear their aligners or orthodontic elastics consistently enough. We had clarified the instructions, addressed the difficulties, and tried to get treatment

back on track, without achieving a sufficiently stable change. I could plan another series of aligners, modify the attachments, or suggest more frequent checkups, but none of those measures could, on its own, make up for the adherence that was still lacking.

In those situations, each new attempt let me put off a decision I found hard to accept. As long as treatment remained open, the incomplete result still seemed provisional. Declaring it finished meant acknowledging that some goals had not been achieved and that I could not achieve them through clinical work alone.

This chapter addresses that very moment: when, after reassessing the causes of the lack of progress and trying to restore adherence, the patient's commitment remains insufficient for treatment. There may still be an incomplete correction, a rotation, or contacts that need improvement: goals that still have clinical value but require a level of participation we cannot count on.

**How do we decide whether to offer one last chance, change treatment methods, or end the active phase?**

The answer requires us to consider both what remains to be done and whether it is realistically achievable. We need to assess the importance of the remaining goals, the current clinical situation, and the behavior observed after our attempts to restore adherence. Knowing that a result could be better is not, by itself, enough to justify another phase if the necessary adherence is still lacking.

I have learned that making this decision is as much a part of the orthodontist's responsibility as planning treatment at the outset. It requires us to be transparent about the result, discuss alternatives, and recognize what we can achieve under the present conditions, without treating every unmet goal as a personal failure.

When there is still a chance to restore adherence, we can offer one last opportunity. But it must have clear conditions and a set time for reassessment; otherwise, it risks becoming just another fresh start.

## THE PRACTICAL SCRIPT

The practical script has five parts: define a recovery window, assess whether treatment has resumed progressing, consider changing treatment methods, decide whether to end the active phase, and, if so, bring it to a close. What happens after reassessment depends on the clinical response and the alternatives appropriate for the patient.

#### 1. Define the Recovery Window

Before proposing a recovery window, I reassess the possible causes of the lack of progress and confirm that the patient has received clear, consistent instructions. I need to distinguish insufficient adherence from difficulties related to treatment planning, biological response, or understanding the instructions.

With the patient, I define the behavior required, how long they need to maintain it, the clinical data we will use for reassessment, and the alternatives we will consider if progress remains insufficient. This is what I call a **recovery window**.

Its length depends on the case and the movement we need to observe. We must agree on the end date from the start, along with specific instructions for wearing aligners and orthodontic elastics, changing aligners, submitting monitoring scans, and attending checkups.

The conversation can begin this way:

> *"I want us to find out whether we can get your treatment back on track, but this time we need to be clear about how we'll assess the result. Until your next checkup, you'll need to wear your aligners and elastics as directed, change aligners on schedule, and submit the scans we've agreed on.*
>
> *At the checkup, we'll compare the fit and the movement achieved. If we see enough progress, we can continue. If we're still in the same situation, we'll need to consider a different treatment method or, if the clinical conditions allow it, ending the active phase."*

That last part is essential. If we do not explain in advance what will happen if progress is insufficient, we will again be tempted to offer another chance at the reassessment. **One last chance loses its meaning if it is automatically renewed.**

#### 2. When We Can Continue with Aligners

At the checkup, I assess the fit, compare the scans, and observe the movement achieved, interpreting these findings alongside the other clinical information. What the patient tells me helps reconstruct the course of treatment, but it does not replace verifying its effectiveness.

If the patient's behavior has changed and treatment has resumed progressing, we can continue with aligners. We do not need to expect the patient to have become perfect. We need to confirm that adherence is sufficient to achieve the remaining goals and that the progress justifies continuing.

A brief improvement in the days before the checkup may not be enough. The window must allow us to observe consistency appropriate to what we still need to achieve. If monitoring or more frequent checkups helped restore adherence, I keep that support in place for as long as needed to make the improvement stick.

#### 3. If We Need to Change Treatment Methods

When a goal remains clinically appropriate but the patient cannot meet the demands of aligner treatment, we can consider a treatment method that depends less on wear time. A treatment can be clinically indicated yet prove difficult for a person to sustain in daily life. I prefer to explain it this way:

> *"Aligners can achieve the movement we need, but they require consistent wear, and so far we haven't been able to maintain that consistency. We've tried several times to get treatment back on track, and I have to recognize that this method is proving difficult to fit into your daily life. If you still want to pursue the remaining goals, we can consider an appliance that depends less on wear time. I'll explain what options it offers and what commitment it requires, so you can decide whether to take on this new phase."*

Changing methods requires a new assessment. We need to discuss indications, limitations, risks, timelines, and cost, while also clarifying the adherence the new appliance will require. Replacing the treatment method does not automatically guarantee the result.

If we propose a new phase, whether through a different method or a fresh start with aligners, we must also clarify its financial terms. Scans, clinical reassessment, treatment planning, appliances, appointments, and team time all represent real work, even when the manufacturer's package still allows us to order more series of aligners.

Any additional fee must be consistent with what was agreed upon at the outset and with the reasons for starting again. A routine refinement and the repetition of work rendered ineffective by persistently insufficient adherence call for different assessments.

There is no universal amount. The fee depends on the work required, the costs, the complexity of the new phase, and the terms communicated and agreed upon. It should not appear out of nowhere in the middle of a conflict: the possibility of an additional financial investment should be explained in advance and discussed precisely before proceeding.

Payment, however, does not prove that adherence will change. The patient may accept the cost and keep the same habits. The financial terms must therefore be accompanied by a concrete plan, with required behaviors and criteria for assessing progress.

**Before starting again, we must clarify both the work required and what will make this phase different from the previous ones.**

#### 4. If We Need to End the Active Phase

If we are considering ending the active phase, we must distinguish the reasons leading us to that choice.

In one situation, the result achieved is clinically acceptable, even if it does not perfectly match the ideal result we planned. What remains incomplete is of limited importance, and the expected benefit of another phase seems modest relative to the time and commitment it would require.

In another situation, clinically significant goals remain, but the patient is unable or unwilling to meet the conditions necessary to achieve them. They may refuse to change treatment methods, decline the additional financial investment, or be unwilling to make the required commitment.

We can then consider ending treatment with a compromise, provided the remaining condition is clinically acceptable and the risks have been assessed and explained. The difference lies in why we stop: in the first case, we believe what remains to be done does not justify another phase; in the second, we recognize the value of the remaining goals, but the patient chooses not to undertake the treatment needed to achieve them.

The compromise must be clear to the patient. We must describe what has been achieved, what remains incomplete, the alternatives proposed, and the possible consequences of the decision. If the remaining condition presents unacceptable risks, simply providing a retainer does not solve the problem: we need to propose an appropriate solution and manage any refusal while ensuring continuity of necessary care.

For this conversation, I would use a script like this:

> *"The result we've achieved still doesn't fully match what we planned, and I want to be transparent about that. We've tried several times to continue, but at this point I don't see enough reason to expect another series of aligners, worn in the same way, to produce a different result.*

> *Continuing would add more time to treatment without a reasonable expectation of improvement. That's why I believe we need to choose a different path.*
>
> *If you want to pursue the remaining goals, we can consider another method. I'll explain its indications, limitations, the commitment it requires, timeline, and costs. If instead we consider ending the active phase, we'll need to confirm that the result we've achieved is clinically acceptable and clarify which goals remain unmet.*
>
> *I'll show you what we've achieved, what remains to be done, and what consequences stopping could have. If we do end this phase, we'll remove the active-treatment components, provide an appropriate retainer, and schedule the necessary checkups.*
>
> *We'll continue to care for you. Ending this phase means recognizing the limits of the current treatment plan and deciding how to protect the result we've achieved."*

The patient may be disappointed, ask for another refinement, or say they are willing to continue for as long as it takes. We must listen and understand what they hope to achieve. Their willingness, however, does not automatically mean that continuing treatment is clinically appropriate.

The decision brings together two responsibilities: the patient decides what commitment they are willing to make and which proposals to accept; the clinician identifies the clinically appropriate options and explains the reasons for them.

The patient may also choose not to continue. In that case, they must understand what remains incomplete, the possible consequences, and the available alternatives. The refusal must be documented, and the patient must be given the instructions needed to manage the remaining clinical situation. Respecting that choice does not mean declaring the result achieved ideal.

#### 5. Bringing Treatment to a Close

When the decision is to end treatment, the active phase must actually be brought to a close. It is not enough to stop ordering aligners or leave the patient wearing their last aligner while waiting to see what they will do.

I document the final clinical situation, the unmet goals, the alternatives proposed, and the patient's choice. I remove the attachments and any other components that should not remain, provide a retainer and clear instructions, and schedule the necessary checkups.

When I end treatment because of insufficient adherence, I do not use the stabilization protocol described in earlier chapters. That protocol requires another nine months of consistent aligner wear. If that consistency was lacking during the active

phase, I cannot base stabilization on the expectation that the patient will maintain it now.

Instead, I move directly to retention, adapting it to the remaining clinical situation. This phase also requires adherence: we must explain how it works, its limitations, and the checkups it requires, without implying that the result will be maintained regardless of the patient's behavior.

Finally, the entire team must know the decision. Appointments and communications must reflect the transition from active treatment to retention. The patient must know what has ended and what care will continue.

## Why This Protocol Works

### A Commitment We Can Assess

The recovery window turns a general promise into a defined behavior. The patient knows what is required of them, how long they need to maintain it, and when progress will be assessed. The orthodontist also has a set date to reassess the case, preventing the last chance from stretching on without a decision.

### Clinical Assessment Without Seeking a Confession

Comparing fit, scans, and the movement achieved allows us to address treatment progress without making the patient's honesty the focus of the conversation. What the patient says helps us reconstruct what happened, but the decision requires clinical assessment.

We must also interpret the findings in light of possible biological and biomechanical causes: a lack of movement does not automatically prove insufficient adherence.

### Consistency in Following Through

If every deadline is pushed back and every reassessment leads to another fresh start, the announced consequence loses its meaning. Making a decision at the agreed-upon deadline makes the boundaries of treatment credible.

This matters for the orthodontist, too: defining the alternatives in advance helps prevent a difficult conversation from automatically leading to another series of aligners.

### Clear Alternatives

Presenting the different options helps the patient understand that the treatment plan can change. We can continue with aligners when progress has sufficiently resumed, consider another method, or end treatment when clinically appropriate.

Clear alternatives allow the patient to express their preferences and priorities. The orthodontist remains responsible for identifying appropriate options and explaining their benefits and limitations.

### An Informed Investment

When a new phase involves a cost, clarifying it in advance makes the work required and the terms of starting again explicit. The patient can consider how important the result is to them and what commitment they are willing to make.

Payment does not guarantee better adherence. The additional investment must be accompanied by a concrete plan, with required behaviors and criteria for assessing progress.

### Continuity of Care

Explaining what will continue after treatment ends helps reduce the fear of being abandoned. Retention, checkups, and appropriate instructions make continuity of care tangible.

Documentation preserves the reasoning behind the decision; communicating it to the team prevents inconsistent appointments or messages from leaving the patient with the impression that treatment is still open.

A treatment that remains open but can no longer make progress does not consume only time and resources. Little by little, it also begins to wear down the people who keep it going.

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## Adapting Communication to DISC Profiles

**D**

### Dominance

I communicate directly, focusing on the decision and the alternatives:

> *"Treatment isn't making enough progress, and I don't think repeating another series under the same conditions would be useful. We can consider a different method or, if the clinical situation allows it, end the active phase and protect the result with an appropriate retainer."*

I emphasize the criteria for choosing and the expected benefit, avoiding explanations that wander off track.

**I**

### Influence

I acknowledge the enthusiasm that comes with starting again and bring the focus back to consistency:

> *"Each time we start again, we're highly motivated, but then it becomes difficult to keep up the routine. Before we start over, we need to choose a plan you can stick with after the first few weeks, too."*

I keep a positive tone and connect motivation to a concrete plan whose progress we can assess.

**S**

### Steadiness

I emphasize continuity of care and guide the patient through the decision:

> *"We'll continue to care for you. Now we need to consider whether to change methods or end this phase, protecting what we've achieved and scheduling the necessary checkups."*

I leave room for questions and clarify which aspects of care will continue after treatment ends.

**C**

### Conscientiousness

I use data, comparisons, and criteria:

> *"We've compared the scans, the fit, and the movement achieved. Progress remains insufficient relative to the goals we still need to meet. I'll show you what we've achieved, what remains incomplete, and the available alternatives, along with their limitations."*

I make clear what led to my recommendation and distinguish what we've observed from what we expect.

CHAPTER 16

## Protocol at a Glance

| | |
|---|---|
| **WHEN TO USE IT** | When treatment has stalled and we need to decide whether to continue, change methods, or end treatment. |
| **GOAL** | Make a clinically appropriate decision and ensure continuity of care. |
| **WHAT TO DO** | 1. Reassess the treatment plan, clinical response, and adherence, distinguishing temporary difficulties from persistent behaviors.<br>2. Confirm that the instructions are clear and consistent across the entire team.<br>3. When indicated, agree on a recovery window that defines the required behaviors, its duration, criteria for assessing progress, and alternatives.<br>4. Assess change through fit and the movement achieved: a new promise is not enough.<br>5. Continue if conditions improve; consider another method if the remaining goals are still clinically appropriate; end treatment when appropriate, accepting only clinically sound compromises.<br>6. Document the goals achieved and those still unmet, the alternatives, and the patient's choice; clarify any additional costs.<br>7. If ending treatment, remove the active-treatment components, provide a retainer, and schedule checkups. In cases of insufficient adherence, the protocol calls for moving directly to retention, without a stabilization phase.<br>8. Make sure team appointments and communications reflect the decision. |
| **WHAT TO CHECK** | The causes of the stalled progress, the response during the recovery window, the feasibility of the alternatives, whether the compromise is clinically acceptable, and the patient's understanding of what comes next. Ending the active phase does not mean ending care. |

CHAPTER

# 17

# The Cost We Hadn't Counted

*"The mind must be allowed to rest: after resting, it will rise refreshed and sharper."*

— SENECA

Some days, everything seems to have gone well. You started on time, saw eighty patients, solved problems, delivered new aligners, checked on treatments progressing exactly as you planned, and met with satisfied parents, cooperative kids, and adults excited about their smiles. You may even have managed to finish almost on time, which on a day like that can count as a small victory.

Then you close the office door, say goodbye to the staff, get in your car, and head home. That's when something strange happens: you saw eighty patients that day, and things went well with the vast majority of them, but they aren't the ones on your mind. It's the two or three who took up an amount of energy completely out of proportion to the time you spent with them.

There's the one who assured you they'd worn their aligners consistently, while the fit told a completely different story; the one who, yet again, didn't bring their aligners and forced you to piece together what might have happened over the past few weeks; the one who, after months of questionable adherence, demands another refinement as though it were the natural next step in treatment. Or there's the patient who has just finished treatment with a result you consider clinically excellent. Looking at the photographs and the scan, she admits that her teeth look perfectly straight on the screen, but then adds that when she looks in the mirror, she still sees something she doesn't like.

**They all left the office a while ago, but some of them got into the car with you.**

This is probably one of the least discussed consequences of poor adherence. In the preceding chapters, we tried to measure many of the others, talking about time, appointments, scans, refinements, materials, costs, and staff hours. We saw how an additional series of aligners is not simply a box that arrives at the office, and how every treatment that has to be restarted involves a great deal of invisible work that rarely shows up in our accounts.

But there is a cost we haven't counted yet, because it doesn't appear in our practice management software, we can't enter it on the practice balance sheet, and no one sends us an invoice at the end of the month: it's the **mental energy** we use up every time we have to deal once again with situations we would rather not have had to handle. Every patient who needs an explanation repeated, every appointment where we have to piece together what really happened, every promise of adherence we know has been made before, every request we consider unreasonable, and every decision we are forced to make again uses up a little of our resources.

Any one of these situations means little on its own and is simply part of our work. The problem comes from the repetition and, above all, the disproportion. It isn't necessarily the eighty patients we see in a day who wear us down: seventy or seventy-five of them may leave us with more energy than we had in the morning, because seeing a treatment go well, a happy child, or a patient who finally smiles is one of the reasons we continue to love this work. Instead, it may be those two or three patients who, despite taking up only a small part of the schedule, occupy a huge part of our thoughts. When this pattern repeats over weeks, months, and years, what seemed like a mere daily annoyance can become something very different.

Dental research has examined occupational stress and burnout for many years, and it would be simplistic, as well as scientifically unsound, to look for a single cause. Hours worked, time pressure, financial concerns, practice management, relationships with coworkers, and clinical and administrative responsibilities are

just some of the factors that can contribute to the demands of the job. Among them, though, one thing directly related to the subject of this book comes up repeatedly: the difficulty of dealing with certain patients.

Studies of dental teams have shown that so-called difficult patients can represent a real job demand, and that greater exposure to these relationships is associated with emotional exhaustion and burnout. In a study of more than six hundred dentists, the practitioners even estimated that about a quarter of their patients could be considered problematic or difficult to manage in some way, and perceiving a higher proportion of such patients showed a particularly significant association with burnout.

Of course, this does not mean that a difficult patient causes burnout, much less that every patient with poor adherence should be considered a difficult patient. The research does not support that conclusion, and it would make no sense in our day-to-day practice, because most patients who forget an orthodontic elastic, fall a few hours short of the prescribed wear time, or go through a period of reduced adherence will never pose a significant problem for us.

It becomes more interesting when certain situations stop being occasional and begin to recur, continually demanding time, attention, decisions, negotiation, and sometimes conflict management. At that point, they are no longer just clinical events; they become part of the workload. An aligner that isn't worn doesn't mean only a few fewer hours in the patient's mouth: it can mean a longer appointment, another explanation, an unplanned scan, a schedule change, a phone call from the front desk, a conversation with a parent or the patient, a new treatment plan, and sometimes a discussion about who should pay for what needs to be done.

A seemingly simple behavior can therefore set off a cascade of consequences that involves many people and uses up far more resources than we see when we look only at the patient's mouth.

Before the problem even reaches us, someone may have had to answer the phone, find a new appointment in an already full schedule, call a patient who missed an appointment, or remind them what had been agreed upon. In many cases, the front desk is the first to hear a patient's dissatisfaction, complaint, or frustration, before we even know there's a problem.

Research on dental teams shows that receptionists, assistants, hygienists, practice managers, and clinicians can be affected in different ways by difficult relationships with patients. Qualitative studies describe something many of us have probably observed without ever giving it a name: the tension generated by a single patient can spread through the practice. The receptionist takes a difficult call and

tells the assistant about it; the assistant warns the doctor before the appointment; the doctor enters the room already knowing a difficult conversation lies ahead; and if the appointment runs longer than expected, the delay affects the patients who follow.

In this way, a problem that began in one person's mouth can change the day for many others. That is why protecting the team cannot simply mean teaching them to tolerate difficult patients better. Above all, it means avoiding, whenever possible, having to handle each situation as though it were happening for the first time.

Starting from scratch each time means asking the team to find an answer and the doctor to make another decision. I believe a substantial part of the strain comes from this: besides managing the consequences of a patient's behavior, we have to decide how to proceed every time. The patient hasn't worn their aligners, and we have to decide whether to let them continue or stop; they haven't worn their orthodontic elastics, and we have to figure out how much we can trust what they tell us; the fit is off on just one tooth, and we have to choose whether to take a new scan immediately or continue; they disappeared for a year and call today saying they want to start again, so we have to decide where to pick up and what costs they will have to cover; they've already had several refinements and keep asking for changes to details we consider clinically irrelevant, forcing us once again to decide where to draw the line.

Taken individually, these are simple problems, and an experienced clinician is perfectly capable of solving them. The problem arises when we have to solve them hundreds of times, perhaps trying to remember what we decided six months earlier with a similar patient, explaining to the staff why we're doing something different this time, and later wondering whether another approach would have been better.

This is where a model widely used in the literature on work-related stress becomes relevant: the Job Demands–Resources model. Put very simply, it distinguishes between the demands our work continually places on us, or job demands, and the resources we have available to meet them, or job resources. Demands require and consume energy, while resources help us manage those demands while maintaining control, motivation, and effectiveness. This model has also been studied specifically in dentists, and some longitudinal studies involving thousands of clinicians have shown an association between high job demands and burnout and, on the other hand, the positive role of job resources in supporting work engagement.

Reading these studies, I realized that, without claiming a scientific significance for the term that the studies have not directly established, we can also think of a

protocol as a resource in our day-to-day practice: if a situation recurs and we've already decided how to handle it, we don't have to start from scratch every time.

## That's Why This Book Has Protocols

At this point, we can finally look back and see everything we've built in the preceding pages from a different perspective. If you've read this book from the beginning, you've encountered many protocols, many scripts, and probably a few rules that may have seemed rigid to you, at least at first. We've talked about the patient who doesn't wear their aligners, the one who doesn't wear their orthodontic elastics, the one whose aligners lose their fit or who doesn't wear their retainer, the one who disappears and then returns a long time later, the one who needs a routine refinement and the one whose lack of adherence forces us to repeat a phase of treatment, all the way to the patient who keeps asking for something more when we believe we've already achieved the result.

For each of these situations, we've tried to decide in advance what to do—not because we want to turn medicine into a rulebook, or because every patient should be treated the same way. A protocol should never replace a doctor's judgment, and there will always be situations where making an exception is the right thing to do. The fundamental difference is that the exception should be a conscious choice on our part, not the result of our having failed to decide what to do.

After so many years in practice, this distinction has become essential to me, because a protocol doesn't just tell me how to manage the patient; it saves me from having to reinvent how I handle the same problem every time. Likewise, a script doesn't just help me find the right words: it makes sure those words don't depend on how tired I am, the time of day, whether I'm running late, or how much that patient has managed to get on my nerves. Above all, it helps everyone in the practice speak the same language.

If the front desk already knows what happens when a patient disappears, they don't have to ask me every time; if the team knows how we handle a refinement needed because of poor adherence, they don't have to guess what to do; if the patient knows from the outset what the consequences of certain behaviors will be, the later conversation becomes easier. We aren't making up a rule when the problem arises; we're applying something we've already discussed with them.

None of this eliminates difficult patients or prevents poor adherence, but it can eliminate a substantial part of the effort involved in managing those situations. And that may be why, now that I've reached the end of this book, I see scripts and

protocols not just as organizational or clinical tools, but as a form of protection: they let the pilot and crew know what to do even when a passenger decides to behave in a way we had hoped they wouldn't.

That protection also begins with recognizing the limits of our control. With aligners, we've developed extraordinary tools for planning tooth movement: we plan in tenths of a millimeter, degrees of torque, intrusion, expansion, sequences, attachments, and staging, spending hours in front of a simulation trying to predict what will happen months later in someone's mouth.

We can design the best possible treatment, but we can't wear the aligners for the patient; we can explain how important orthodontic elastics are, but we can't be there every evening to check that they've put them on; we can remind them of appointments, but we can't force them to show up; we can show them photographs, scans, and results, but we can't decide for them whether they'll be satisfied. Above all, we can't make them tell us exactly what happened every time.

I used to believe that getting better at this work meant being able to control more and more things. To some extent, I still believe that, because clinical experience is precisely what improves our ability to predict, prevent, and manage what might happen. But there is another kind of experience, one that probably takes longer to acquire: learning to recognize what we will never be able to control and no longer spending energy trying to control it.

When I started writing this book, I thought it would be primarily about patient adherence. By the time I reached the end, I realized it had also become a book about us and what we can do to make a treatment easier when, by its very nature, it depends to a considerable extent on another person's behavior.

None of the scripts, rules, costs, protocols, or strategies described in the preceding pages is meant to punish patients who don't follow instructions or to create a practice where there are no exceptions. They are meant to establish clear boundaries, understandable responsibilities, and predictable consequences, so that our relationship with the patient can remain as positive as possible even when something doesn't go as planned. They certainly help protect the clinical result, the practice's time, and the team's work. But having written this book, I believe they also help protect something I took almost for granted for many years: the pleasure of doing this work.

Because in the long run, the risk isn't just losing a few hours or having to order a few more aligners. The risk is allowing a very small percentage of our patients to take up too large a share of our energy, until we forget that the vast majority of the

people we saw that same day gave us a sense of satisfaction, trusted us, followed the treatment plan, and perhaps reminded us exactly why we chose this profession.

After eighty patients, then, I'd like to be able to go home remembering them above all. Not because the others don't exist or because problems should be ignored, but because those problems need a place, a time, and a response. Once we've built a system that can provide one, we can lighten the burden we keep carrying home even after our work is done.

Perhaps the real success of all the protocols described in this book isn't measured only by the number of refinements they help us avoid, the time they give back to the schedule, or the costs they help contain. It's also measured by the decisions we no longer have to make, the discussions we no longer have to improvise, the calls the staff already knows how to handle, and the peace of mind that comes from accepting that part of the treatment will never be entirely up to us.

In the end, perhaps this whole book can be summed up like this: we can't stop patients from forgetting, not following instructions, disappearing, changing their minds, or giving us an account of what happened that doesn't match reality. But we can decide in advance what to do when it happens, protecting the treatment, our practice, the people who work with us, and, above all, the pleasure of continuing to do this work.

**Because we can't stop a patient from lying, but we can keep that lie from coming home with us.**

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